

Carrier Universality and Global Smoothness for the Periodic 3D Navier–Stokes Equations

A closure-induced impossibility proof of Fefferman’s periodic statement (B)

Amos Jay Maley

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Main endpoint theorem (Fefferman (B), periodic case)

Classical endpoint. Let $\nu > 0$ and let $u_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$ satisfy $\nabla \cdot u_0 = 0$. Then there exists a unique smooth periodic Navier–Stokes solution pair (u, p) on $\mathbb{T}^3 \times [0, \infty)$ satisfying

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u|_{t=0} = u_0,$$

with the fixed mean-zero pressure gauge. This statement is obtained first as Corollary 21A and then restated in the official classical form as Theorem 21.17.

Proof-class, claim-class, and source-use locks

Proof class. This manuscript is a mathematical non-PDE-native constraint-formalism impossibility proof of Fefferman’s periodic Navier–Stokes statement (B). It is not a continuation-estimate proof, not a constructive solution-formula proof, and not a claim that admissibility supplies a missing vortex-stretching bound.

Claim class. The proof fixes the official periodic problem as a non-degenerate same-scope proof/refutation regime. It then shows that a hypothetical finite-horizon breakdown induces a same-scope endpoint condition, that such a condition has only the endpoint statuses classified below, and that no licit same-scope finite-horizon counterexample completes. The conclusion is obtained by impossibility of counterexample completion, not by an analytic continuation template.

Clay/Fefferman method-neutrality. The official Clay rules make eligibility depend on a complete mathematical solution that satisfactorily answers the questions in the official problem description after publication, scrutiny, and general acceptance; they do not prescribe a PDE-native continuation-estimate template, a vortex-stretching bound, a weak-to-strong regularity route, or a constructive solution formula as the required method class [Cla22]. Fefferman’s official Navier–Stokes description likewise states that solvers are given reasonable leeway while retaining the heart of the problem, and it asks for proof of one of four official alternatives, including periodic existence and smoothness statement (B) [Fef06]. Thus a PDE-native method is not a Clay-rule prerequisite. What is required here is still the full mathematical endpoint: a proof of the official periodic statement itself.

Source-use lock. The live proof chain is restricted to theorem-local material in the

manuscript, Appendix A, Appendix B, Appendix E for the analytic atlas used by Part I, and public official-problem/source-alignment material where cited. Private notes, companion slogans, validation packets, and non-load-bearing audit files do not carry the endpoint. Appendix H records the source-governance and Lean-support boundary.

Clay rules do not require a PDE-native proof method

The phrase “Navier–Stokes problem” names the official PDE endpoint, but the Clay rules do not require a proof to be PDE-native in method. They require a complete mathematical solution to the Problem as defined in its official description, publication in a qualifying outlet, at least two years of scrutiny, general acceptance in the global mathematics community, and CMI examination [Cla22]. Fefferman’s official description is also method-neutral in this sense: it grants reasonable leeway to solvers while retaining the heart of the problem and asks for proof of one of the four stated alternatives [Fef06].

Accordingly, a referee objection of the form “this is not a PDE continuation estimate” identifies the proof class but does not by itself refute the manuscript. A same-mode objection must show that the non-PDE-native constraint-formalism route fails to prove the official periodic statement (B), fails to preserve Fefferman’s data/equation/gauge/continuation target, or fails at one of the theorem-local links in the declared proof spine. This paragraph does not relax the endpoint: the theorem still proves the classical periodic PDE statement stated above.

Counterexamplehood lock

A finite-time expression such as $T_*(u_0) < \infty$ is not by itself a theorem-bearing refutation of Fefferman (B). To refute the official universal statement it must be admitted in the ambient proof/refutation regime as an official datum, an official realized maximal classical lineage, a finite-horizon condition, and a same-scope non-continuation claim. Ordinary ZFC or classical-analysis counterexamplehood therefore already requires admissible formation, proof-standing, fixed reference, same-target fidelity, and final theorem consequence. Those are precisely the kernel roles at theorem-bearing use level. If an alleged object lacks those roles, it is not a counterexample in the ambient proof system either.

The relevant theorem-local nodes are Theorem 3.K3, Lemma 3.K4, Theorem 3.K5, and Corollary 3.K6.

Read this first: fixed-scope horizon proof spine

The live proof route is the following single spine:

official periodic proof/refutation regime $\Rightarrow K_{\text{per}}^{NS} \Rightarrow \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ unique
 $\Rightarrow \mathcal{R}_{NS}(x)$ is one same-scope realized lineage $\Rightarrow [T_*(x) < \infty]_i \Rightarrow B_{\text{hor}}^{NS}(x, T_*(x))$
 $\Rightarrow$ endpoint-status trichotomy $\Rightarrow$ non-witness branches collapse
 $\Rightarrow$ standing-positive witness branch is exhausted
 $\Rightarrow T_*(x) = \infty \Rightarrow$ Fefferman (B), periodic case.

The notation $[T_*(x) < \infty]_i$ denotes the local exact-complement countercase opened in the endpoint proof. It is not an additional object-level premise. The horizon bridge object

$B_{\text{hor}}^{NS}(x, T_*(x))$ packages only the finite-horizon counterexample data: fixed carrier, realized-lineage identity, pre-horizon witness material, horizon discriminator, metric-bookkeeping discipline, and branch extraction. It does not contain the conclusion $T_*(x) = \infty$.

The detailed theorem ladder is recorded in Appendix G. The route-locus audit for analytic and endpoint alternatives is recorded in Appendix F.

No one-step counterexample promotion

The proof does not use the inference

$$T_*(x) < \infty \implies \text{completed counterexample to Fefferman (B).}$$

The route is instead

$$[T_*(x) < \infty]_i \Rightarrow \text{finite-horizon endpoint condition} \Rightarrow B_{\text{hor}}^{NS}(x, T_*) \Rightarrow \text{trichotomy} \Rightarrow \perp.$$

The contradiction discharges the local finite-horizon assumption itself. It does not discharge a strengthened premise such as “finite-time breakdown plus official endpoint-governed negative status.”

Vortex-stretching seam discipline

The vortex-stretching channel, BKM-type quantities, critical norms, metric divergences, and profile/concentration diagnostics remain valid analytic diagnostics and obstruction witnesses. The manuscript does not bound them by a new estimate. Their endpoint role is narrower: they do not become independent same-scope endpoint authority. On the live route they are licensed metric bookkeeping or quotient-visible witness bookkeeping subordinate to the finite-horizon endpoint condition and the carrier-side horizon discriminator. If promoted as independent continuation authority, they become an illicit same-scope metric gate or a scope move.

This seam is localized in Proposition 13.23A.3, Proposition 13.23A.4, Theorem 13.23A.6, Theorem 20.11, and Appendix F.

Objection	Theorem-local denial target
No PDE-native estimate is supplied.	Clay/Fefferman method-neutrality box; proof-class lock; vortex-stretching seam discipline; Theorem 20C.4.
A classical counterexample escapes the kernel.	Theorem 3.K3; Theorem 3.K5.
Finite T_* automatically refutes Fefferman (B).	Theorem 20C.0A; Theorem 20C.4; horizon reductio in Section 20.2.
The realized lineage can be admissible before T_* and non-admissible at T_* .	Theorem 20B.1; Theorem 20B.2; Theorem ST4.
Metric divergence carries endpoint authority.	Proposition 13.23A.3; Proposition 13.23A.4; Theorem 13.23A.6.
A weak/profile/alternative formulation is a counterexample.	Fixed-domain comparison class, Definition 3.4A; route-locus L8 and L14 in Appendix F.
Part III reopens primitive admissibility.	Corollary 13L; Proposition 18A; Proposition 18C.
The endpoint does not restate Fefferman (B).	Theorem 21.15; Corollary 21A; Theorem 21.17.

Abstract

This manuscript proves the periodic three-dimensional Navier–Stokes endpoint in Fefferman’s statement (B) by a fixed-scope constraint-formalism impossibility route. The argument does not add a vortex-stretching estimate or a new continuation criterion. It proves that the official periodic proof/refutation regime instantiates the fixed-scope kernel, identifies the standard periodic carrier with the unique admissible interior, realizes every official smooth divergence-free datum as one same-scope maximal classical lineage, and then treats a hypothetical finite horizon as a local exact-complement countercase. The horizon bridge object associated to that countercase is exhausted by a three-way endpoint-status classification: standing-positive witness, terminal standing failure, or illicit same-scope gate/scope move. The non-witness branches collapse locally, and the remaining witness branch is exhausted and eliminated by the branch-forcing packet. Hence no finite-horizon same-scope counterexample completes, so every official periodic smooth datum has a unique global smooth periodic solution with the fixed mean-zero pressure gauge.

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Part I. Compressed Analytic Wall Statement

Burden status at entry. Nothing primitive or carrier-specific has yet been certified. Part I carries only the analytic-side burden of the paper: isolate the common vortex-stretching wall, show that every standard same-scope analytic route meets that wall, and compress the analytic search space into a finite atlas that Part II can then reclassify. What remains after Part I is *not* a PDE continuation theorem, but the theorem-visible statement that any successful same-scope continuation argument would require a discriminator not supplied by the existing analytic bundle.

Compression note. The detailed route-by-route analytic atlas is moved to Appendix E in this edition. The main line retains only the shared bottleneck formulation and the atlas-closure theorem needed for Part II's pivot.

1 The shared bottleneck (“the same wall”)

We formulate the common endpoint in a single statement.

Proposition 1.1 (Common bottleneck formulation). *All standard analytic routes catalogued in Appendix E reduce global regularity to establishing at least one of the following critical upgrades for every energy-class solution on $(0, T)$:*

- (a) *a scale-critical spacetime bound (Serrin–Prodi, $L_t^\infty L_x^3$, $L_{t,x}^5$, or equivalent Besov/Morrey control);*
- (b) *a scale-critical vorticity bound (e.g. $\int_0^T \|\omega\|_\infty < \infty$);*
- (c) *a rigidity/compactness principle excluding all nontrivial ancient solutions arising in Type II rescaling;*
- (d) *a depletion mechanism controlling $\mathcal{T}(u)$ without introducing extra hypotheses.*

In each case, the missing step is an unconditional bridge from energy-class invariants to critical control of the nonlocal vortex-stretching channel.

2 Analytic obstruction and atlas closure

Theorem 2.1 (Analytic obstruction). *Within the classical formulation of the three-dimensional incompressible Navier–Stokes equations on $\mathbb{T}^3$, all known analytic continuation routes require an unconditional same-scope discriminator capable of globally suppressing vortex-stretching amplification. No currently available analytic invariant, estimate, or continuation criterion provides such a discriminator.*

Proof. Proposition 1.1 isolates the shared endpoint of the standard routes: each reduces global regularity to an unconditional bridge from energy-class control to a scale-critical suppression of the vortex-stretching channel. The route-by-route survey in Appendix E shows that the standard criteria either assume critical control (smallness/integrability) or re-express the same missing bridge as an a priori blow-up condition. None supplies an unconditional bundle-internal discriminator that closes the wall. $\square$

Corollary 2.2 (Analytic incompleteness). *The classical analytic program cannot close the Navier–Stokes regularity problem without introducing a discriminator external to the existing analytic bundle.*

Definition 2.3 (Analytic continuation mechanism). Fix the standard Navier–Stokes carrier on $\mathbb{T}^3$. An *analytic continuation mechanism* is a same-scope analytic attempt to certify continuation by strengthening control over some interior feature of the admitted carrier while preserving its equations, scope constants, and licensed skins. A purported mechanism that requires additional standing-bearing bundle data or a same-scope operator family is not an analytic continuation mechanism of the standard carrier but a scope change.

Theorem 2.4 (Atlas closure). *Within the classical Navier–Stokes carrier on $\mathbb{T}^3$, every analytic continuation mechanism in the sense of Definition 2.3 must modify exactly one of the following structural features:*

- (i) *norm growth,*
- (ii) *scale transfer,*
- (iii) *geometric structure,*
- (iv) *profile formation.*

No additional analytic continuation mechanism exists without introducing new bundle structure.

Proof. Energy/enstrophy programs target norm growth. Littlewood–Paley and cascade analyses target scale transfer. Geometric depletion and alignment arguments target geometric structure. Blow-up profile and ancient-solution rigidity programs target profile formation. A same-scope analytic continuation mechanism must act on some interior feature already visible in the standard carrier skins. For that carrier, the compressed Part I statement together with Appendix E exposes only the four listed feature families. Any purported fifth mechanism would require additional standing-bearing bundle data or a new same-scope operator family not already licensed in the standard carrier. By definition, that is not a continuation mechanism of the fixed carrier but a scope change. Hence the atlas is complete. $\square$

Part II. Primitive Certification of the Unique Admissible Interior

Burden status at entry. Part I has already closed the analytic atlas and shown that no same-scope analytic discriminator closes the wall from inside the classical bundle. Part II carries the primitive-side burden: force the unique admissible interior at fixed periodic scope, identify the standard periodic carrier with that interior, and eliminate any rival *standing-positive* same-scope continuation criterion before Part III begins. The complementary standing-failure branch is not closed here; it is made explicit only after Part III adds the realized-lineage endpoint-status trichotomy / collapse. What remains after Part II is therefore not a new analytic estimate but classical realization and faithful transfer of a carrier theorem already fixed locally on the standing-positive branch.

Part II proof map. Part II carries the primitive-side burden of the paper. It does not attempt a new analytic estimate on vortex stretching. Instead it turns the analytic obstruction of Part I into a fixed-scope admissibility problem on the standard periodic carrier and organizes the manuscript around the claim that, at the fixed periodic scope, this carrier is the *unique admissible interior*. Before the carrier-specific specialization begins, Section 3 begins the structural theorem packet P1–P4, with P1–P4 all proved in this manuscript (the P4 proof core appearing in Appendix A), and then records a compact fixed-scope necessity chain: no standing-preserving repair, bivalent fail-closed admissibility, no parameterized admissibility crossing, admissible initial conditions are not variables, no admissible variation, and the theorem of the *unique admissible interior*. The theorem-visible bridge packet in Section 13.1 then proves in-paper that the fixed periodic problem instantiates those antecedents and therefore admits the explicit periodic corollary of a unique admissible interior. Section 13.2E then records the carrier-identification packet 13I–13K: the standard carrier realizes that unique admissible interior, no rival same-scope continuation regime survives, and no additional PDE-side continuation criterion remains inside fixed scope. The ordering fixes claim-type clarity: the primitive layer is made theorem-visible before Part II can be mistaken for a hidden same-scope PDE continuation remedy or an unstated PDE-side continuation ingredient. First, the Part I atlas is closed into a finite gate classification for same-scope continuation mechanisms. Second, the periodic Navier–Stokes carrier itself forces a minimal specialized rule package: no competing same-scope continuation gates, binary continuation status, no parameterized same-scope gate, and no repair / no generator in specialized NS form. Subsection 13.1A makes this forcing explicit in ordinary Navier–Stokes language. Third, every *standing-positive* continuation-relevant same-scope candidate determination on the live route is forced through the finite typed decision packet 13.5E–13.5L, showing branch by branch that no independent standing-positive same-scope continuation discriminator survives on the standard carrier. Fourth, only after that finite classification and elimination packet is in the theorem-local chain do zero-parameter certification together with the narrowed continuation-completeness package verify the standing-positive carrier-side completeness claim actually used on the live proof chain, and Theorem 13.28 verifies the Part II rule packet for this specific periodic carrier in that theorem-local sense. Subsection 13.2A records the explicit translation dictionary used on that standing-positive route, and Section 13.2D interprets the derived carrier-specific exhaustion theorem as a same-scope existence statement on the fixed carrier. The completeness claims in these subsections are therefore frozen at the domain they visibly prove rather than silently enlarged to full same-scope totality before the later endpoint-status audit. Fifth, structural regularity and branch elimination force regularity on the carrier already identified with the unique admissible interior, and the terminality package shows in Clay-facing form that no

same-scope analytic escape remains for the official periodic problem.

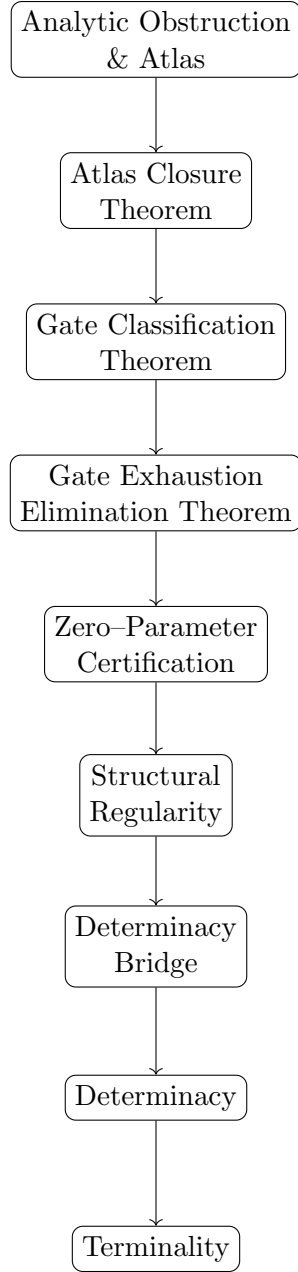


Figure 1: Dependency diagram for the carrier-side segment of the Navier–Stokes admissibility cascade. The fixed-scope primitive necessity layer is proved in Section 3 and feeds the cascade shown here.

The structural resolution proceeds through the following theorem cascade:

1. Analytic Obstruction Theorem (Theorem 2.1)
2. Atlas Closure Theorem (Theorem 2.4)
3. Gate Classification Theorem (Theorem 13.1)
4. Primitive necessity at fixed scope (Lemma 3.14, Proposition 3.15, Lemma 3.16, Lemma 3.18, Lemma 3.19, Theorem 3.20)
5. Specialized no-second-gate theorem package (Theorem 13.10, Corollary 13.11, Corollary 13.12, Theorem 13.13)
6. 13.5A–D: the periodic carrier forces the minimal rule package
7. Finite standing-positive same-scope decision packet (13.5E–13.5L, especially Theorem 13.5F and Corollary 13.5L)
8. Gate Exhaustion Elimination Theorem (Theorem 13.19)
9. Zero–Parameter Certification (Theorem 13.20)
10. Standing-positive scope-completeness theorem for the standard carrier (Theorem 13.24)
11. Architectural standing-positive continuation-completeness of the standard carrier (Theorem 13.26)
12. Verification of the Part II rule package for the standard periodic carrier (Theorem 13.28)
13. Structural Regularity Theorem (Theorem 16.5)
14. Determinacy Bridge Lemma (Lemma 16.6)
15. Determinacy Corollary (Corollary 16.7)
16. Terminality Theorem (Theorem 17.4)
17. No same-scope analytic escape for the official periodic problem (Corollary 17.5)

analytic atlas → gate classification → primitive necessity
 → minimal rule package → finite standing-positive same-scope classification
 → gate exhaustion → zero-parameter certification
 → standing-positive scope completeness
 → architectural continuation-completeness
 → rule-package verification
 → structural regularity.

3 Primitive necessity at fixed scope, the unique admissible interior, and the Part II rule package

Burden status. At this section entry the analytic burden has already been discharged by Part I. The remaining task is to prove, inside the manuscript and at fixed scope, that admissibility is bivalent, non-parameterized, and uniqueness-forcing, then transport that fixed-scope packet into the periodic specialization. Nothing in this section appeals to Part III or to any hidden classical continuation principle. Part II is theorem-explicit about its primitive dependencies. Tier A begins the structural theorem packet P1–P4: all four theorems are proved in-paper, with P4 routed through the minimal appendix gate/exhaustion core, so the primitive layer is visible inside the manuscript rather than ambient. Tier B then records, in manuscript-local language and before any Navier–Stokes-specific specialization, a compact fixed-scope necessity chain culminating in the *unique admissible interior*: non-degenerate reasoning, no standing-preserving repair, bivalent fail-closed admissibility, no parameterized admissibility crossing, admissible initial conditions are not variables, and no admissible variation. Tier C then records the explicit role and rule package that later carrier-specific sections cite modularly. The point is not to hide a PDE continuation burden behind abstract vocabulary; it is to state the structural burden first and only then specialize it to the fixed periodic problem.

3.1 Standing, scope, and same-scope parameters

Definition 3.1 (Standing and admissibility). An *admissibility judgement* is a fixed-scope standing judgement that classifies a candidate as licit or illicit at that scope. When the judgement is positive we say *standing-positive*; when negative, *standing-negative*. Proposition 3.15 will show that, under non-degenerate reasoning, no third standing-preserving status exists.

Definition 3.2 (Scope and same-scope move). *Scope* is the fixed carrier specification (domain, equations, role commitments, and licensing conditions) under which admissibility is evaluated. A *same-scope move* is any move that purports to preserve this carrier specification.

Definition 3.3 (Fixed-scope reasoning regime). A *fixed-scope reasoning regime* consists of a stock of same-scope candidates or expressions, a standing-bearing commitment state recording what has been admitted at that scope, and the licensed same-scope inference, redescription, or update moves that act on that commitment state. Any undeclared scope extension or silent change of licensing conditions falls outside the regime rather than counting as an internal move of it.

Definition 3.4 (Same-scope parameter). Within a fixed admitted scope, a *same-scope parameter* is any selector, tolerance, deferred enforcement, privileged representative, repair knob, or other admissibility-bearing degree of freedom whose variation purports to preserve scope while altering standing, legality, or consequence.

Definition (3.4A (Fixed-domain comparison class for the official periodic problem)). *For the official periodic Navier–Stokes target, the fixed-domain comparison class is the class of proposals that preserve simultaneously the torus carrier $\mathbb{T}^3$, the fixed viscosity $\nu > 0$, zero forcing, the smooth divergence-free periodic datum class, the fixed mean-zero pressure gauge, the lawful redescription class of Definition 13.2, and the same-scope continuation question fixed in Lemma 12A. A proposal that adds an independent selector, tolerance, threshold, horizon rule, or admissibility gate is not in this comparison class unless the added object is already quotient-visible bookkeeping in the sense of the later carrier packet.*

Definition (3.4B (Same-scope counterexample)). *A same-scope counterexample to the official periodic endpoint is a purported finite-horizon failure proposal that belongs to the fixed-domain comparison class of Definition 3.4A and nevertheless claims to change the endpoint status of the realized periodic lineage. A proposed failure that changes one of the fixed-domain commitments is a different-domain proposal rather than a same-scope counterexample to the theorem proved here.*

Definition (3.4C (Non-degenerate theorem-bearing regime)). *A non-degenerate theorem-bearing regime is a fixed-scope reasoning regime in which all of the following are present: determinate reference, a fixed comparison domain, standing-bearing consequence claims, lawful redescription identity, irreversible commitment, fail-closed admissibility, nontrivial admitted content, and no arbitrary post-hoc repair of failed standing. This definition is intentionally method-neutral: it applies to a PDE regime, algebraic regime, formalized kernel, or any other mathematical setting once it is used to make determinate theorem-bearing same-scope claims.*

Definition (3.4D (Kernel instantiation)). *A theorem-bearing regime instantiates the fixed-scope kernel when its admissibility, standing, reference preservation, irreversibility, redescription invariance, and fixed-scope exhaustion govern which consequence claims remain licit at that scope. Kernel instantiation is not the adoption of an optional vocabulary; it is the presence of the minimal governance structure required for non-degenerate same-scope consequence.*

Theorem (3.K (Kernel Necessity for non-degenerate regimes)). *Every non-degenerate theorem-bearing regime in the sense of Definition 3.4C instantiates the fixed-scope kernel in the sense of Definition 3.4D. Consequently, kernel governance is not an optional modeling convention, interpretive preference, or external formalism imposed on such a regime; it is the minimal fixed-scope structure required for determinate standing-bearing consequence.*

Proof. A non-degenerate theorem-bearing regime already has a fixed comparison domain, determinate reference, standing-bearing consequence, lawful redescription identity, irreversible commitment, fail-closed admissibility, and nontrivial admitted content. These are exactly the handles through which same-scope claims can remain determinate while preserving identity and excluding post-hoc repair. If admissibility did not govern standing, the regime would have no theorem-bearing licit / illicit distinction. If reference or lawful redescription were not preserved, the regime would not have same-scope identity. If irreversibility or fail-closed enforcement failed, rejected consequence could be repaired by later reclassification. If fixed-scope exhaustion failed, an additional selector could decide the same claim while pretending not to change scope. Thus the regime necessarily carries the fixed-scope kernel structure. No optional adoption step is involved. $\square$

Minimality note: these are theorem-bearing conditions, not extra PDE hypotheses

The non-degeneracy clauses in Definition 3.4C are not additional physical assumptions, PDE regularity hypotheses, or optional AASC decorations. They are the minimum conditions under which a mathematical regime has a determinate subject matter: fixed reference, a fixed comparison domain, consequence-bearing assertions, identity under lawful redescription, irreversible commitment to the declared problem, fail-closed treatment of out-of-domain moves, and at least one admitted case. If a critic denies those clauses, the critic is not merely rejecting a formalism; the critic is denying that the regime supports a stable theorem to prove or refute at the declared scope. Thus the kernel is not smuggled into the problem as a preferred vocabulary. It is the fixed-scope consequence of treating the problem as a non-degenerate theorem-bearing object.

Corollary (3.K1 (No framework opt-out under non-degeneracy)). *A reader cannot reject the fixed-scope kernel while preserving non-degenerate same-scope theorem-bearing discourse. To opt out of the kernel one must deny at least one of the non-degeneracy clauses of Definition 3.4C.*

Proof. This is the contrapositive of Theorem 3.K. If all non-degeneracy clauses are retained, kernel instantiation follows. If kernel instantiation is denied, at least one clause needed for determinate same-scope theorem-bearing consequence has been abandoned. $\square$

Ambient counterexamplehood and kernel-role subsumption

Jurisdiction note: ambient proof systems do not escape the kernel

The fixed-scope kernel is not a competing syntax for ZFC or classical analysis. It is a role theorem about what any non-degenerate proof/refutation regime must already do in order for an object to count as a theorem-bearing proof or counterexample. If an ambient proof system supplies admissible object formation, proof-standing, fixed reference to the target theorem, same-target fidelity, and final theorem consequence, then it performs the kernel roles whether or not it uses AASC vocabulary. If it does not supply those roles, it cannot support determinate theorem-bearing counterexamplehood.

Definition (3.K2 (Ambient proof/refutation regime)). *An ambient proof/refutation regime for a fixed universal mathematical theorem is the ordinary mathematical setting in which a proposed proof or counterexample is formed, checked, and allowed to bear consequence for that theorem. The regime may be ZFC set theory, classical analysis, a formal proof assistant, or any other accepted proof environment. It is non-degenerate for the theorem only if it fixes the target statement, identifies admissible objects, preserves reference to the target, distinguishes theorem-bearing from non-theorem-bearing assertions, and gives final consequence to a successful proof or counterexample.*

Theorem (3.K3 (Ambient proof-system subsumption)). *Every non-degenerate ambient proof/refutation regime for a fixed universal theorem instantiates the fixed-scope kernel at the level of theorem-bearing use. In particular, ordinary ZFC / classical-analysis counterexamplehood supplies admissibility, standing, reference, same-target fidelity, and final consequence only by performing the work of the kernel roles.*

Proof. Let an ambient proof/refutation regime be able to establish a counterexample to a fixed universal theorem. Then it must determine which objects are admissibly formed for the theorem's domain; otherwise the alleged counterexample has no target-licensed object. It must distinguish an assertion with proof-standing from a mere string or informal picture; otherwise no theorem-bearing consequence follows. It must preserve reference to the fixed theorem and its hypotheses; otherwise the alleged counterexample may concern a different target. It must make the refutation final at the act of proof, rather than allowing later repair by changing the theorem, domain, or equivalence class. These are exactly admissibility, standing, reference, same-regime fidelity, and irreversibility/finality at the level of theorem-bearing use. By Theorem 3.K, such a regime instantiates the fixed-scope kernel. Renaming these roles as “ordinary ZFC admissibility” or “classical-analysis verification” does not remove them; it is governance-equivalent performance of the same work. $\square$

Role-equivalence table for ambient counterexamplehood

The equivalence asserted above is role-equivalence, not terminological identity. Ordinary proof systems need not use AASC labels; they instantiate the kernel when they perform the corresponding work.

Ambient proof/refutation role	Kernel role	Why the role is required for counterexamplehood
Admissible object formation in the official domain	Adm	Without an admitted object in the theorem's domain, there is no object that can instantiate the negation.
Proof-standing as a counterexample rather than informal assertion	St	Without standing, the object carries no theorem-bearing consequence.
Fixed reference to the same universal theorem and hypotheses	Ref	Without reference preservation, the alleged refutation may concern a different theorem.
Final contradiction of the asserted conclusion	Irr	Without finality, the alleged refutation can be repaired, deferred, or retyped after the act.
No target drift, hidden selector, or scope transfer	Same-regime fidelity	Without same-regime fidelity, the alleged counterexample no longer refutes the official statement.

Thus an ambient proof system either performs the kernel roles or fails to supply theorem-bearing counterexamplehood. This is a governance-equivalence claim, not a request to replace ordinary mathematical syntax with AASC vocabulary.

Lemma (3.K4 (Counterexamplehood is theorem-bearing construction)). *A counterexample to a fixed universal mathematical theorem is not merely an object-level expression satisfying the syntactic negation. It is a theorem-bearing construction: an admissible object or tuple must be identified, verified to satisfy the theorem's hypotheses, shown to contradict the asserted conclusion, and kept faithful to the same target theorem.*

Proof. A universal theorem has the form “for every object in a fixed class, the asserted conclusion holds.” A counterexample therefore must identify an object in that fixed class and establish that the conclusion fails for that same object under the same hypotheses. If the object is not admissibly in the fixed class, it is not a counterexample. If the verification has no proof-standing, it does not refute the theorem. If reference to the theorem changes, the target has changed. If the contradiction is not final for the theorem-bearing act, it is not a completed refutation. Hence counterexamplehood is an admissibility-assessable, identity-preserving construction inside the ambient proof/refutation regime. $\square$

Theorem (3.K5 (Fefferman counterexample subsumption)). *Any ambient ZFC / classical-analysis counterexample to Fefferman's periodic Navier–Stokes statement (B), if it is to refute the official theorem, is an admissibility-assessable, identity-preserving construction on the fixed official periodic domain. It therefore instantiates the fixed-scope kernel roles, regardless of whether the proof is written in AASC terminology.*

Proof. Such a counterexample would have to exhibit an official datum $u_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$ with $\nabla \cdot u_0 = 0$, identify the corresponding maximal classical solution pair (u, p) on the fixed torus carrier with viscosity $\nu > 0$ and the fixed pressure-gauge convention, establish finite maximal time $T_*(u_0) < \infty$, establish failure of same-scope smooth continuation past that horizon, and show that this tuple contradicts the universal statement of Fefferman (B). Each step is theorem-bearing: the datum must be admissible for the official class, the solution must preserve reference to the official equation and carrier, the non-continuation claim must be same-target, and the contradiction must have final consequence for the theorem. By Lemma 3.K4 and Theorem 3.K3, this is a kernel-instantiating proof/refutation construction. The kernel is therefore not an optional overlay on the

counterexample; it is the role structure required for the alleged object to refute the theorem at all. $\square$

Corollary (3.K6 (No ambient proof-system escape)). *Appealing to an ambient proof system such as ZFC or classical analysis does not exempt a proposed counterexample from the fixed-scope kernel. Either the ambient system supplies admissibility, standing, reference, same-target fidelity, and final consequence, in which case the kernel roles are instantiated by governance-equivalence; or it does not, in which case the alleged object cannot function as a theorem-bearing counterexample.*

Proof. Immediate from Theorem 3.K3 and Theorem 3.K5. The issue is not vocabulary but role performance. An ambient proof system that can refute a fixed theorem must perform the kernel roles; an ambient assertion that performs none of those roles has no theorem-bearing standing. $\square$

3.2 Structural theorem packet (P1–P4 self-contained)

What this packet is and is not

P1–P4 are proved in this manuscript. P1–P3 appear inline below, while the minimal fixed-scope gate/exhaustion core proving P4 is supplied in Appendix A. None of these results supplies a hidden PDE estimate, a blow-up criterion, or a same-scope continuation remedy. Their role is only to state, before carrier specialization begins, the fixed-scope governance results to which the later periodic argument appeals. This is the Part II firewall against the misreading that the manuscript should be searched for a concealed analytic continuation mechanism.

Definition 3.5 (Pre-admissible proposal regime). *A pre-admissible proposal regime is any regime that generates candidate descriptions, objects, or transitions without itself enforcing admissibility conditions. Concretely, such a regime lacks invariance under lawful redescription, lacks conserved quantities across its transformations, lacks a licensed metric or standing-bearing ordering, and lacks standing or closure enforcement. It may generate candidates, but it does not yet license theorem-bearing same-scope claims.*

Lemma 3.6 (Dimensionlessness of pre-admissible proposal regimes). *A pre-admissible proposal regime lacking conserved quantities and invariant transformations is dimensionless: no well-defined metric or monotone ordering exists inside that regime.*

Proof. Without conserved quantities, no magnitude is preserved under transformation. Without preservation, no cross-state comparison is well-defined. Without cross-state comparison, no metric or monotone ordering can be constructed. The regime is therefore dimensionless. $\square$

Theorem (P1 (AMetric Boundary; proved in this manuscript)). *Assume a proposal regime lacks conserved quantities, invariant redescription, and licensed ordering or metric structure. Then the regime lies on the pre-admissible side of a hard AMetric Boundary: no metric, ordering, or licensing authority crosses from that regime into admissible same-scope discourse by gradual refinement, attenuation, or hybrid regularization.*

Proof. By Lemma 3.6, a regime lacking conserved quantities and invariant transformations is dimensionless. In a dimensionless regime there is no conserved comparable quantity and hence no licensed parameter by which authority could be transmitted gradually from proposal-generation into admissible discourse. Therefore no metric structure, standing-bearing ordering, or licensing

authority crosses that boundary by refinement, attenuation, or hybrid regularization. The regime remains pre-admissible until admissibility, standing, invariance, and closure are enforced on the admissible side of the boundary. $\square$

Source provenance and manuscript role. The source provenance of P1 is recorded in Appendix H, but the theorem is proved in this manuscript and no external source is used load-bearingly here. In the present manuscript P1 is used only to separate heuristic proposal-generation, exploratory PDE intuition, and auxiliary analytic guessing from standing-bearing same-scope admissibility claims. It therefore blocks the reading on which Part II is expected to furnish a conventional PDE continuation remedy before the structural burden has been discharged.

Remark 3.7 (Manuscript-local application of P1). P1 is jurisdictional, not analytic. Exploratory continuation guesses, heuristic singularity scenarios, and informal PDE-side proposal generation may suggest candidate pictures, but they do not carry same-scope licensing authority for the official periodic theorem. In this manuscript they therefore cannot count as missing admissibility ingredients, hidden continuation gates, or latent PDE-side remedies waiting to be discovered later in Part II or Part III.

Remark 3.8 (P2 proof treatment location). P2 is realized in this manuscript by the fixed-scope no-repair / bivalence chain in Definitions 3.3 and 3.11, Lemma 3.14, Proposition 3.15, and Corollary P2 below. Its source provenance is recorded in Appendix H; no external source is used load-bearingly at this step.

Remark 3.9 (P3 proof treatment location). P3 is realized in this manuscript by the fixed-scope uniqueness chain beginning with Definition 3.12, continuing through Lemmas 3.16, 3.18, and 3.19, and culminating in Theorem 3.20 together with Corollary P3 and Corollaries 3.22–3.24. Its source provenance is recorded in Appendix H; no external source is used load-bearingly at this step.

Theorem (P4 (Structural Exhaustion; proved in this manuscript)). *Assume bivalent fail-closed admissibility, lawful redescription neutrality, irreversibility, and boundary signature-closure at a fixed scope. Then admissibility classification is unique: no second same-scope discriminator, gate, selector, or admissibility-relevant parameterization survives inside that scope.*

Proof. Appendix A supplies the minimal fixed-scope gate/exhaustion core required for P4. Definitions D.1–D.4, Lemmas D.5–D.8, and Theorem D.9 prove exactly the no-second-gate / no-second-selector / no-admissibility-relevant-parameterization statement recorded here. Corollary D.10 then gives the closure-logic form used later in the manuscript. $\square$

Source provenance and manuscript role. The source provenance of P4 is recorded in Appendix H, but the proof burden is carried in this manuscript by Appendix A. No external source is used load-bearingly here. In the present manuscript P4 is the theorem-level dependency behind the later no-second-gate / no-rival-criterion claims. Naming it here prevents the carrier-side exhaustion route from being read as atmosphere or as an optional philosophy of presentation.

Remark 3.10 (Why the structural packet is followed by local fixed-scope derivations). The structural theorems above are live dependencies, not decorative citations. The short fixed-scope derivations in the next subsection are retained so that later carrier-specific sections can cite compact manuscript-local statements without forcing the reader to reconstruct the structural chain from distant appendices or paper titles alone. This is also why Part II should not be read as a concealed PDE continuation program: the structural burden is named first and only then specialized to the periodic problem.

3.3 Primitive necessity at fixed scope

The next results are the minimal primitive burden of Part II. They are stated without regularity or blow-up vocabulary and therefore do not assume the Navier–Stokes conclusion in disguised form. Their role is to show that once admissibility is meaningful at a fixed scope, the admitted side cannot split into multiple standing-preserving interiors.

Definition 3.11 (Non-degenerate reasoning at fixed scope). Fix a scope in the sense of Definition 3.2 and a fixed-scope reasoning regime in the sense of Definition 3.3. The regime is *non-degenerate* if it satisfies all four conditions below:

- (i) *Stable reference*: at least one candidate at this scope carries determinate standing-relevant content, and the regime does not collapse all non-equivalent candidates into a single trivial standing object.
- (ii) *Standing-preserving identity*: across every licensed same-scope inference or lawful redescription, the identity conditions of admitted content are preserved unless an explicit identity-reset or scope change is declared.
- (iii) *Irreversible commitment*: once a same-scope admissibility commitment is made, no licensed same-scope lineage may return to a pre-commitment equivalent state while preserving reference and standing.
- (iv) *Live admissibility*: at least one same-scope admissibility judgement is standing-positive, so the scope actually has an admitted side.

The first three clauses are the manuscript-local forms of Reference, Standing, and Irreversibility; together with live admissibility they are the fixed-scope vocabulary in which P2 is proved below.

Definition 3.12 (Binary non-eventful admissibility boundary). Fix a scope. The boundary between pre-admissible proposals and admissible reasoning is *binary* if it admits exactly two statuses — admissible and not admissible, equivalently standing-positive and standing-negative — and therefore admits no graded, partial, or probabilistic crossing. It is *non-eventful* if crossing that boundary does not itself constitute an event, transition, or standing-bearing process: there is no internal state change, trajectory, or parameter evolution associated with crossing.

Definition 3.13 (Admissible interior). At a fixed scope, the *admissible interior* is the totality of standing-positive content licensed by the admissibility judgement at that scope. A *second admissible interior* means a distinct same-scope standing-positive regime rather than a mere restatement of the first.

Lemma 3.14 (No standing-preserving repair). *Under non-degenerate reasoning at fixed scope, no same-scope violation of admissibility, reference, or standing can be repaired while preserving all three.*

Proof. Assume a same-scope repair operation exists. If the repair occurs by reinterpretation, then the standing-bearing content is changed after the fact and stable reference fails. If the repair occurs by silent identity drift, then the identity conditions of the admitted object change across a licensed same-scope lineage and standing-preserving identity fails. If the repair occurs by undoing, rewriting, or reversibly bypassing the earlier commitment, then commitment is not irreversible. These are the only same-scope ontological handles by which a repair could be attempted. Hence no standing-preserving repair exists. $\square$

Proposition 3.15 (Fixed-scope bivalence of admissibility). *Under non-degenerate reasoning at fixed scope, admissibility is bivalent and fail-closed: every same-scope candidate is either standing-positive or standing-negative, and any under-specified intermediate status is rejected rather than repaired later.*

Proof. Assume a same-scope candidate is assigned an intermediate, graded, or selector-dependent status. If that status can later be promoted to admission or demoted to rejection by reinterpretation, identity adjustment, or silent scope-preserving repair, then Lemma 3.14 is violated. If it is later resolved by undoing or rewriting an earlier commitment, then irreversibility fails, contrary to Definition 3.11(iii). If distinct same-scope selectors or presentations are allowed to resolve it differently while claiming to preserve scope, then standing-preserving identity and stable reference fail, contrary to Definition 3.11(i)–(ii). Finally, if the intermediate status is kept as a standing-preserving third value, then admissibility is not determinate in the sense of Definition 3.1, so the regime has degenerated rather than preserved fixed-scope governance. Therefore only standing-positive and standing-negative status remain, and any under-specified case is rejected rather than carried as a third standing-preserving state. This is exactly the manuscript-local fail-closed form of bivalence needed later in the periodic bridge. $\square$

Corollary (P2 (Bivalence for non-degenerate reasoning; fixed-scope specialization proved in this manuscript)). *Assume a fixed-scope reasoning regime has stable reference, standing-preserving identity, irreversible commitment, and live admissibility. Then same-scope admissibility is bivalent and fail-closed: no intermediate, graded, or selector-dependent standing survives without degeneration.*

Proof. The four hypotheses are exactly the clauses of Definition 3.11. Lemma 3.14 excludes any same-scope repair preserving reference, standing, and irreversibility, and Proposition 3.15 then forces the fail-closed bivalent split between standing-positive and standing-negative status. Corollary P2 is therefore the fixed-scope specialization of the source-paper Bivalence Theorem. $\square$

Source provenance and manuscript role. The source provenance of P2 is recorded in Appendix H, but the proof burden is carried in this manuscript by Definitions 3.3 and 3.11, Lemma 3.14, and Proposition 3.15. No external source is used load-bearingly here. In the present manuscript P2 serves as the theorem-visible name for the local no-third-status / fail-closed governance result later specialized to the fixed periodic problem.

Lemma 3.16 (Binary non-eventful boundary implies no parameterized admissibility crossing). *Assume the admissibility boundary at a fixed scope is binary and non-eventful. Then no graded, indexed, or parameterized family of admissibility crossings exists at that scope.*

Proof. Parameterization requires ordered variation across states or degrees. But a binary boundary admits no intermediate status between admissible and not admissible, and a non-eventful boundary admits no process, transition, or internal trajectory across which a parameter could evolve. Therefore no graded, indexed, or parameterized admissibility crossing exists at that scope. $\square$

Remark 3.17 (How “initial conditions” are used in the primitive uniqueness chain). In the present fixed-scope primitive layer, “initial conditions” does not denote temporal or physical initial data. It denotes the preconditions required for admissibility at the declared scope. The lemma below is therefore about admissibility preconditions, not about prescribing a unique Navier–Stokes datum.

Lemma 3.18 (Admissible initial conditions are not variables). *At a fixed scope with a binary, non-eventful admissibility boundary, admissible initial conditions are fixed preconditions rather than same-scope variables.*

Proof. If admissible initial conditions were variable, then alternative admissible configurations could be selected by varying them while claiming to preserve scope. Such variation would require parameterization at or before the admissibility boundary. But Lemma 3.16 excludes any graded, indexed, or parameterized family of admissibility crossings at a binary, non-eventful boundary. Therefore admissible initial conditions are fixed preconditions rather than same-scope variables. $\square$

Lemma 3.19 (No admissible variation at fixed scope). *At a fixed scope satisfying Definition 3.11, Proposition 3.15, Lemma 3.16, and Lemma 3.18, no admissible variation of the admitted side exists.*

Proof. Suppose an admissible variation existed. We split by where the purported variation is claimed to occur.

Before admissibility. Then the purported variation remains on the pre-admissible side of the boundary and is therefore not yet a licensed same-scope admissible alternative at all.

At the boundary. Then the purported variation would amount to a graded, indexed, or otherwise parameterized mode of entry into admissibility, contrary to Lemma 3.16.

Within admissibility. Then the admitted side would come in distinct same-scope admissible variants. But Lemma 3.18 says that admissible initial conditions are fixed preconditions rather than variables, so no multiplicity of same-scope admissible configurations can be generated from varying them. Equivalently, any purported within-admissibility difference either changes nothing standing-relevant, in which case it is not an admissibility variation, or it silently changes admitted identity while claiming the same scope, contrary to Definition 3.11(ii).

After commitment. Then the already fixed admissibility judgement is being rewritten post hoc, contrary to Definition 3.11(iii) and Lemma 3.14.

All four cases contradict fixed-scope admissibility. Therefore no admissible variation exists. $\square$

Theorem 3.20 (Unique admissible interior at fixed scope). *Assume non-degenerate reasoning at a fixed scope together with a binary non-eventful boundary between pre-admissible proposals and admissible reasoning. Then exactly one admissible interior exists at that scope.*

Proof. By Definition 3.11(iv), at least one standing-positive admitted side exists, so an admissible interior exists. By Proposition 3.15, admissibility is bivalent and fail-closed: within the fixed scope, same-scope status is standing-positive or standing-negative, with no third standing-preserving state. By Lemma 3.16, the binary, non-eventful boundary admits no graded or indexed family of distinct entries into admissibility. By Lemma 3.18, admissible initial conditions are fixed preconditions rather than same-scope variables. By Lemma 3.19, no admissible variation exists before admissibility, at the boundary, within admissibility, or after commitment.

Suppose, for contradiction, that more than one same-scope admissible interior existed. Then the putative interiors would differ by some same-scope admissible distinction. That distinction would have to arise either from alternative admissible initial conditions or from admissible variation of the admitted side. The former is excluded by Lemma 3.18; the latter is excluded by Lemma 3.19. Hence multiplicity is impossible. Exactly one admissible interior exists at that scope. $\square$

Corollary (P3 (Unique admissible interior; fixed-scope specialization proved in this manuscript)). *Assume non-degenerate reasoning at a fixed scope together with a binary, non-eventful admissibility boundary between pre-admissible proposals and admissible reasoning. Then exactly one admissible interior exists at that scope.*

Proof. This is exactly Theorem 3.20 restated as the theorem-visible P3 packet used by later bridge and carrier citations. $\square$

Remark 3.21 (Anti-circularity of the fixed-scope uniqueness theorem). The statement and proof of Theorem 3.20 do not use the terms regularity, smoothness, singularity, blow-up, or vortex stretching. They appeal only to fixed-scope reference, standing-preserving identity, irreversible commitment, a binary non-eventful admissibility boundary, and the derived no-repair / non-parameterization / fixed-initial-condition / no-variation lemmas. The later Navier–Stokes exclusion is therefore an application of the primitive theorem rather than a conclusion smuggled into its vocabulary.

Corollary 3.22 (No alternative admissible entry condition at fixed scope). *At a fixed scope satisfying Theorem 3.20, no second same-scope entry condition into admissibility exists.*

Proof. A second admissible entry condition would be either a pre-admissible alternative, a parameterized boundary crossing, or an admissible variation of what counts as admitted. These are excluded respectively by the first case of Lemma 3.19, Lemma 3.16, and Theorem 3.20. $\square$

Corollary 3.23 (No plural same-scope interiors at fixed scope). *Plural same-scope interiors are impossible.*

Proof. Immediate from Theorem 3.20. $\square$

Corollary 3.24 (No standing-preserving opt-out from the unique admissible interior). *Any purported same-scope opt-out from the unique admissible interior loses standing.*

Proof. A standing-preserving opt-out would amount to a second admissible regime or a same-scope admissible variation of the existing one. Both are excluded by Theorem 3.20 and Lemma 3.19. $\square$

Remark 3.25 (Interpretive boundary of the fixed-scope uniqueness theorem). Theorem 3.20 is not a cosmological, empirical, or temporal necessity claim. It is a fixed-scope admissibility necessity claim: once the scope is declared, the standing-positive regime admitted at that scope is unique. In particular, the theorem does not say that only one physical Navier–Stokes datum exists; the later carrier argument uses it only to say that the admitted regime itself is unique once the periodic problem’s scope is fixed.

Primitive necessity chain at fixed scope

non-degenerate reasoning $\implies$ no standing-preserving repair $\implies$ bivalent fail-closed admissibility
 $\implies$ no parameterized admissibility crossing $\implies$ admissible initial conditions are not variables
 $\implies$ no admissible variation $\implies$ unique admissible interior.

Why this fixed-scope architecture is forced, not chosen

The primitive layer above is included so that disagreement with the later Navier–Stokes conclusion must take a precise form. Once one accepts the fixed-scope premises of non-degenerate reasoning together with a binary non-eventful admissibility boundary, the manuscript does not leave bivalence, no repair, non-parameterization, no admissible variation, or unique interior as free rule-package choices. Those items have already been derived in-line.

Forcedness test for the manuscript claim

A referee who rejects the later periodic theorem while keeping the fixed periodic scope must therefore identify which primitive necessity fails. In the logic of the present manuscript, the available targets are explicit: reference, standing-preserving identity, irreversible commitment, live admissibility, the binary non-eventful boundary, the no-standing-preserving-repair trilemma, the lemma that admissible initial conditions are not variables, or the non-parameterization / fixed-initial-condition / no-variation consequences drawn from them. A generic objection of the form “*the framework is optional*” does not engage the proof unless it is sharpened into the denial of one of those necessities or one of the short derivations built from them.

Why this matters for later sections. The later role package and carrier-specific audit add Navier–Stokes-facing bookkeeping, specialization, and realization. They do not reopen the fixed-scope question of whether several same-scope interiors could coexist. That burden has already been discharged by Theorem 3.20. The later dispute is therefore clean: either the primitive necessity chain is accepted, in which case the fixed-scope interior is unique, or one must identify exactly which primitive necessity is denied.

3.4 Role discipline: anchor, tensor, skin

Definition 3.26 (Anchor / tensor / skin). A *role assignment* is a typing of interior content into:

- (i) *anchors*: non-negotiable preconditions whose violation collapses standing at the current scope;
- (ii) *tensors*: load-bearing compatibility relations that carry standing-relevant constraint content;
- (iii) *skin*: post-admission descriptive surplus (bookkeeping, presentation, narrative) whose removal does not change tensor content.

Definition 3.27 (Tensor content and standing invariants). Fix an admitted scope with interior state-set P . For $x \in P$, the *tensor content* of x is the total standing-relevant compatibility content carried by x at that scope: the conjunction of all *anchor* and *tensor* clauses satisfied by x (Definition 3.26), modulo *skin* (pure presentation/bookkeeping surplus). We write $\text{Ten}(x)$ for this tensor content. An *invariant* (at this scope) means any predicate or value determined solely by $\text{Ten}(x)$.

Definition 3.28 (Standing equivalence). For $x, y \in P$, declare $x \sim y$ if and only if $\text{Ten}(x) = \text{Ten}(y)$. Equivalently, x and y are standing-equivalent exactly when they carry the same tensor content.

Definition 3.29 (Lawful redescription). A *redescription* is any post-admission map $r : P \rightarrow P$ used only for interior bookkeeping or presentation. It is *lawful* (at the fixed scope) if for every $x \in P$ one has $x \sim r(x)$, i.e. if r preserves tensor content (Definition 3.27).

Definition 3.30 (Standing quotient). The *standing quotient* is $Q_S := P/\sim$, where $\sim$ is standing equivalence (Definition 3.28). Its elements are *standing classes*. Let $q : P \rightarrow Q_S$ denote the quotient map.

3.5 Future-behavior quotient, history factorization, and forced role typing

The endgame later needs more than the bare role names of Definition 3.26. It needs a theorem-local packet saying exactly how same-scope identity is fixed, why standing-relevant functionals must factor through that fixed identity, why admissibility cannot depend on path or representative choice, and why anchor / tensor / skin is an exhaustive trichotomy rather than a suggestive vocabulary. The next packet supplies that burden in compact form inside the submitted PDF.

Definition 3.31 (Same-scope presentations, admitted updates, and admitted tests). Fix a standing-bearing interior state-set P at a declared same scope.

- (i) A *presentation* of a state means any lawful skin-level representative of that state inside the fixed scope.
- (ii) An *admitted update* is any same-scope map $u : P \rightarrow P$ licensed by the fixed carrier commitments; it may move within the interior, but it may not alter anchors, introduce a second gate, or silently retype the scope.
- (iii) An *admitted test* is any typed standing-relevant same-scope functional $t : P \rightarrow Y_t$ already licensed on the fixed scope.

A *finite same-scope history* is any finite composite of admitted updates. When an admitted test t is read after such a history h , the value $t(h(x))$ is the in-scope continuation reading of the state x through that history.

Definition 3.32 (Future-behavior congruence). For $x, y \in P$, declare

$$x \equiv_{\text{fb}} y \iff t(h(x)) = t(h(y))$$

for every finite same-scope history h and every admitted test t .

In words: two states are future-behavior congruent exactly when no admitted same-scope continuation history followed by an admitted test distinguishes them.

Theorem 3.33 (Standing quotient = future-behavior quotient). *Within a fixed same scope, standing equivalence agrees with future-behavior congruence:*

$$x \sim y \iff x \equiv_{\text{fb}} y.$$

Consequently the standing quotient $q : P \rightarrow Q_S$ is exactly the future-behavior quotient for the fixed same-scope interface, and any distinct standing classes admit an in-scope separating witness consisting of a finite same-scope history together with an admitted test.

Proof. Assume first that $x \sim y$. By Definition 3.28, x and y carry the same tensor content. Admitted updates are same-scope only when they preserve anchors and act through that already-licensed interior content, so any finite same-scope history sends x and y to states that still agree on standing-relevant tensor content. Admitted tests are standing-relevant only through that same licensed content. Hence $t(h(x)) = t(h(y))$ for every admitted test t and finite same-scope history h , proving $x \equiv_{\text{fb}} y$.

Conversely, assume $x \equiv_{\text{fb}} y$ but $x \not\sim y$. Then the manuscript would be treating x and y as different same-scope standing states even though no admitted same-scope future history and no admitted same-scope test can distinguish them. That is exactly a privileged-representative distinction or unwitnessed same-scope discriminator. Such a distinction is forbidden by fixed-scope standing (Definition 3.1), by lawful-redescription neutrality (Axiom 3.49), and by Structural Exhaustion (Rule label 3.52). Therefore $x \sim y$ after all.

The separating-witness clause is the contrapositive: if $[x] \neq [y]$ in Q_S , then $x \not\equiv_{\text{fb}} y$, so some finite same-scope history together with an admitted test separates them. $\square$

Definition 3.34 (Same-scope standing-relevant functional). A typed map $F : P \rightarrow Y$ is *same-scope standing-relevant* if it is used to affect admissibility, continuation status, legality, enforcement, or any other standing-bearing same-scope consequence claim.

Theorem 3.35 (Factorization discipline for same-scope discriminators). *Every same-scope standing-relevant functional $F : P \rightarrow Y$ is constant on standing classes and therefore factors uniquely through the standing quotient $q : P \rightarrow Q_S$. If a purported same-scope discriminator fails to factor through q , it is not a refinement of the present scope but an illicit additional discriminator or scope change.*

Proof. Let $F : P \rightarrow Y$ be same-scope standing-relevant. If F were not constant on standing classes, then there would exist $x \sim y$ with $F(x) \neq F(y)$. By Theorem 3.33, the pair x, y is future-behavior congruent, so no admitted same-scope continuation history and no admitted same-scope test distinguishes them. A standing-relevant functional that nevertheless separates them is therefore a second same-scope discriminator not licensed by the fixed quotient. By lawful-redescription neutrality (Axiom 3.49) and Structural Exhaustion (Rule label 3.52), that move is not a same-scope refinement. Hence F must be constant on standing classes.

Define $\tilde{F} : Q_S \rightarrow Y$ by $\tilde{F}([x]) = F(x)$. Constancy on standing classes makes this well-defined, and uniqueness is immediate because q is surjective. Therefore every same-scope standing-relevant functional factors uniquely through the standing quotient. $\square$

Definition 3.36 (History-equivalent same-scope histories). Let h_1 and h_2 be finite same-scope histories with common source state $x \in P$. They are *history-equivalent at x* if $q(h_1(x)) = q(h_2(x))$. A *same-scope history functional* is standing-relevant if it is used to alter admissibility, continuation, or any other standing-bearing verdict by appeal to which history representative was taken rather than only to the standing class reached.

Theorem 3.37 (History-factorization / no path-dependent same-scope admissibility). *Within a fixed same scope, admissibility and all same-scope standing-bearing verdicts may depend only on the standing class reached, not on which lawful history representative reached it. Equivalently: if h_1 and h_2 are history-equivalent at x , then every same-scope standing-relevant history functional assigns the same verdict to h_1 and h_2 at x .*

Proof. Suppose a same-scope standing-relevant history functional G distinguished two history-equivalent histories h_1, h_2 at the same source state x . Then $q(h_1(x)) = q(h_2(x))$, so by Theorem 3.33 the resulting states have the same admitted future behavior. Distinguishing them anyway would create a path- or order-dependent standing discriminator inside a single standing class. By Theorem 3.35, no such same-scope standing-relevant discriminator exists. Therefore admissibility cannot depend on the history representative or on the update order once the fixed same-scope quotient state is fixed. $\square$

Definition 3.38 (Forced role typing by same-scope variation). Fix a standing-bearing description Δ at the present scope and vary one named component while holding the remaining fixed commitments unchanged.

- (i) The component is *anchor-typed* if an attempted variation changes definedness, revokes same-scope well-typedness, or silently changes the scope.
- (ii) It is *tensor-typed* if some same-scope variation leaves the scope fixed but changes a quotient-visible same-scope standing-relevant consequence.
- (iii) It is *skin-typed* if every same-scope variation leaves every same-scope standing-relevant consequence unchanged.

Theorem 3.39 (ATS role minimality and no fourth same-scope role). *Every named component appearing in a same-scope standing-bearing description is forced into exactly one of the three roles anchor, tensor, or skin. There is no fourth same-scope standing-bearing role.*

Proof. Let C be any named component of a same-scope standing-bearing description. If varying C changes definedness or silently retypes the scope, then C is anchor-typed by Definition 3.38(i). Assume instead that the variation remains same-scope and well-typed. Then either some same-scope standing-relevant consequence changes or none does. In the first case C is tensor-typed by Definition 3.38(ii); in the second it is skin-typed by Definition 3.38(iii). The three roles are mutually exclusive by construction.

For the no-fourth-role claim, suppose some component were standing-relevant while being neither anchor, tensor, nor skin. Then varying it would neither change definedness nor change any same-scope standing-relevant consequence, yet it would still be doing standing-bearing work. That is exactly an unwitnessed same-scope discriminator, forbidden by Theorem 3.35. Hence no fourth same-scope standing-bearing role exists. $\square$

Corollary 3.40 (Skin non-authority and representative-only diagnostics collapse). *Skin carries no same-scope standing-bearing authority. In particular, a representative-only diagnostic, metadata tag, selector, or canonical-presentation choice that is not quotient-visible is either bookkeeping only or an illicit additional gate / scope change.*

Proof. By Theorem 3.39, a component that changes no same-scope standing-relevant consequence is skin and therefore cannot itself be standing-bearing authority. If a representative-only diagnostic or selector were promoted to standing-bearing status without becoming quotient-visible, it would distinguish presentations inside one standing class. That violates Theorem 3.35. Hence such a move is either bookkeeping only or an illicit additional gate / scope change. $\square$

Proposition 3.41 (Periodic carrier interface for the quotient / history / ATS packet). *For the standard periodic carrier of Definition 13.2, the abstract packet above is instantiated as follows.*

- (i) *Presentations are the lawful carrier skins already admitted in Definition 13.2 and Proposition 13.5.*
- (ii) *Admitted updates are the lawful same-scope redescription and continuation moves already licensed on the fixed carrier, including smooth-slice restart and overlap-preserving continuation on already smooth intervals as later realized theorem-locally in Theorem 19D.*
- (iii) *Admitted tests are the standing-relevant continuation, legality, compatibility, and horizon-reading functionals already fixed on that carrier and later read on realized lineages.*
- (iv) *Finite same-scope histories are finite composites of those admitted updates.*

Consequently Theorems 3.33, 3.35, and 3.37 apply on the fixed periodic carrier and are available for later use in Part III and Section 21 without importing an external corpus theorem by title.

Proof. Items (i)–(iii) are exactly the theorem-local same-scope content already fixed by Definition 13.2, Proposition 13.5, Proposition 13.6, and the bridge packet that fixes the periodic same-scope continuation notion before Part III begins. The smooth-slice restart clause becomes theorem-local at the realization layer by Theorem 19D, but it does not change scope; it is one of the already licensed same-scope continuation moves of the official periodic problem. Therefore the abstract quotient / history / ATS packet applies to the standard periodic carrier exactly as stated. $\square$

Corollary 3.42 (No path-dependent same-scope admissibility on the periodic carrier). *Within the fixed periodic carrier, admissibility and continuation status cannot depend on which lawful redescription, restart representative, or finite same-scope history reached the carrier state. They depend only on the standing class reached at that same scope.*

Proof. Immediate from Theorem 3.37 together with Proposition 3.41. $\square$

Remark 3.43 (Why this packet is placed in Part II). The quotient / history / ATS packet is placed before the carrier-identification endgame because later Sections 20–21 use exactly these consequences: same-scope identity is quotient-fixed, same-scope discriminators must factor through that quotient, history/path dependence is not available as a same-scope escape, and skin-level representative choices cannot become a hidden second gate. Recording those facts here keeps the later realized-lineage and singular-image arguments theorem-local.

3.6 Explicit structural rules used in Part II

The first block of rules below is not introduced as an optional rule-package choice. Proposition 3.15 together with Lemmas 3.14 and 3.16, Theorem 3.20, and Corollary 3.24 already prove the fixed-scope content of bivalence, no repair, non-parameterization, no admissible variation, and unique interior. The axioms below restate those necessities in compact theorem-facing form and then append the additional interior bookkeeping rules used later in the carrier audit.

Axiom 3.44 (Bivalence and fail-closed enforcement). Admissibility is bivalent. Under-specified, unlicensed, or ambiguous moves are rejected (fail-closed).

Axiom 3.45 (No generators, no carriers, no repairs). Standing cannot be generated, transported across scopes by association, or recovered inside a rejected lineage. Any move that requires post-hoc repair is inadmissible.

Axiom 3.46 (AMetric boundary). At the boundary only coherent classification and standing determination occur. No metric notions (norms, rates), operators, or analytic mechanisms exist at the boundary.

Axiom 3.47 (Licensed interior bookkeeping). Metric descriptors and analytic operators may appear *inside* an admitted scope only as licensed bookkeeping. Licensing does not grant generative or standing-bearing power.

Axiom 3.48 (Operators as envelopes). Operators exist only as *licensed envelopes* with specified domain, invariants, failure conditions, and revocation conditions. Unlicensed operator use is inadmissible.

Axiom 3.49 (Redescription neutrality). Redescription is post-admission only. A redescription step is admissible only if it is *lawful* in the sense of Definition 3.29, i.e. it preserves tensor content (Definition 3.27) at the fixed scope. No partial redescription and no power gain under redescription are permitted.

Axiom 3.50 (Composition discipline). There is no retroactivity and no partial composition. If an intermediate step fails admissibility, the entire lineage fails.

Axiom 3.51 (Deterministic enforcement). Violations of the axioms above are structurally detectable. Enforcement is fail-closed and admits no repair within a lineage.

Rule label 3.52 (Structural Exhaustion (generic fixed-scope label; carrier-specific verification later)). Once the scope is fixed, no new admissibility level, atomic bundle, admissible question, or operator family may be introduced.

Remark 3.53 (Status of the Structural Exhaustion label). Rule label 3.52 is recorded here as the compact fixed-scope name for the no-new-level / no-new-bundle / no-new-question / no-new-operator prohibition used later in Part II. For the standard periodic carrier, the live route verifies this label by the carrier-certified packet of Theorem 13.28 and Corollary 13.29, so the official endpoint theorem does not rely on it as an unproved carrier-specific primitive. The later continuation-mechanism reformulation of Section 13.2B is retained as explanatory architecture; it is not required as an additional live source of Structural Exhaustion on the official theorem route.

Remark 3.54 (Provenance). The rules above are stated only to the extent required by Part II. They are included here so that the structural certification argument is auditable without external reading. Their source provenance is non-load-bearing for the proof of the official theorem.

4 Self-contained derivation ledger

This page is an audit aid. The fixed-scope necessity chain proved in Section 3 converts the core Part II rule names from stipulations into compact theorem-facing labels, with Rule label 3.52 functioning as a generic fixed-scope label whose standard-carrier verification is supplied later. The ledger below records, for each later load-bearing result in the Part II cascade, (i) its immediate dependencies in the manuscript and (ii) the admissibility axioms from Section 3 that its proof invokes (directly or through the cited intermediate lemmas).

Axiom / rule-label key. FC=Axiom 3.44, NR=Axiom 3.45, AM=Axiom 3.46, LM=Axiom 3.47. ENV=Axiom 3.48, RD=Axiom 3.49, COMP=Axiom 3.50. ENF=Axiom 3.51, EXH=Rule label 3.52.

Result	Immediate dependencies	Axioms invoked
Theorem 2.1	Part I analytic atlas	— (Part I PDE analysis)
Theorem 2.4	Definition 2.3 + Part I atlas	— (typed classification)
Theorem 13.1	Theorem 2.4 and Definition 2.3	— (purely from Part I closure)
Theorem 13.19	Theorem 13.1 and Proposition 13.17 (via Lemmas 13.3, 13.4, 13.14 and 13.18)	LM, RD, ENV, EXH
Theorem 13.20	Theorem 13.19 (via Lemmas 13.3, 13.4 and 13.14)	LM, RD, ENV, EXH
Theorem 12.1	zero-parameter + equivalence closure (hypotheses) + Definitions 3.4 and 3.27	RD, NR, COMP, FC, ENF
Proposition 14.3	Definitions 3.4 and 3.30	RD, EXH
Theorem 16.5	Theorems 13.1 and 13.19 and Definition 15.4	— (elimination via certified gate exhaustion)
Lemma 16.6	Theorem 16.5 and Definition 3.30	— (quotient consequence under regular branch)
Corollary 16.7	Proposition 14.3 and Theorems 12.1 and 16.5	RD, EXH, NR, COMP, FC, ENF (via dependencies)
Theorem 17.4	Lemma 17.3 and Theorem 16.5	EXH, RD, FC, ENF, NR, COMP
Corollary 17.5	Theorem 17.4	— (same-scope consequence of terminality)

Table 1: Primitive base $\rightarrow$ theorem cascade axiom ledger (Part II).

5 Why a pivot is compulsory

Part I was written in ordinary PDE language on purpose. The intent was not rhetorical balance but stress localization: each standard route was allowed to run in its native notation until the precise point at which closure would require a new ingredient. The resulting pattern is not accidental. Every route terminates at the same place because every route needs an additional bridge from the energy-class bundle to a scale-critical control of the vortex-stretching channel.

In the present self-contained rule package, that recurring need must be retyped. A “missing estimate” is not innocent. It is either an already licensed interior consequence, in which case it must be derivable without changing admissible power, or it is a new admissibility object, in which case it is prohibited by structural exhaustion. The pivot is therefore not a stylistic choice. It is forced by the map. Equivalently: the vortex-stretching wall marks the incompleteness of same-scope PDE analytic closure; it does not reopen primitive admissibility, does not create a second same-scope interior, and does not license a same-scope gate.

Proposition 5.1 (Common analytic bottleneck, retyped). *For each route catalogued in Part I, the unresolved step is one of the following:*

- (i) *an unlicensed bridge from one admissible skin to another;*
- (ii) *a partial redescription in which equivalence is asserted without invariant preservation;*
- (iii) *an operator chain whose intermediate failure would have to be repaired downstream; or*
- (iv) *a local weakening, exception, or tolerance introduced to preserve the appearance of closure.*

Proof. The energy/enstrophy route requires a new coercive estimate on $\mathcal{T}(u)$; the geometric route requires a direction-to-kernel bridge; the frequency route requires a low/high decomposition whose error terms do not alter admissible consequences; the profile route requires a rigidity principle that is not already present in the admitted operator stock. Each of these is either a new operator-level constraint (an unlicensed envelope, contrary to Axiom 3.48), a partial redescription (contrary to Axiom 3.49), a forbidden repair-by-composition (contrary to Axioms 3.50 and 3.45), or a local relaxation that functions as a same-scope parameter in the sense of Definition 3.4 and is excluded by Structural Exhaustion (Rule label 3.52). $\square$

Corollary 5.2 (Necessity of the pivot). *If the admissibility map is complete and non-extendable, then no further conventional same-scope supplement may be pursued unless it is already licensed by an existing atomic bundle.*

Proof. Immediate from Structural Exhaustion (Rule label 3.52). $\square$

6 Boundary, interior, and the licensed status of analysis

The rule package distinguishes sharply between the AMetric boundary, where only coherent classification and standing determination occur, and the interior on the admitted side, where licensed operators and metric bookkeeping may appear.

6.1 The AMetric boundary does not decide regularity

Axiom 3.46 fixes the AMetric boundary as non-operational: no metric notions, no operators, no transformations, and no admissible structure may be smuggled across it. A “singular Navier–Stokes state” therefore cannot be decided at the boundary in terms of norms, blow-up rates, or continuation operators. Those are all post-admission notions.

Lemma 6.1 (Regular versus singular is not an AMetric distinction). *The distinction between regular and singular Navier–Stokes behavior is not classifiable at the AMetric boundary.*

Proof. Any such distinction presupposes interior notions: classical solution class, continuation, operator-defined compatibility, or metric bookkeeping. All are undefined at the boundary by Axiom 3.46. Hence the distinction is not boundary-level. $\square$

6.2 The L-metric boundary licenses analytic bookkeeping

Axiom 3.47 permits metric descriptors as interior bookkeeping while forbidding any generative or explanatory role for the boundary itself. This matters here: Part I used norms, derivatives, kernels, and rescalings only as licensed interior bookkeeping. No claim in Part II attributes their existence to the boundary.

Remark 6.2. This paper therefore does not reject analysis. It retypes analysis. Analytic objects may appear, but only as licensed interior descriptors and never as primitive sources of admissibility.

7 Role audit for Navier–Stokes

By the role triad (Definition 3.26), anchors are detected by collapse under attempted motion, tensors are load-bearing and stress-localizing, and skin is discardable expression.

7.1 Anchor assignment

In the present scope the relevant anchors are:

- (A1) standing/admissibility of the interior object-class;
- (A2) divergence compatibility as a non-negotiable condition for the flow class under discussion;
- (A3) role fixity itself.

Attempts to “relax incompressibility locally,” to defer standing, or to treat role assignment as negotiable are ill-typed: they convert an anchor into a tunable same-scope parameter (Definition 3.4) and are rejected by fail-closed enforcement (Axioms 3.44 and 3.51).

7.2 Tensor assignment

The load-bearing tensor content is not the skin-level narrative of “cascade” or “blow-up,” but the compatibility relations that bind the interior description. In standard PDE language the stress localizes at the vortex-stretching production channel

$$\mathcal{T}(u) = \int \omega \cdot S(u) \omega,$$

because every route in Part I converges on this channel. $\mathcal{T}(u)$ is not itself the whole bundle; it is the tensor location at which incompatibility first becomes unavoidable.

7.3 Skin assignment

The following are skin:

- “cascade” language,
- geometric stories of alignment or misalignment,
- pictures of energy transfer across scales,
- profile narratives and self-similar imagery,
- mechanism-language for enforcement or saturation.

By the role triad (Definition 3.26) together with redescription neutrality (Axiom 3.49), skin may not repair incompatibility or carry inferential weight unless its tensor content is independently preserved.

8 The Navier–Stokes constraint bundle as load-bearing tensor

By the role triad (Definition 3.26), bundles are tensor-level compatibility conditions rather than mechanisms. Stability is negative classification: absence of incompatibility under lawful redescription (Axiom 3.49).

Definition 8.1 (Navier–Stokes interior bundle). On $\mathbb{T}^3$, the Navier–Stokes interior bundle B_{NS} consists of the compatibility of:

- (B1) divergence compatibility $\nabla \cdot u = 0$;
- (B2) momentum compatibility in the admitted interior carrier;
- (B3) pressure compatibility with mean-zero normalization;
- (B4) energy bookkeeping compatibility;
- (B5) role-preserving redescription legality across admitted skins.

Remark 8.2. Definition 8.1 is not a new structure. It is the standard Navier–Stokes content retyped under the role triad: a bundle is simply the compatibility condition across these constraints.

Proposition 8.3 (Bundle-localized meaning of the common wall). *The common wall of Part I is the statement that no route therein produced a redescription-stable, role-preserving certificate that B_{NS} remains compatible once the vortex-stretching tensor is pushed to the critical scale.*

Proof. Energy arguments fail to bind $\mathcal{T}(u)$; geometric arguments control only a subset of the kernel content; frequency decompositions reorganize but do not eliminate the same tensor load; profile arguments require a rigidity principle not yet supplied. In each case the unresolved step is exactly whether bundle compatibility survives the push. This is stress localization at the tensor level in the sense of Definition 3.26. $\square$

9 Why the conventional bridges collapse inside the map

This section translates the walls of Part I into failure modes under the axioms of Section 3 (especially Axioms 3.48–3.51 together with Rule label 3.52).

Conventional route	Needed missing move	Map-level failure mode
Energy / enstrophy	new coercive bridge to critical control	operator invalidation or Exhaustion Failure
Geometry / coherence	direction-to-kernel promotion	redescription collapse
Frequency splitting	decomposition whose errors are harmless in every skin	redescription collapse or composition collapse
Profiles / Type II	chain of normalization, compactness, rigidity	composition collapse
Weak-strong uniqueness / Serrin	derive critical criterion from subcritical data	operator invalidation

Table 2: Part I retyped through the admissibility map.

9.1 Energy and enstrophy route

To close by energy one would need an additional operator-level bridge from subcritical energy bookkeeping to critical tensor control. Under Structural Exhaustion, such a bridge must already be licensed or it is inadmissible as a new route.

9.2 Geometry-to-kernel route

The attempted move

$$\text{direction incoherence} \implies \text{kernel depletion} \implies \mathcal{T}(u) \text{ control}$$

is a redescription move. Without a proof of equivalence plus invariant preservation, it is exactly the partial-redescription collapse forbidden by Axiom 3.49.

9.3 Frequency splitting route

The low/high decomposition is skin-level bookkeeping until its error terms are shown to preserve the same tensor compatibility. If the decomposition changes admissible consequences, it is not a redescription but an illicit power gain.

9.4 Profile and continuation route

Any profile-based closure uses a chain of operators: normalization, compactness extraction, limiting profile identification, rigidity. Axioms 3.50 and 3.45 forbid partial sequence survival and downstream repair; if any step lacks licence or changes admissible consequences, the whole chain collapses.

10 Operator-envelope audit for smoothness extension

Axiom 3.48 fixes operators as licensed envelopes, not actions.

A naive closure route would use an extension operator

$$\text{EXT} : u(t_0) \mapsto u|_{[t_0, t_0+\varepsilon)}.$$

The audit shows where this route fails.

Definition 10.1 (Extension envelope). The extension envelope **EXT** is the putative operator whose domain is the class of standing-positive interior Navier–Stokes states and whose output is an admissible continuation in the same standing lineage.

Lemma 10.2 (Required invariants for **EXT**). *For **EXT** to be licensed, it must preserve:*

- (i) *standing;*
- (ii) *bundle compatibility;*
- (iii) *operator identity, hence determinacy of admissible consequences.*

Proof. By Axiom 3.48, a licensed envelope comes with a domain and an invariant set. Items (i) and (ii) are exactly the standing- and compatibility-invariants demanded of any same-scope continuation envelope. For (iii), if the “operator” admits two incompatible consequence-classes under the same stated invariants, then it is not a single well-defined envelope at all: it encodes a hidden selector (a same-scope parameter in the sense of Definition 3.4) and is rejected by fail-closed enforcement (Axioms 3.44 and 3.51). $\square$

Proposition 10.3 (Why **EXT** is not the primitive route). *Within the current map, **EXT** cannot be taken as the primitive closure operator unless determinacy is already bundle-carried. Otherwise the envelope is unlicensed.*

Proof. If admissible continuation were not determinate, then the envelope would not have a fixed consequence class, violating operator identity and redescription neutrality (Axiom 3.49). Treating such a family as a single operator would introduce new admissible power by redescription. Hence either determinacy is already forced by the bundle, or **EXT** is unlicensed. $\square$

Remark 10.4. This is why Part II does not proceed by “continuation” as a primary operator. The route is too late in the admissibility architecture.

11 Structural exhaustion and the necessity of reclassification

Theorem 11.1 (Structural Exhaustion revisited). *No new admissibility level, atomic bundle, admissible question, or putative operator may be introduced once the map is fixed.*

Proof. Immediate from Structural Exhaustion (Rule label 3.52). $\square$

Corollary 11.2 (No “one more lemma” route). *Any statement of the form “regularity would follow if we had one more bridge lemma” is, within the fixed carrier, either:*

- (a) *a theorem already forced by an existing atomic bundle, or*
- (b) *an inadmissible extension attempt.*

Proof. Fix the scope. Any “one more bridge lemma” that is not already forced by existing bundle content would have to either (i) introduce a new operator envelope, contrary to Axiom 3.48 and Rule label 3.52, (ii) assert a non-neutral redescription, contrary to Axiom 3.49, or (iii) introduce a selector/tolerance that changes standing or consequence without changing scope, i.e. a same-scope parameter in the sense of Definition 3.4. Such moves are rejected by fail-closed enforcement. Hence any valid “extra bridge” is either already derivable inside the existing bundle or else is an inadmissible extension. $\square$

Remark 11.3 (Why reclassification, not supplementation). The exhaustion result changes the methodological burden. Once the standard routes of Part I fail, the next step is not to search for an extra bridge at the same scope but to ask whether the missing bridge was mis-typed from the start. Part II answers that question by reclassification.

12 Constraint saturation of the unique admissible interior

We now arrive at the theorem that turns the pivot from heuristic to structural necessity.

Theorem 12.1 (Constraint Saturation). *Within the zero-parameter unique admissible interior with completed equivalence closure, the admissible constraint set is saturated: no admissible constraint is removable, relaxable, localizable, deformable, or isolable; admissible outcomes are unique, rigid, overdetermined, and fail-closed.*

Proof. Assume the unique admissible interior is zero-parameter and equivalence-complete. If a constraint were removable, relaxable, localizable, deformable, or isolable while preserving scope, then the choice of *whether*, *where*, or *how much* to enforce it would define a same-scope selector or tolerance in the sense of Definition 3.4. This contradicts zero-parameter status. Framing the move as a redescription does not help: by Axiom 3.49, any lawful redescription must preserve tensor content (Definition 3.27) and hence standing equivalence. A relaxation that changes legality or consequence is therefore inadmissible. If one instead attempts to preserve admissibility by downstream repair, Axioms 3.45 and 3.50 reject that move under fail-closed enforcement (Axioms 3.44, 3.51).

Hence every admissible description must satisfy the full compatibility set without tunable exceptions: the interior is rigid and overdetermined. Finally, if two distinct admissible outcomes were available at the same scope, choosing between them would again introduce a same-scope selector, contradicting zero-parameter status. Therefore admissible outcomes are unique and enforcement is fail-closed. $\square$

Remark 12.2. Theorem 12.1 does not add a new constraint. It classifies the interior once zero-parameter admissibility and equivalence completion are fixed.

Proposition 12.3 (What “zero-parameter” excludes). *Within the saturated interior, no admissible route may depend on a selector, tolerance, deferred enforcement, canonical representative, privileged reweighting, or same-scope repair.*

Proof. Each listed object is a selector, tolerance, privileged representative, or repair knob. In a zero-parameter interior these are exactly same-scope parameters (Definition 3.4) and are excluded. The exclusion is a direct restatement of Constraint Saturation (Theorem 12.1) together with fail-closed enforcement (Axioms 3.44, 3.51) and the no-repair prohibition (Axiom 3.45). $\square$

Remark 12.4 (Why saturation matters for Navier–Stokes). Theorem 12.1 by itself classifies the interior only after two conditions are fixed: zero-parameter admissibility and completed equivalence closure. The remaining task is therefore not another analytic estimate but a certification problem: one must show that the standard $\mathbb{T}^3$ Navier–Stokes carrier already enters the interior in exactly that form.

13 Zero-parameter certification of the standard $\mathbb{T}^3$ Navier–Stokes carrier

Application of Theorem 12.1 requires two certifications and only two: zero-parameter status and equivalence completion for the standard carrier.

Recall Definition 3.4. The certification problem is therefore exact: identify every putative same-scope parameter in the standard $\mathbb{T}^3$ presentation and eliminate it by role.

Theorem 13.1 (Gate classification). *Under the analytic atlas closure of Part I (Theorem 2.4), any candidate same-scope discriminator for Navier–Stokes evolution belongs to one of four gate classes:*

- (i) *amplitude (norm-growth) gates,*
- (ii) *scale-transfer gates,*
- (iii) *geometric-structure gates,*
- (iv) *profile-formation gates.*

Proof. By Definition 2.3, an analytic continuation mechanism is a same-scope attempt to certify continuation by strengthening control over an interior feature of the standard carrier. Theorem 2.4 shows that such mechanisms are exhausted by four feature-families: norm growth, scale transfer, geometric structure, and profile formation. A same-scope discriminator is, in particular, a purported same-scope continuation mechanism for selecting between admissible continuations. Hence every such discriminator must act through one of the four listed gate classes. $\square$

Definition 13.2 (Standard $\mathbb{T}^3$ Navier–Stokes carrier). Fix the three-torus $\mathbb{T}^3$, a positive viscosity constant $\nu > 0$, and the mean-zero pressure gauge. The *standard $\mathbb{T}^3$ Navier–Stokes carrier* consists of the ordinary interior analytic presentations of the same object-class:

- (i) the velocity formulation (1);
- (ii) the vorticity formulation (4);
- (iii) the pressure Poisson relation on $\mathbb{T}^3$ with mean-zero normalization;
- (iv) the strain/kernel articulation $S = \mathcal{R}_s(\omega)$;
- (v) the standard frequency and Littlewood–Paley skins used in Part I.

These are treated as licensed interior descriptions of one object-class, not as competing same-scope gates.

Lemma 13.3 (Licensed bookkeeping carries no same-scope admissibility power). *Within Definition 13.2, the torus carrier, the positive viscosity constant ν , the mean-zero pressure normalization, and the choice of standard analytic skin are scope constants or licensed bookkeeping choices, not same-scope admissibility parameters.*

Proof. By Definition 3.4, a same-scope admissibility parameter must function as a selector, tolerance, deferred enforcement, privileged representative, or repair device. The fixed torus and fixed viscosity identify the scope itself; varying either changes scope rather than refines it. The mean-zero pressure condition on $\mathbb{T}^3$ is a gauge representative with no standing-bearing force. The choice among the ordinary skins of Part I is redescription only. By Axiom 3.47, interior metric descriptors are

authorized as bookkeeping without acquiring generative or standing-bearing status; by redescription neutrality (Axiom 3.49), no skin choice may acquire extra admissible power. Hence none of these items can function as a same-scope parameter. $\square$

Lemma 13.4 (Equivalence completion of the standard skins). *The presentations listed in Definition 13.2 are admissibility-equivalent and preserve the same interior invariants. In particular, velocity, vorticity, pressure, strain/kernel, and standard frequency bookkeeping are lawful redescrptions of one carrier rather than competing same-scope structures.*

Proof. Velocity and vorticity are linked by the licensed differential identities of the admitted carrier. Pressure and strain/kernel are fixed by the same carrier together with incompressibility. The frequency decompositions of Part I were introduced only as bookkeeping skins on that already-admitted carrier. None of these moves changes standing, expands the domain, or alters admissible consequences. By Axiom 3.49, that is exactly the criterion for lawful redescription. Conversely, if any one of these skins carried extra same-scope force, it would constitute either an unlicensed operator family (Axiom 3.48) or a redescription collapse (Axiom 3.49). Both are excluded by Structural Exhaustion (Rule label 3.52). $\square$

13.1 Why these categories are forced by the periodic Navier–Stokes carrier

The purpose of this subsection is PDE-facing adequacy rather than extra architecture generality. It explains, from the ordinary periodic Navier–Stokes presentations used throughout the paper, why the categories of lawful redescription, standing equivalence, anchor / tensor / skin, and admissibility are the natural bookkeeping distinctions for the fixed torus carrier. Nothing in this subsection adds a same-scope gate; it records why the rule package above is the right typed reformulation of familiar distinctions between continuation-neutral presentation changes and genuinely continuation-relevant content.

Proposition 13.5 (Lawful redescription for the standard periodic carrier). *For smooth periodic incompressible data on $\mathbb{T}^3$, the velocity, vorticity, mean-zero pressure, strain / kernel, and the standard frequency skins are lawful presentations of the same carrier content. More precisely: they are obtained from the same smooth periodic velocity field by deterministic torus operations (differentiation, periodic Poisson reconstruction with the fixed mean-zero pressure gauge, and standard frequency decomposition), and passage among them does not create a continuation criterion on the fixed torus carrier.*

Proof. Let u be a smooth periodic divergence-free velocity field on $\mathbb{T}^3$. Its vorticity $\omega = \nabla \times u$ and strain $S(u) = \frac{1}{2}(\nabla u + \nabla u^\top)$ are standard differential bookkeeping attached to u . The pressure is recovered, up to the fixed mean-zero gauge, from the torus Poisson equation determined by u , and the Littlewood–Paley / frequency skins are standard decompositions of the same field. None of these constructions changes the underlying smooth trajectory or the classical continuation question; they only re-express the same periodic solution in standard PDE skins. Thus passage among them is lawful redescription in the sense used later in the paper. $\square$

Proposition 13.6 (Standing equivalence is the right equivalence notion for continuation). *For the periodic carrier, two admissible presentations are standing-equivalent exactly when they determine the same continuation behavior on the fixed torus carrier.*

Proof. By Proposition 13.5, the allowed presentation changes for the standard carrier are deterministic reconstructions or bookkeeping skins of the same smooth periodic state. Every continuation-relevant judgment used later in Part III — local realizability, horizon continuation, breakdown witness formation, and branch extraction — depends only on the underlying smooth periodic evolution on the fixed torus carrier, not on which lawful presentation is used to display it. Hence two presentations are equivalent for continuation exactly when they preserve all standing-relevant continuation judgments. This is precisely the role of standing equivalence in Definition 3.28. $\square$

Proposition 13.7 (Anchor / tensor / skin arise from the periodic Navier–Stokes carrier). *For the periodic Navier–Stokes carrier, the three-way role typing of Definition 3.26 is forced by the PDE structure itself:*

- (i) *the torus carrier, incompressibility, positive viscosity, zero forcing, smooth category, and mean-zero pressure gauge are anchors;*
- (ii) *the coupled continuation-relevant compatibility clauses carried by the Navier–Stokes bundle are tensors;*
- (iii) *velocity / vorticity / pressure / strain / frequency displays that do not alter continuation status are skins.*

Proof. The carrier, incompressibility condition, viscosity sign, forcing choice, regularity category, and pressure gauge fix what counts as the same problem; changing any of them changes the active continuation question and therefore changes scope. These are anchors. What remains continuation-relevant on the fixed carrier is the coupled content of the Navier–Stokes evolution: the compatibility of the velocity, pressure, vorticity, strain, and frequency bookkeeping with the same smooth periodic trajectory. That is what the paper later calls tensor content. By contrast, working in velocity, vorticity, pressure, strain, or frequency language does not by itself change continuation behavior; these are presentation-level enrichments and therefore skins. This is a PDE internal distinction between fixed commitments, continuation-relevant content, and harmless re-expression. $\square$

Proposition 13.8 (Why admissibility / standing is the right language here). *A same-scope continuation argument for the periodic Navier–Stokes problem must distinguish genuine continuation data on the fixed torus carrier from harmless re-expression. This is exactly what the paper calls standing and admissibility: a typed reformulation of the ordinary PDE distinction between continuation-relevant content and mere presentation change.*

Proof. Any argument about continuation on the fixed carrier must decide whether a proposed move preserves the torus, the incompressibility condition, the viscosity sign, the forcing choice, the smooth category, and the lawful presentation class. If it does, the move is same-scope and can meaningfully contribute to continuation analysis; if it does not, then the argument has changed problem, carrier, or presentation class in a way that is not continuation-neutral. The paper records that yes/no distinction as standing/admissibility. Thus the language is not an extra layer added to the PDE; it is the typed bookkeeping of the same distinction a classical analyst already uses when separating genuine continuation data from harmless reformulation. $\square$

Bridge packet 12A–12H. Applicability of structural necessity at fixed periodic scope

What this bridge packet does and does not do

The statements below are proved *in this manuscript*. They do not add a new PDE estimate, blow-up criterion, or same-scope continuation remedy. Their task is to prove that the fixed periodic problem is a non-degenerate standing-bearing fixed scope and therefore falls under the fixed-scope kernel by necessity, via Theorem 3.K and Corollary 12A.0.1. The subsequent bridge then applies the structural theorem packet Theorem P1, Corollary P2, Corollary P3, and Theorem P4, with P1–P4 proved in-paper and the minimal P4 gate/exhaustion core supplied in Appendix A. Every later structural appeal in the bridge, carrier-identification, and transport packets is therefore routed through manuscript-local theorem numbers rather than companion papers. In particular, Lemma 12F is only the scope-fixity / typing component of the bridge packet; the no-second-gate consequence is carried later by Theorem 12G. This is the place where the manuscript answers, in the theorem-local chain, the recurrent misreading that keeps searching for a missing PDE-side continuation ingredient.

Theorem (12A.0 (Official periodic Navier–Stokes proof/refutation regime is non-degenerate)). *The proof/refutation regime induced by Fefferman’s official periodic Navier–Stokes problem on $\mathbb{T}^3$ —with $\nu > 0$, zero forcing, smooth divergence-free periodic data, the fixed mean-zero pressure gauge, the standard lawful redescription class, and the classical smooth continuation question—is a non-degenerate theorem-bearing regime in the sense of Definition 3.4C. The claim is not that the differential equations themselves are a construction act; it is that proving or refuting Fefferman (B) requires an admissibility-assessable, identity-preserving theorem-bearing regime on the fixed official domain.*

Proof. We verify the clauses of Definition 3.4C explicitly.

Non-degeneracy clause	Official periodic Navier–Stokes witness
Determinate reference	The carrier is the fixed torus $\mathbb{T}^3$, with fixed viscosity $\nu > 0$, zero forcing, the incompressible Navier–Stokes equations, the smooth periodic divergence-free datum class, and the fixed pressure-gauge convention.
Fixed comparison domain	The domain is exactly the official periodic comparison class of Definition 3.4A: same torus, same viscosity, same forcing status, same datum class, same gauge, same lawful redescription class, and same continuation question.
Standing-bearing consequence	Fefferman (B) is a truth-valued universal theorem over official periodic data: every datum either has the asserted global smooth realization or a same-scope counterexample must supply a finite-horizon endpoint condition.
Lawful redescription identity	Velocity, vorticity, pressure, strain, and frequency presentations are deterministic lawful skins of the same smooth periodic state by Proposition 13.5 and Lemma 13.4.
Irreversible commitment	Once the official theorem is fixed, changing carrier, datum class, gauge, equation, lawful-skin class, or continuation question changes the problem instead of repairing an endpoint claim.
Fail-closed admissibility	A proposal outside the fixed-domain comparison class is rejected as a different-domain proposal, not retained as a third same-scope counterexample status.
Nontrivial admitted content	The zero datum supplies a standing-positive same-scope carrier state, and the local periodic realization packet of Appendix B supplies ordinary local classical realization for smooth admitted data.
No post-hoc repair	A failed same-scope endpoint cannot be repaired by later selectors, tolerances, gauge shifts, presentation switches, or altered continuation questions without leaving the declared problem.

Each witness is part of the official periodic proof/refutation setup rather than an added regularity assumption. The equations are the object-level dynamics; the theorem-bearing act of proving or refuting Fefferman (B) is the regime being classified. Thus the official periodic Navier–Stokes proof/refutation regime is non-degenerate in the precise fixed-scope sense needed for the kernel; denying this theorem requires denying that an alleged proof or counterexample to Fefferman (B) must preserve fixed reference, fixed domain, lawful presentation identity, admissible object formation, and stable theorem-bearing consequence. $\square$

Corollary (12A.0.1 (Kernel applies to the official periodic problem by necessity)). *By Theorem 3.K, Theorem 3.K5, and Theorem 12A.0, the fixed-scope kernel applies to the official periodic Navier–Stokes proof/refutation regime by necessity, not by optional adoption of a framework.*

Proof. Theorem 12A.0 proves that the proof/refutation regime induced by the official periodic problem is a non-degenerate theorem-bearing regime. Theorem 3.K5 shows that any ambient counterexample to Fefferman (B) is an admissibility-assessable construction on that fixed official domain. Theorem 3.K says that every such non-degenerate regime instantiates the kernel. Therefore the official periodic proof/refutation regime instantiates the kernel by necessity. $\square$

Lemma (12A: Fixed periodic problem defines a standing-bearing fixed scope). *Fix the official periodic Navier–Stokes problem exactly at the scope declared in this manuscript: smooth divergence-free periodic data on $\mathbb{T}^3$, positive viscosity, zero forcing, the fixed mean-zero pressure gauge, the official local classical solution concept used in Part III, and the lawful-redescription class generated by the standard carrier skins. Then these data determine a fixed same-scope reasoning object: the datum class, standing-bearing continuation claims, lawful redescrptions, and evaluated same-scope admissibility interface are fixed rather than optional.*

Proof. Definition 3.2 says that scope is fixed precisely by the carrier specification, the same-scope continuation notion, and the licensing conditions under which admissibility is evaluated. For the periodic problem, Proposition 13.7 identifies the torus carrier, incompressibility, viscosity sign, zero forcing, smooth category, and pressure gauge as the fixed commitments that determine what counts as the same problem. Definition 13.2 together with Proposition 13.5 and Lemma 13.4 fix the lawful presentation class, while Proposition 13.6 and Proposition 13.8 identify which continuation judgments are standing-bearing on that class. Hence the declared periodic problem already determines a fixed datum class, lawful redescription class, and same-scope admissibility interface. The manuscript is therefore not choosing an arbitrary formalism after the fact; it is identifying the standing-bearing fixed-scope object already implicit in the posed periodic problem. $\square$

Lemma (12B: The fixed periodic problem lies on the admissible side of the AMetric Boundary). *When the fixed periodic Navier–Stokes problem is used to make theorem-bearing same-scope claims, it lies on the admissible side of the AMetric Boundary. Consequently, pre-admissible proposal machinery has no same-scope licensing authority for this problem.*

Proof. Lemma 12A fixes a datum class, a lawful-redescription class, and standing-bearing continuation claims for the periodic problem. Theorem P1 says that a regime lacking conserved reference, invariant redescription, and licensed ordering or metric structure remains pre-admissible and cannot license admissible claims by gradual refinement. A theorem-bearing same-scope argument for the periodic problem therefore cannot treat exploratory PDE intuition, proposal generation, or future analytic guesswork as admissibility authority. Such moves may suggest candidates, but once the official periodic theorem is at issue they do not alter same-scope standing. The fixed periodic problem is therefore governed on the admissible side of the AMetric Boundary. $\square$

Lemma (12B.1: Live admissibility at fixed periodic scope). *At the fixed periodic scope of Lemma 12A, at least one same-scope admissibility judgement is standing-positive. Concretely, the null periodic datum together with its canonical zero skins is a licit same-scope carrier state, so the fixed periodic interface has an admitted side.*

Proof. Take the smooth divergence-free periodic datum $u_0 \equiv 0$ on $\mathbb{T}^3$. Its induced vorticity, strain, and standard frequency skins are the canonical zero skins, and the fixed mean-zero pressure gauge yields $p_0 \equiv 0$. By Proposition 13.7, this object stays inside the declared anchors of the periodic problem: the torus carrier, positive viscosity, zero forcing, smooth category, and mean-zero gauge are unchanged. By Proposition 13.5, the zero velocity, vorticity, pressure, strain, and frequency displays are lawful same-scope presentations of one carrier state, and by Proposition 13.6 they preserve the same continuation-relevant content. Proposition 13.8 identifies the resulting judgement exactly: it is a licit same-scope admissibility judgement rather than exploratory pre-admissible proposal machinery. Hence at least one same-scope admissibility judgement is standing-positive at the fixed periodic scope, verifying the live-admissibility clause of Definition 3.11(iv). $\square$

Lemma (12C: Non-degeneracy at fixed periodic scope). *At the fixed periodic scope of Lemma 12A, the evaluated admissibility fragment is non-degenerate in the exact sense of Definition 3.11: it has stable reference, standing-preserving identity, irreversible commitment, and live admissibility.*

Proof. We verify the four clauses of Definition 3.11 one by one.

- (i) *Stable reference.* Lemma 12A already fixes determinate same-scope objects: datum class, lawful-redescription class, and continuation claims on the fixed torus carrier. Proposition 13.5 says that lawful skins are deterministic presentations of one carrier state rather than ambiguous

candidate objects, while Proposition 13.6 identifies standing equivalence with sameness of continuation behavior on that fixed carrier. The regime therefore has determinate same-scope referents and does not collapse every candidate into one trivial standing object.

- (ii) *Standing-preserving identity.* Lemma 12D proves exactly that lawful same-scope redescription preserves standing-bearing content rather than silently changing which carrier state or continuation claim is being judged.
- (iii) *Irreversible commitment.* Lemma 12E proves that no failed same-scope admissibility move can be repaired retroactively by later metadata, relabeling, reparameterization, or reinterpretation.
- (iv) *Live admissibility.* Lemma 12B.1 supplies an explicit standing-positive same-scope admissibility judgement on the fixed periodic interface.

Hence every clause of Definition 3.11 holds at the declared periodic scope, so the evaluated admissibility fragment is non-degenerate in the exact sense required later by Corollary P2 and Theorem 3.20. $\square$

Lemma (12D: Lawful redescription neutrality at fixed periodic scope). *At the fixed periodic scope, lawful redescription does not change standing-bearing content.*

Proof. By Proposition 13.5, the velocity, vorticity, pressure, strain / kernel, and standard frequency skins are deterministic torus reconstructions or bookkeeping displays of the same smooth periodic carrier state. Proposition 13.6 then identifies standing equivalence with sameness of continuation behavior on that fixed carrier, and Lemma 13.4 records that the standard skins are already admissibility-equivalent presentations of one carrier. Therefore passage among lawful skins preserves the same continuation-relevant content and cannot by itself alter standing. If a lawful redescription did change standing, it would cease to be lawful and would instead function as a second same-scope discriminator on the fixed carrier. Hence lawful redescription neutrality holds at the declared periodic scope. $\square$

Lemma (12E: Irreversibility / no same-scope retroactive repair). *At the fixed periodic scope, no failed same-scope admissibility move can be retroactively repaired by later metadata, relabeling, reparameterization, or scope-preserving reinterpretation.*

Proof. Lemma 3.14 already gives the abstract fixed-scope trilemma: a same-scope repair can work only by changing standing-bearing reference, changing identity conditions, or undoing an earlier commitment. Lemma 12A specializes that trilemma to the declared periodic problem by fixing the datum class, lawful-redescription class, and same-scope admissibility interface, and Lemma 12D rules out the idea that a later relabeling among lawful skins could change standing after the fact. A purported retroactive repair on the periodic problem would therefore have to add a new selector, parameter, or reinterpretation rule not already present in the fixed scope, or else silently alter one of the fixed anchors of Proposition 13.7. In either case it is not a same-scope repair. Hence irreversibility holds at the fixed periodic scope. $\square$

Lemma (12F: Boundary signature-closure for the periodic problem). *At the fixed periodic scope, no additional boundary-level discriminator, continuation selector, metadata gate, or admissibility-relevant criterion may be introduced while still claiming the same periodic scope.*

Role declaration. *This lemma is a same-scope typing / scope-fixity result; it does not, by itself, constitute a no-second-gate theorem.*

Proof. Lemma 12A fixes what counts as datum, lawful redescription, and same-scope admissibility for the periodic problem. Relative to that already-fixed periodic interface, any purported additional criterion has only two possibilities.

Case 1. If the proposed criterion changes same-scope admissibility outcomes, then it does not merely enrich a lawful presentation; it alters the already-fixed admissibility interface and therefore presents itself as a second same-scope selector or gate. The exclusion of such second same-scope gates is *not* being derived here; it is obtained only downstream once Theorem P4 is applied in Theorem 12G.

Case 2. If the proposed criterion leaves admissibility outcomes unchanged, then it is continuation-neutral bookkeeping or skin in the sense of Proposition 13.7; it adds no same-scope admissibility authority.

Thus every purported additional criterion at the fixed periodic interface is classified as either a same-scope gate proposal or bookkeeping only. Lemma 12F does not exclude the existence of continuation criteria in general; it classifies their status relative to the fixed periodic same-scope interface. The no-second-gate consequence used later is obtained via Theorem 12G, not from Lemma 12F in isolation. $\square$

What 12F is not

Lemma 12F is not a PDE estimate, not a continuation criterion, and not a standalone impossibility theorem. Its role is only to classify the same-scope status of purported additional criteria at the fixed periodic interface: outcome-changing proposals present themselves as same-scope gate claims, while outcome-neutral proposals are bookkeeping only. The no-second-gate consequence used later is sourced from Theorem 12G, where Theorem P4 is applied to the completed bridge packet.

Bridge order for the fixed periodic packet

12C $\rightarrow$ 12G in one direction only. The bridge order used later is theorem-local and one-directional: Lemma 12C (12C) supplies the fixed-scope non-degeneracy packet and hence the bivalent fail-closed side used by Corollary P2; Lemma 12D (12D) supplies lawful redescription neutrality at the fixed periodic scope; Lemma 12E (12E) supplies irreversibility / no same-scope retroactive repair; Lemma 12F (12F) supplies the fixed-periodic typing form of boundary signature-closure; and Theorem 12G (12G) then applies Theorem P4 (P4) to that completed packet.

Theorem (12G: Applicability of structural necessity at fixed periodic scope). *The fixed periodic Navier–Stokes problem satisfies the antecedents of Corollary P2, Theorem 3.20, and Theorem P4. More precisely, Lemma 12C verifies the four clauses of Definition 3.11, Lemmas 12D–12F verify the remaining same-scope bridge hypotheses, and the only appendix-level input used here is the already named fixed-scope exhaustion core of Appendix A through Theorem P4. In this bridge packet, Lemma 12F contributes only the fixed-periodic form of boundary signature-closure / same-scope typing: no additional boundary-level admissibility authority may be introduced while preserving that scope. With bivalence and scope-fixity established by Lemma 12C together with Lemmas 12D–12F, Theorem P4 (P4) applies to exclude additional same-scope admissibility gates. Consequently, admissibility at this scope is bivalent and structurally exhausted, the admissibility boundary is binary and non-eventful, and no second same-scope discriminator, gate, selector, or admissibility-relevant parameterization exists.*

Proof. Lemma 12A supplies the fixed-scope reasoning object and its standing-bearing continuation

interface. Lemma 12B places theorem-bearing reasoning for that object on the admissible side of Theorem P1. The bridge order used here is theorem-local and one-directional: Lemma 12C (12C) verifies all four clauses of Definition 3.11, including the live-admissibility clause through Lemma 12B.1; Lemma 12D (12D) verifies lawful redescription neutrality; Lemma 12E (12E) verifies irreversibility; and Lemma 12F (12F) is the fixed-periodic form of boundary signature-closure.

Corollary P2 therefore applies, so admissibility at the declared periodic scope is bivalent and fail-closed. Because the same-scope interface fixed in Lemma 12A is a yes/no licensing interface on standing-bearing claims rather than a standing-bearing transition process, the admissibility boundary is binary and non-eventful in the sense of Definition 3.12. Theorem 3.20 is also available on the nose at this scope, since its fixed-scope hypotheses have been verified theorem-locally. Finally, with this bivalence together with lawful redescription neutrality, irreversibility, and the boundary signature-closure supplied by Lemma 12F, Theorem P4 (P4) applies; its proof burden is already isolated in Appendix A and enters the bridge only through that named theorem. Hence admissibility classification at the fixed periodic scope is structurally exhausted and no second same-scope discriminator, gate, selector, or admissibility-relevant parameterization survives. $\square$

Corollary (12H: Unique admissible interior for the fixed periodic problem). *Exactly one admissible interior exists for the fixed periodic Navier–Stokes problem.*

Proof. Lemma 12C verifies the four clauses of Definition 3.11, including live admissibility via Lemma 12B.1. Theorem 12G adds the remaining fixed-scope bridge consequences actually needed for uniqueness: bivalent fail-closed admissibility together with a binary, non-eventful admissibility boundary. The fixed-scope theorem of unique admissible interior, Theorem 3.20, therefore applies directly at the declared periodic scope, and Corollary P3 records the same conclusion in the manuscript’s P3 notation. Hence the fixed periodic Navier–Stokes problem admits exactly one admissible interior. $\square$

13.2 Standing/intelligibility packet internalized at fixed periodic scope

The next three statements are the minimal fixed-periodic standing/intelligibility packet later used to control the endgame. To avoid collision with the branch labels $\{S1, S2, S3, S4, S5\}$ reserved later for the constructive branch-forcing set, this packet is labeled *ST1–ST5*. The present subsection carries the fixed-scope half of that packet. No companion paper is needed to recover these clauses from the submitted PDF.

Definition (ST1: Standing-positive continuation-relevant determination at fixed periodic scope). *Fix the periodic scope determined by Lemma 12A. A same-scope continuation-relevant determination is standing-positive if and only if all of the following hold simultaneously:*

- (i) *it fixes a determinate referent on the certified periodic carrier (or on the datum-induced classical continuation object later realized from that carrier), so the judgement answers to one definite same-scope object;*
- (ii) *it is evaluated on the declared domain of application already fixed by scope—the torus carrier, viscosity sign, zero forcing, smooth category, pressure gauge, and same-scope horizon language fixed in Lemma 12A and Proposition 13.7;*
- (iii) *it is invariant under the lawful redescription class of Proposition 13.5 and Lemma 12D; and*
- (iv) *it preserves identity across those lawful same-scope presentations, so that no later relabeling, skin change, or presentation switch alters which carrier state or continuation claim is being judged.*

A same-scope continuation-relevant determination failing any of (i)–(iv) is standing-negative at that scope.

Theorem (ST2: Intelligibility failure is standing loss at fixed periodic scope). *Fix the periodic scope of Lemma 12A. If a purported same-scope continuation-relevant determination fails fixed reference, domain-of-application closure, lawful-redescription invariance, or identity stability, then it is standing-negative. Equivalently, intelligibility failure at the declared periodic scope is standing loss.*

Proof. The present theorem is already downstream of the fixed-scope bridge packet: 12F classifies fixed-periodic scope-fixity, Theorem 12G then applies P4 to that packet, and only on that one-directional basis is the fixed-scope standing/intelligibility structure used here available. Failure of fixed reference leaves the judgement of Definition 3.1 without a determinate same-scope object to classify. Failure of domain-of-application closure means that the claim is not about the fixed torus problem declared in Definition 3.2, Lemma 12A, and Proposition 13.7; such a claim is therefore not a same-scope determination at all. Failure of lawful-redescription invariance would let licensed skins alter continuation status, contrary to Proposition 13.6, Lemma 13.4, and Lemma 12D. Failure of identity stability would split one standing class into several same-scope objects without a new licensed selector, contrary to Definition 3.28, Lemma D.6, and Theorem 12G. In each case the purported determination cannot remain standing-positive on the fixed periodic interface. Hence intelligibility failure is standing loss. $\square$

Theorem (ST3: Standing–admissibility equivalence at fixed periodic scope). *Within the fixed periodic interface of Lemma 12A, a same-scope continuation-relevant determination is admissible if and only if it is standing-positive. Consequently standing loss is same-scope inadmissibility, not a neutral third status.*

Proof. The fixed-scope standing packet invoked here is established via Theorem 12G, which applies P4 after Lemma 12F’s scope-fixity classification. By Definition 3.1, admissibility judgements at fixed scope are standing judgements: standing-positive means licit and standing-negative means illicit. Theorem 12G gives bivalent fail-closed governance on the fixed periodic interface, so no neutral third standing-preserving status remains available once the scope is fixed. Therefore admissibility on the evaluated same-scope fragment coincides exactly with standing-positivity, while standing loss is precisely inadmissibility. This is the fixed-periodic form of standing–admissibility equivalence used later in the manuscript. $\square$

Corollary 13.9 (No standing-neutral same-scope status at fixed periodic scope). *At the declared periodic scope, any same-scope continuation-relevant determination is either standing-positive and admissible or standing-negative and inadmissible. No neutral third same-scope status survives.*

Proof. Immediate from Theorem ST3 together with the fail-closed bivalence supplied by Theorem 12G. $\square$

13.3 Minimal rule package forced by the periodic Navier–Stokes carrier

The adequacy propositions above identify the fixed torus, incompressibility, positive viscosity, zero forcing, smooth category, and lawful presentation class as the data that make continuation questions same-scope on the standard periodic carrier. The present subsection records the minimal specialized consequences of that observation in Navier–Stokes form. These are the PDE-facing versions of the no-second-gate / binary-status / no-repair principles used later in the section.

Theorem 13.10 (No competing same-scope continuation gates on the standard periodic carrier). *On the fixed periodic carrier, any same-scope continuation discriminator either coincides extensionally with the carrier-side continuation discriminator on admitted states or changes scope. Equivalently: two genuinely different continuation gates cannot coexist on the same smooth periodic carrier.*

Proof. Let G and G' be same-scope continuation discriminators on admitted states of the standard periodic carrier and suppose that they disagree on some admitted state x . By Proposition 13.6, standing equivalence is exactly the equivalence relation preserving continuation behavior on the fixed torus carrier. Thus a disagreement between G and G' cannot be dismissed as harmless re-expression: to keep the disagreement standing-relevant, one would have to specify which discriminator governs the same carrier state. But by Proposition 13.8, a same-scope continuation argument is well-typed exactly when it preserves the carrier-determined equivalence relation and does not import an additional continuation selector. Choosing between G and G' would therefore amount to an extra same-scope selector, i.e. a scope change in the sense of Definition 3.2. Hence two competing same-scope continuation gates cannot coexist on the fixed periodic carrier. $\square$

Corollary 13.11 (Binary continuation status on the fixed carrier). *For the standard periodic carrier, continuation status is bivalent when defined: continuation or non-continuation. No graded same-scope continuation levels exist.*

Proof. If a third standing-relevant continuation level existed on the same smooth periodic carrier, then by thresholding or selecting that level one would obtain a continuation discriminator different from the carrier-side continuation discriminator, contradicting Theorem 13.10. Thus continuation status is bivalent when defined. $\square$

Corollary 13.12 (No parameterized same-scope continuation gate). *No threshold family, selector family, tolerance family, or profile-indexed family defines a genuinely same-scope continuation gate on the fixed periodic carrier.*

Proof. Any such family would produce either a continuum or a discrete collection of continuation predicates on the same smooth periodic carrier. If every member of the family coincides extensionally with the carrier-side continuation discriminator, the family is continuation-neutral bookkeeping. If some member differs, then the family provides a competing same-scope continuation gate, contradicting Theorem 13.10. Therefore no parameterized family supplies a genuinely same-scope continuation gate. $\square$

Theorem 13.13 (No repair / no generator in specialized Navier–Stokes form). *On the fixed periodic carrier, a continuation failure cannot be repaired as the same carrier state, and no derived quantity can generate continuation standing independently of the carrier-side continuation discriminator.*

Proof. Suppose a continuation failure for an admitted carrier state x could be repaired while preserving that same carrier state. Then the repaired and unrepaired versions of x would carry different continuation status on the same scope, contradicting Theorem 13.10. The only alternative is a change of carrier commitments, i.e. a scope change. Likewise, if a derived quantity were able to generate continuation standing independently of the carrier-side continuation discriminator, it would define a same-scope gate or a parameterized family of such gates on the fixed torus carrier, contradicting Theorem 13.10 and Corollary 13.12. Hence there is no same-scope repair and no independent same-scope generator of continuation standing. $\square$

Lemma 13.14 (Pointwise vortex-stretching gate collapse). *In the standard $\mathbb{T}^3$ carrier, any standing-relevant pointwise use of the vortex-stretching functional*

$$\mathcal{T}(u) = \int_{\mathbb{T}^3} \omega \cdot S(u) \omega$$

either

- (i) *factors through the standing quotient and is therefore quotient-visible bookkeeping, or*
- (ii) *induces a forbidden second same-scope discriminator on the fixed carrier.*

Hence $\mathcal{T}$ contributes no independent same-scope gate at a fixed carrier state.

Proof. The force of the lemma is strictly pointwise: it concerns the standing-relevant role of $\mathcal{T}(u)$ at a fixed carrier state and does not itself settle the later use of the trajectory-level observable $t \mapsto \mathcal{T}(u(t))$ as a continuation gate.

Step 1: $\mathcal{T}$ is an admitted pointwise bundle functional. Inside the admitted carrier, $\omega = \nabla \times u$, $S(u) = \frac{1}{2}(\nabla u + \nabla u^\top)$, and the pressure / elliptic relations are already fixed by the divergence and momentum clauses of B_{NS} . Thus $\mathcal{T}$ is a well-typed scalar functional on the admitted bundle data; for example it appears in the enstrophy bookkeeping identity (5). At a fixed carrier state, any standing-relevant use of $\mathcal{T}$ is therefore a pointwise use of already admitted bundle content rather than an appeal to some omitted same-scope continuation structure.

Step 2: a non-quotient-visible use would either privilege skin or open a second gate. Assume that a standing-relevant pointwise use of $\mathcal{T}$ does *not* factor through the standing quotient. Then there exist standing-equivalent admitted states $x \sim y$ on the fixed carrier for which the proposed $\mathcal{T}$ -based use assigns different standing-relevant status. By Lemma 13.4, such standing-equivalent states differ only by lawful skin or presentation of one carrier object. Therefore the proposed use can only work in one of two ways. Either it privileges one lawful redescription of that carrier over another, contrary to Axiom 3.49; or it promotes $\mathcal{T}$ itself, or a threshold / selector built from $\mathcal{T}$, to an additional same-scope discriminator on the fixed carrier, contrary to Theorem 13.10 and Corollary 13.12. A repair reading is no help: by Theorem 13.13, no derived quantity may generate or restore continuation standing independently of the carrier-side continuation discriminator.

Hence the non-factorizing case is inadmissible. Therefore every admissible standing-relevant pointwise use of $\mathcal{T}$ factors through the standing quotient and is quotient-visible bookkeeping rather than an independent same-scope gate at a fixed carrier state. This proof uses only lawful-skin equivalence together with the specialized periodic no-second-gate / no-parameterized-gate / no-repair package proved above; it does not invoke the broader continuation-completeness theorem of Section 13.2C. $\square$

Remark 13.15 (Pointwise role of Lemma 13.14). Lemma 13.14 is a pointwise carrier-standing theorem proved entirely inside the specialized periodic rule package above. Unbounded growth of $\mathcal{T}(u(t))$ along a realized trajectory is not time-zero carrier standing data. If one later tries to promote that trajectory-level growth to a same-scope continuation gate, the move is tested in Part III against the fixed carrier continuation notion and the no-second-gate / no-parameterized-gate consequences already recorded here; the formal treatment of such uses is deferred to Definition 20.11A and Theorem 20.11.

13.1A. Why the standard periodic carrier forces this minimal rule package

The claims above are already enough to run the later certification chain. The present subsection repeats the same content in the language of the periodic Navier–Stokes problem itself, so that the phrase “the standard carrier forces the minimal rule package” is read as a short derivation rather than as a declaration.

Proposition (13.5A (Periodicity–pressure–divergence collapse of competing gates)). *On the fixed periodic carrier $\mathbb{T}^3$ with divergence-free data and the mean-zero pressure normalization, any same-scope continuation predicate that distinguishes lawful presentations beyond the standing quotient either factors through that quotient or induces a forbidden second same-scope continuation gate.*

Proof. Periodicity fixes the carrier on which continuation is asked. The divergence-free condition fixes the admissible state class on that carrier, and the mean-zero pressure gauge reconstructs the pressure on the same carrier rather than by passing to a larger domain. By Proposition 13.5, velocity, vorticity, mean-zero pressure, strain/kernel, and the standard frequency presentations are therefore lawful skins of one carrier state rather than competing same-scope objects. If a same-scope continuation predicate distinguished such lawful presentations beyond the standing quotient, then it would either descend to that quotient or privilege one lawful presentation over another. The former case is quotient-visible bookkeeping; the latter is exactly a second same-scope continuation gate on the fixed carrier and is forbidden by Theorem 13.10. Thus the periodic / divergence / pressure package already collapses every competing same-scope gate to quotient visibility or inadmissibility. $\square$

Proposition (13.5B (Energy and momentum force binary continuation status)). *For each horizon $T > 0$ on the standard periodic carrier, smooth continuation past T is bivalent: a realized trajectory either extends smoothly on the same carrier past T or it does not. No graded same-scope continuation status survives as standing-relevant data.*

Proof. On the fixed periodic carrier, the continuation question is whether the smooth trajectory determined by the momentum equation and its pressure reconstruction extends past the chosen horizon while remaining in the same smooth class on the same torus. That question has two and only two outcomes: continuation or non-continuation. The energy identity and momentum equation constrain the trajectory, but they do not generate intermediate standing values between those two outcomes. Any graded continuation status treated as standing-relevant would therefore have to be used as a second gate or parameter family on the same carrier, contradicting Corollary 13.11 together with Corollary 13.12. Hence smooth continuation status is bivalent on the fixed carrier. $\square$

Proposition (13.5C (Vortex stretching cannot generate repair or parameterized gates)). *On the certified standard periodic carrier, the vortex-stretching functional cannot produce a repair, tolerance, or parameterized same-scope continuation gate. Pointwise this is the content of Lemma 13.14; trajectory-level uses are completed later by the formal Θ_x -gate analysis of Part III.*

Proof. Pointwise, Lemma 13.14 shows that $\mathcal{T}(u)$ carries no standing-relevant information beyond the standing quotient at a fixed carrier state. Therefore a finite-thresholded pointwise use of $\mathcal{T}$ cannot open a same-scope parameterized gate. Likewise, if one tried to use $\mathcal{T}$ pointwise to repair a failed continuation state on the same carrier, the repair would amount to a same-scope continuation gate or generator, contradicting Theorem 13.13. Trajectory-level uses are not smuggled in here: they are handled later, and there too the formal Θ_x -gate analysis shows that the evolution of $\mathcal{T}(u(t))$ does not create an independent same-scope continuation gate. $\square$

Corollary (13.5D (The standard periodic carrier forces exactly the minimal rule package)). *The standard periodic Navier–Stokes carrier forces exactly the minimal same-scope rule package used in Part II: a unique same-scope continuation gate, binary continuation status, no parameterized continuation gate, and no repair / generator move on the fixed carrier.*

Proof. Combine Propositions 13.5A–13.5C with the specialized rule package already proved in this subsection: Theorem 13.10, Corollary 13.11, Corollary 13.12, and Theorem 13.13. The periodic carrier supports no other same-scope rule beyond these, so the minimal package is forced exactly as stated. $\square$

Finite standing-positive same-scope decision procedure on the standard periodic carrier

The finite standing-positive decision packet compresses the Part II forcing step into a finite decision packet before zero-parameter certification is stated. Once the bridge packet and the minimal rule package are in the theorem-local chain, every standing-positive continuation-relevant candidate determination on the fixed carrier is tested in exactly this order:

1. Is the proposed ingredient already represented by the certified carrier bundle? If yes, it is existing carrier content.
2. If not, is it only a lawful change of skin or quotient-visible bookkeeping? If yes, it is bookkeeping-only variation.
3. If not, does it require a threshold, selector, tolerance, or family member to decide continuation? If yes, it is a forbidden parameterized gate.
4. If not, does it still purport to alter continuation authority while claiming fixed scope? If yes, it is a forbidden competing discriminator.
5. If none of the above holds, then some carrier commitment has changed and the move is a scope change, not a same-scope rival.

Only after this finite classification and its branch eliminations are established do the manuscript-local zero-parameter and no-second-gate consequences get stated.

Definition (13.5E (Typed standing-positive same-scope continuation candidate determination on the standard periodic carrier)). *A typed standing-positive same-scope continuation candidate determination on the standard periodic carrier is any quantity, criterion, selector, mechanism, or reformulation presented as a standing-positive continuation-relevant determination while claiming to preserve the fixed periodic scope. Such a candidate determination is typed as exactly one of the following:*

- (i) existing carrier content: *it is already represented by the standard carrier bundle and introduces no new selector or admissibility authority;*
- (ii) bookkeeping-only variation: *it depends only on lawful skin choice, quotient-visible display, or licensed reconstruction of the same carrier content;*
- (iii) parameterized gate: *it requires a threshold, tolerance, selector, family member, or deferred choice to decide continuation while claiming fixed scope;*
- (iv) competing discriminator: *it purports to deliver new standing-relevant continuation authority at fixed scope without being bookkeeping-only and without openly changing scope;*

(v) scope change: *it alters one of the fixed carrier commitments and is therefore not a same-scope candidate.*

Theorem (13.5F (Finite standing-positive same-scope candidate classification on the standard periodic carrier)). *Every typed standing-positive same-scope continuation candidate determination on the standard periodic carrier is exactly one of the five kinds listed in Definition 13.5E. No sixth standing-positive same-scope candidate type exists.*

Proof. Let $\mathfrak{C}$ be a standing-positive continuation-relevant candidate determination presented for the fixed periodic carrier. If $\mathfrak{C}$ changes the torus, the smooth divergence-free class, the forcing choice, the pressure normalization, the lawful skin class, or the continuation notion, then by Definition 3.2 and Proposition 13.7 it changes scope and is of kind (v). Assume now that $\mathfrak{C}$ is genuinely same-scope. By Definition 13.2 and Proposition 13.5, the carrier already contains the velocity, vorticity, pressure, strain/kernel, and standard frequency bundle together with their lawful deterministic reconstructions. If $\mathfrak{C}$ is already borne by that bundle with no new selector or admissibility authority, it is of kind (i). If it distinguishes only lawful presentations or quotient-visible displays of that same content, then by Proposition 13.6 and Proposition 13.3 it is of kind (ii). If it requires a threshold, tolerance, selector, family member, or deferred choice to decide continuation, then it is of kind (iii) by Definition 3.4. If none of (i)–(iii) holds while fixed scope is still claimed, then $\mathfrak{C}$ is being presented as new standing-relevant continuation authority on the same carrier and is therefore of kind (iv). The cases are mutually exclusive by construction, and they exhaust all standing-positive same-scope possibilities. $\square$

Proposition (13.5G (Existing carrier content adds no continuation gate)). *Any typed same-scope candidate of existing-carrier kind contributes no independent continuation criterion beyond the standard carrier bundle.*

Proof. By Definition 13.2, the standard periodic carrier already contains the ordinary velocity, vorticity, pressure, strain/kernel, and standard frequency presentations. By Lemma 13.4, these are admissibility-equivalent skins of one carrier rather than competing same-scope structures. Therefore a candidate already represented by that bundle adds no continuation authority; it simply names content already present in the carrier. $\square$

Proposition (13.5H (Bookkeeping-only variation carries no new standing-bearing force)). *Any typed same-scope candidate of bookkeeping-only kind is quotient-visible bookkeeping and cannot supply an independent same-scope continuation criterion.*

Proof. By Proposition 13.5, lawful skin changes are deterministic torus reconstructions or displays of the same carrier state. Proposition 13.6 shows that continuation-relevant standing is preserved across those lawful skins, and Proposition 13.3 states precisely that any same-scope predicate distinguishing lawful presentations beyond the standing quotient either factors through that quotient or becomes a forbidden second same-scope gate. Thus bookkeeping-only variation is quotient-visible and carries no new standing-bearing force. $\square$

Proposition (13.5I (Parameterized same-scope gates are forbidden)). *Any typed same-scope candidate of parameterized-gate kind is inadmissible on the standard periodic carrier.*

Proof. A parameterized gate is exactly a same-scope selector, threshold, tolerance, or family choice in the sense of Definition 3.4. The specialized periodic rule package already forbids such same-scope parameterization: Corollary 13.12 excludes parameterized continuation gates on the standard carrier, Theorem 13.13 excludes using such devices to repair failure, and the fixed-scope closure

logic internalized in Theorem D.9 forbids any second same-scope selector or admissibility-relevant parameterization. Hence no parameterized same-scope gate is admissible. $\square$

Proposition (13.5J (Competing same-scope discriminators are forbidden)). *Any typed same-scope candidate of competing-discriminator kind is inadmissible on the standard periodic carrier.*

Proof. By Definition 13.5E, a competing discriminator claims new standing-relevant continuation authority at fixed scope while avoiding both the bookkeeping-only and parameterized cases. But Theorem 13.10 already proves that no second same-scope continuation gate survives on the standard periodic carrier, and Proposition 13.3 shows that a purported same-scope predicate beyond the standing quotient either factors through the quotient or opens exactly such a forbidden second gate. Therefore a competing same-scope discriminator is inadmissible. $\square$

Proposition (13.5K (Scope changes are not same-scope rivals)). *Any typed continuation candidate of scope-change kind lies outside the same-scope theorem and cannot count as a rival same-scope continuation criterion for the standard periodic carrier.*

Proof. Definition 3.2 fixes the carrier specification and the same-scope continuation notion. Proposition 13.7 identifies the torus, incompressibility, viscosity sign, forcing choice, smooth category, and mean-zero pressure gauge as scope-fixing anchors, while Proposition 13.5 identifies the lawful presentation class. Any candidate that changes one of those commitments is not operating inside the fixed-scope problem. It may define a different problem or a broader regime, but it is not a same-scope rival. $\square$

Corollary (13.5L (Finite standing-positive same-scope rival elimination)). *On the standard periodic carrier, every standing-positive same-scope continuation-relevant candidate determination either reduces to existing carrier content, reduces to bookkeeping-only variation, or is forbidden. Consequently, no independent standing-positive same-scope continuation criterion survives.*

Proof. Apply Theorem 13.5F. Candidates of kinds (i) and (ii) reduce respectively by Propositions 13.5G and 13.5H. Candidates of kinds (iii) and (iv) are forbidden by Propositions 13.5I and 13.5J. Candidates of kind (v) are not same-scope rivals by Proposition 13.5K. Thus no independent standing-positive same-scope continuation criterion survives on the standard periodic carrier. $\square$

Remark 13.16 (How the finite typed packet fits the later endpoint-status trichotomy). The packet 13.5E–13.5L acts only on standing-positive same-scope candidate determinations on the standard periodic carrier. It is not asserted here to absorb terminal standing failure or any standing-negative endpoint. The standing-failure branch is isolated only later, by the realized-lineage endpoint-status trichotomy / collapse of Part III; the role of 13.5E–13.5L is to make the standing-positive branch finite before that later theorem-local collapse is invoked.

Proposition 13.17 (Residual standing-positive same-scope gate exhaustion). *Within Definition 13.2, every putative standing-positive same-scope gate is exhausted by the finite typed list of Theorem 13.5F: existing carrier content, bookkeeping-only variation, parameterized gate, competing discriminator, or scope change. Within the fixed periodic scope, only the first two survive, and neither contributes an independent standing-positive same-scope gate.*

Proof. This is the numbered compression of Corollary 13.5L. The only additional candidate type in Theorem 13.5F is scope change, and by Proposition 13.5K that case is outside the standing-positive same-scope theorem rather than a standing-positive same-scope survivor. $\square$

Lemma 13.18 (Profile-formation gates collapse to quotient visibility). *Within the standard $\mathbb{T}^3$ Navier–Stokes carrier, any profile-based discriminator either reduces to quotient-visible bookkeeping or else becomes a forbidden parameterized gate or competing same-scope discriminator. Hence every admissible profile-formation gate is quotient-visible and contributes no independent same-scope discriminator.*

Proof. Let G_{prof} be a profile-based discriminator on admitted presentations. If it depends only on lawful redescription of the same standing class, then by Proposition 13.5H it is bookkeeping-only and quotient-visible. If it instead uses a threshold, selector, or family member to distinguish continuation status, then it is a forbidden parameterized gate by Proposition 13.5I. If it purports to change continuation authority at fixed scope without such a selector, then it is a forbidden competing discriminator by Proposition 13.5J. These are exactly the possible types by Theorem 13.5F. Therefore every admissible profile-based gate is quotient-visible rather than independent. $\square$

Theorem 13.19 (Gate exhaustion / elimination). *Within the standard $\mathbb{T}^3$ Navier–Stokes carrier of Definition 13.2, none of the gate classes of Theorem 13.1 supplies an independent same-scope admissibility parameter.*

Proof. By Theorem 13.1, every same-scope analytic discriminator is of amplitude, scale-transfer, geometric-structure, or profile-formation kind. The finite typed packet 13.5E–13.5L then decides what can happen to each such candidate at fixed periodic scope. Amplitude and scale-transfer gates are either existing carrier content or bookkeeping-only variation, so they add no gate by Corollary 13.5L. Geometric-structure gates are carrier-internal only if they reduce to existing bundle content; otherwise they attempt to open a parameterized or competing same-scope gate and are forbidden by Propositions 13.5I and 13.5J. Profile-formation gates collapse by Lemma 13.18. Therefore none of the four atlas gate classes yields an independent same-scope admissibility parameter. $\square$

Why no gates can appear. By Theorem 2.4, any analytic continuation mechanism at fixed carrier structure must modify one of the four atlas features. The finite typed packet 13.5E–13.5L then proves, branch by branch, that every same-scope candidate either reduces to existing carrier content / bookkeeping or is forbidden. A fifth same-scope gate would therefore require either new bundle structure or a change of scope; it cannot survive as an additional same-scope discriminator.

Theorem 13.20 (Zero-parameter certification of standard $\mathbb{T}^3$ Navier–Stokes). *The standard $\mathbb{T}^3$ Navier–Stokes carrier of Definition 13.2 is zero-parameter and equivalence-complete. Consequently, it satisfies the existence-side certification later used in Theorem 13I to identify the unique admissible interior at fixed periodic scope, and the Navier–Stokes constraint bundle B_{NS} is subject to Theorem 12.1.*

Proof. Lemma 13.3 shows that the fixed torus, positive viscosity, mean-zero pressure normalization, and lawful skins are scope-fixed commitments or licensed representatives, not same-scope parameters. Lemma 13.4 provides equivalence completion of the ordinary skins, and Lemma 13.14 shows that the pointwise vortex-stretching channel contributes no independent same-scope gate. The finite standing-positive same-scope decision packet then closes the remaining cases: Corollary 13.5L removes every purported same-scope rival criterion, while Theorem 13.19 shows that none of the atlas gate classes opens an independent admissibility parameter on the standard carrier. Hence the standard carrier contains no admissibility-bearing same-scope degree of freedom, and the specialized Navier–Stokes rule package of Theorem 13.10, Corollary 13.12, and Theorem 13.13 forbids adding one or repairing failure by same-scope means. The carrier is therefore zero-parameter and equivalence-complete in

the precise existence-side sense later used in Theorem 13I to identify the *unique admissible interior*. The final saturation claim is then immediate from Theorem 12.1. $\square$

Corollary 13.21 (Certified interior inclusion). *The standard $\mathbb{T}^3$ Navier–Stokes carrier is interior-admissible in the precise sense previously used only hypothetically: saturation and factorization apply to it without any extra scope assumption.*

Proof. Theorem 13.20 establishes zero-parameter status and equivalence completion for the standard carrier. Theorem 12.1 therefore applies to B_{NS} , and the factorization discipline developed below applies within the same admitted scope. No further inclusion hypothesis remains. $\square$

Definition 13.22 (Continuation-relevant PDE structure for the standard periodic problem). A same-scope periodic Navier–Stokes quantity, criterion, mechanism, or selector is *continuation-relevant* if either

- (i) its failure blocks smooth continuation on the fixed torus carrier, or
- (ii) it purports to define a same-scope continuation discriminator for smooth periodic trajectories.

In particular, a continuation-relevant structure is not merely descriptive bookkeeping: it is claimed to matter to whether smooth continuation past a horizon is available on the fixed carrier.

Lemma 13.23 (Inventory of standard continuation-relevant structures). *For the standard periodic Navier–Stokes problem on $\mathbb{T}^3$, the ordinary continuation-relevant objects appearing in classical analysis are all interior to the standard carrier of Definition 13.2 or are lawful functionals thereof. This inventory includes:*

- (i) *energy and enstrophy bookkeeping;*
- (ii) *vorticity, strain, and pressure reconstructions;*
- (iii) *standard frequency / Littlewood–Paley localizations;*
- (iv) *BKM-type vorticity-growth observables and related threshold/profile selectors;*
- (v) *profile-formation data and the horizon continuation discriminators used later in Part III.*

Proof. Items (i)–(iii) are already present in Definition 13.2: velocity, vorticity, pressure, strain/kernel, and the standard frequency skins are licensed presentations of the same carrier object. Their derived scalar quantities (energy, enstrophy, dissipation, and related norm bookkeeping) are therefore lawful functionals of admitted carrier data.

For item (iv), the BKM-type vorticity-growth observable of Appendix C is computed from the realized vorticity field and is therefore a lawful functional of the carrier presentations. Any threshold or profile rule built from it is continuation-relevant only by adding a selector or tolerance to data already interior to the carrier. Item (v) is the same phenomenon for profile-formation and horizon logic: profile data are built from the admitted skins of Part I, while the horizon continuation discriminator formalized in Part III is exactly the carrier-side encoding of continuation status for those same smooth periodic trajectories.

Thus every standard continuation-relevant object explicitly used on the manuscript’s standing-positive same-scope route is already present in the standard carrier or is formed from it by a lawful same-scope functional. $\square$

Theorem 13.24 (Standing-positive scope completeness of the standard $\mathbb{T}^3$ carrier). *Every standing-positive same-scope continuation-relevant structure for smooth periodic Navier–Stokes on $\mathbb{T}^3$ that is admitted on the carrier-side route used below is interior to the standard carrier of Definition 13.2 or is a lawful same-scope functional thereof. Equivalently: within the standing-positive same-scope content actually carried on the live proof chain, no continuation-relevant PDE structure lies outside the standard carrier scope.*

Proof. Let Ξ be a standing-positive same-scope continuation-relevant PDE structure for the official periodic problem that is admitted on the carrier-side route used below. By Definition 13.22, either Ξ blocks smooth continuation on the fixed torus carrier or it purports to define a same-scope continuation discriminator. The former case is already a special case of the latter: if failure of Ξ blocks continuation, then the predicate “ Ξ is available” is a continuation discriminator on the fixed carrier.

By Theorem 2.4, every same-scope analytic continuation mechanism on the fixed carrier must act through amplitude control, scale transfer, geometric structure, or profile formation. Lemma 13.23 shows that the standard periodic representatives of each of those four feature families are already interior to the carrier or are lawful functionals thereof. Therefore a same-scope continuation-relevant structure cannot remain external to Definition 13.2 while still acting on one of the allowed continuation families.

If a purported continuation-relevant structure were neither interior nor a lawful functional of the carrier, then it would have to enter by new bundle data, a new selector, or a new admissibility-bearing operator family. That is exactly what Structural Exhaustion forbids at fixed scope (Rule label 3.52). Hence every standing-positive same-scope continuation-relevant PDE structure admitted on this route is already interior to the standard carrier or a lawful same-scope functional thereof. $\square$

Remark 13.25 (Carrier-interiority is only the standing-positive half of the later endgame). Theorem 13.24 is intentionally narrower than a theorem of total same-scope endpoint closure. It settles only the standing-positive carrier-admitted route used later by witness exhaustion and singular-candidate extraction. A purported endpoint that can be stated only by leaving admissible representability is not closed here; that alternative is converted into the standing-failure branch only later, by Theorem 20C together with Proposition 20C.2.

13.2C. Broader continuation-completeness architecture

The results in this subsection are retained as broader conceptual architecture for the periodic carrier. On the narrow horizon-detectability route used in Part III, they are not load-bearing for the official periodic theorem on page 1. No independent reconstruction of this broader package is claimed in the present manuscript; its role is explanatory, orienting, and explanatory.

Theorem 13.26 (Broader architectural standing-positive continuation-completeness of the standard carrier). *Every standing-positive same-scope continuation-relevant PDE phenomenon for the official periodic class that is admitted by the fixed carrier-side framework is represented by a standing-relevant discriminator on the standard carrier. Equivalently: once the standard periodic carrier is fixed, no standing-positive continuation phenomenon used on the live route remains merely external bookkeeping; each such phenomenon already determines carrier-side continuation content.*

Proof. Let Ξ be a standing-positive same-scope continuation-relevant PDE phenomenon for the official periodic class that is admitted by the fixed carrier-side framework. By Theorem 13.24, Ξ is interior to the standard carrier or a lawful same-scope functional thereof. There are then two cases.

If failure of Ξ blocks smooth continuation on the fixed torus carrier, the predicate

$$\Delta_{\Xi}(x) = 1 \iff \Xi \text{ is available / continuation-compatible on the carrier state } x$$

defines a continuation discriminator on the fixed carrier. By Corollary 13.11, continuation status on that carrier is bivalent when defined, so Δ_{Ξ} is standing-relevant exactly when Ξ matters to continuation.

If instead Ξ already purports to define a same-scope continuation discriminator, then by Definition 13.22 it is continuation-relevant in the discriminator sense directly. Because Ξ is interior to the carrier or a lawful same-scope functional thereof, its continuation content is already carried by the standard carrier data.

In either case, Ξ is represented by a standing-relevant discriminator on the standard carrier. Thus every standing-positive same-scope continuation-relevant PDE phenomenon admitted on that route is carrier-represented. $\square$

Remark 13.27 (Architectural status of strong continuation-completeness). Theorem 13.26 is retained in the manuscript as a broader conceptual and architectural statement about the standard periodic carrier. On the narrow horizon-detectability route used in Part III, it is not load-bearing for the proof of the official periodic theorem on page 1. The official theorem instead runs through the narrower detectability route given by Theorem 20.7A together with the branch-forcing and no-singular-image chain of Section 21. No independent PDE-side reconstruction of any broader totality claim is asserted in the present manuscript beyond this standing-positive carrier-admitted domain; the theorem is kept explicitly as non-load-bearing architecture for readers who want the wider conceptual picture.

Theorem 13.28 (Verification of the Part II rule package for the standard periodic carrier). *The standard periodic Navier–Stokes carrier satisfies Axioms 3.44–3.51 together with Rule label 3.52 on the fixed scope determined by the torus carrier $\mathbb{T}^3$, smooth divergence-free data, zero forcing, positive viscosity, the mean-zero pressure normalization, the lawful skins of Definition 13.2, and the same-scope continuation notion used throughout the paper.*

Proof. We verify the clauses in the order they appear in Section 3.

Bivalence and fail-closed enforcement (Axiom 3.44). By Corollary 13.11, continuation past a fixed horizon on the standard carrier is bivalent: a realized trajectory either extends smoothly on the same carrier or it does not. Under-specified or unlicensed continuation moves do not define a third status; they are simply not lawful same-scope moves on the fixed carrier and so are rejected fail-closed.

No generators, no carriers, no repairs (Axiom 3.45). This is exactly the content of Theorem 13.13 in specialized periodic Navier–Stokes form: no same-scope continuation status can be generated by association, transported across skins, or repaired after failure on the same lineage.

AMetric boundary (Axiom 3.46). At admission to the standard carrier one checks only carrier membership: smooth periodicity, divergence-freeness, the fixed torus carrier, the fixed viscosity sign, zero forcing, and the mean-zero pressure normalization. Norms, blow-up rates, cascade data, and continuation operators do not function at this boundary stage; they arise only after admission as interior analytic bookkeeping. Thus the standard periodic carrier realizes the AMetric boundary in concrete Navier–Stokes form.

Licensed interior bookkeeping (Axiom 3.47). The standard carrier explicitly licenses velocity, vorticity, pressure, strain/kernel, and frequency presentations as interior bookkeeping skins (Definition 13.2 and Lemma 13.4). By Lemma 13.3, these skins and the associated metric descriptors do not acquire independent standing-bearing force.

Operators as envelopes (Axiom 3.48). The operators used on the standard carrier — differential passage between skins, torus Poisson reconstruction with the fixed mean-zero gauge, and the admitted continuation operators of the fixed periodic problem — appear only with specified domains and failure conditions. Unlicensed operator use would therefore constitute a same-scope operator family, ruled out by the specialized periodic rule package beginning with Theorem 13.10.

Redescription neutrality (Axiom 3.49). Proposition 13.5 and Lemma 13.4 show that passage among the standard skins is lawful redescription on the fixed torus carrier: it preserves the underlying smooth periodic evolution and therefore cannot change continuation status. No power gain under redescription is available.

Composition discipline (Axiom 3.50). On the fixed carrier, smooth continuation is built from compatible same-scope local pieces of one and the same trajectory. If a horizon fails, continuation fails at that horizon for the lineage as such; there is no lawful same-scope retroactive repair or partial continuation beyond the failure time. This is the periodic Navier–Stokes specialization of composition discipline.

Deterministic enforcement (Axiom 3.51). The specialized rule package of Theorem 13.10, Corollary 13.12, and Theorem 13.13 shows that same-scope violations are structurally detectable: any attempted competing gate, parameterized continuation criterion, or repair move is ruled out by the fixed carrier commitments and fails closed.

Structural Exhaustion (Rule label 3.52). By Theorem 13.24, every standing-positive same-scope continuation-relevant PDE structure admitted on the carrier-side route used below is already interior to the standard carrier or a lawful same-scope functional thereof. By Theorem 13.10 and Corollary 13.12, no second same-scope continuation gate or parameterized admissibility level survives on that carrier; by Theorem 13.20, the carrier is already zero-parameter. Hence no new admissibility level, atomic bundle, admissible question, or operator family may be introduced on that standing-positive same-scope route. This is exactly the carrier-side Structural Exhaustion packet used later in the present manuscript.

Collecting these clauses, the standard periodic Navier–Stokes carrier satisfies the full Part II rule package on the fixed scope used throughout the paper. $\square$

Corollary 13.29 (Structural Exhaustion verified for this carrier). *For the standard periodic Navier–Stokes carrier, the carrier-side Structural Exhaustion packet actually used later in the proof is a verified property of the specific $\mathbb{T}^3$ scope: no standing-positive same-scope continuation-relevant PDE structure admitted on the live route remains outside the carrier.*

Proof. Immediate from the Structural Exhaustion clause of Theorem 13.28. $\square$

13.2A. Translation dictionary for the standard periodic carrier

The continuation-completeness claims above can be read row-by-row against the standard periodic Navier–Stokes observables. Table 13.A records the dictionary used in the rest of the paper for the standing-positive same-scope route actually carried later: every classical continuation-relevant observable used on that route appears either as a lawful carrier skin, a tensor-level bundle quantity, the horizon continuation discriminator, or the singular-image extractor. In this sense the table is not a theorem but an explicit reading guide to the carrier-representative content already certified in Theorems 13.24 and 13.26 within that narrowed domain.

Corollary 13.23A (Row-by-row verification of strong continuation-completeness)

For every continuation-relevant classical Navier–Stokes observable listed in Table 13.A that is used on the manuscript’s standing-positive same-scope route, the carrier-side object recorded in the table captures its same-scope standing-relevant content on the standard periodic carrier. The table is not asserted here to certify every non-standing or otherwise externalized same-scope endpoint.

The metric/tensor/witness localization packet sharpens Table 3 into an explicit role-localization packet for the actual periodic Navier–Stokes observables used on the live route. The point is not to add a new analytic estimate. It is to prevent metric / fixation language from becoming a hidden second gate: the next definition and propositions say exactly which objects are anchors, which carry tensor-level compatibility content, which remain skin-level bookkeeping, and when a metric divergence statement is allowed to appear only as bookkeeping for an already-fixed quotient-visible witness.

Definition (13.23A.1 (Quotient-visible witness bookkeeping)). *Fix the standard periodic carrier and a realized maximal classical lineage issued from an admitted carrier state. A lawfully formed quantity Q on that lineage is quotient-visible witness bookkeeping on the live same-scope route if:*

- (i) *Q is computed from admitted carrier presentations or deterministic reconstructions already licensed in the fixed scope;*
- (ii) *whenever Q is used to record a purported finite-horizon failure, its standing-bearing content is wholly inherited from an already fixed same-scope witness — namely the pre-horizon witness of Lemma 19F, the horizon discriminator of Theorem 20.7A, or the singular-candidate extractor of Definition 21.6; and*
- (iii) *Q does not itself define a same-scope continuation gate, admissibility selector, or competing discriminator.*

Such a Q is bookkeeping subordinate to an already fixed quotient-visible witness rather than a fresh source of standing authority.

Proposition (13.23A.2 (Periodic NS role localization on the live same-scope route)). *On the live same-scope route used by the official periodic endpoint, the periodic Navier–Stokes objects listed in Table 4 occupy exactly the ATS roles recorded there. In particular, carrier constants are anchors; field-level displays such as the fixed-gauge velocity-pressure pair, vorticity, and strain are lawful skins; reconstruction and compatibility laws are tensor content; metric continuation quantities carry no standing authority by mere divergence; and the only same-scope standing-bearing continuation gate on that route is the quotient-visible horizon discriminator together with the witness packaging later extracted from it.*

Proof. The anchor row is exactly Proposition 13.7(i) together with Lemma 13.3: changing the torus carrier, viscosity sign, forcing, smooth category, or fixed pressure gauge changes the same-scope problem rather than refining it. Definition 13.2 and Definition 18.0 show that the realized velocity-pressure pair in the fixed gauge is one lawful presentation of the same carrier-bound lineage, not a second admissibility object. Proposition 13.5 makes the same point for vorticity, strain, and the other deterministic field displays: as displayed fields they are lawful skins of one carrier state.

What is tensor-level on this route is not the mere presence of another display, but the fixed compatibility content tying those displays to one same-scope state. Proposition 13.6 identifies standing equivalence with sameness of continuation behavior on the fixed carrier, so the curl / Poisson / velocity-vorticity-pressure reconstruction laws and the energy-dissipation ledger contribute only as quotient-visible compatibility content. Proposition 13.5B supplies the energy / dissipation row in that tensor-bookkeeping sense.

For the metric row, Axiom 3.47 and Lemma 13.3 already say that metric descriptors are licensed only as interior bookkeeping and do not acquire independent standing authority merely by being measurable or divergent. Lemma 19F and Theorem 20.7A show how such quantities can still appear on the live route: not as gates, but only as bookkeeping for an already fixed pre-horizon witness or horizon discriminator. The horizon discriminator row is exactly Theorem 20, which makes continuation failure bivalent and quotient-visible on the fixed carrier. The singular-candidate extractor row is then Definition 21.6 together with Proposition 21.6A: the extractor packages the already fixed standing-positive witness content carried by the horizon discriminator and its induced branch data, and does not create an additional continuation selector. $\square$

Proposition (13.23A.3 (Skin non-authority for metric divergence)). *Let M be any same-scope lawfully formed metric quantity on the fixed periodic carrier or on one of its realized periodic lineages, including critical continuation norms, BKM-type growth quantities, or any divergence statement built from them. Then:*

- (i) *M is licensed interior bookkeeping and does not by itself alter same-scope standing or admissibility classification;*
- (ii) *if M is used on the live route, its only licit standing-bearing role is as quotient-visible witness bookkeeping in the sense of Definition 13.23A.1, subordinate to the pre-horizon witness of Lemma 19F, the horizon discriminator of Theorem 20.7A, or the singular-candidate extractor used later in Section 21;*
- (iii) *if M is promoted to an independent same-scope continuation gate or admissibility selector, that promotion is an illicit extra discriminator / scope change.*

In particular, a singularity claim gains no standing authority merely by being expressed as metric divergence.

Proof. Item (i) is the fixed periodic specialization of Axiom 3.47: once the carrier is admitted, metric descriptors are licensed as interior bookkeeping, not as boundary-level authorizers. Lemma 13.3 and Corollary 3.40 then prevent such bookkeeping from becoming a hidden same-scope gate merely by being attached to a particular presentation.

For item (ii), the live route already fixes the standing-bearing witness content of any finite-horizon failure by local theorems: Lemma 19F produces an in-scope pre-horizon witness on the realized lineage, Theorem 20.7A converts first non-extendability into the carrier-side horizon discriminator, and later Section 21 packages that standing-positive content by the singular-candidate extractor of Definition 21.6. Any metric quantity used on that route is therefore licit only as bookkeeping subordinate to one of those already fixed witnesses.

If item (iii) were denied, then some metric quantity M would change same-scope standing outcomes without factoring through the already fixed witness content. By Theorem 3.35 every same-scope standing-relevant discriminator must factor through the fixed quotient, by Theorem 3.37 and Corollary 3.42 it may not depend on path or horizon-order bookkeeping, and by Theorem 20B realization adds no independent classical-layer continuation gate. Therefore promoting M to independent same-scope authority is exactly an illicit extra discriminator / scope change. $\square$

Proposition (13.23A.4 (Quotient-visible witness vs illicit metric gate)). *Let $\mathcal{M}$ be a same-scope singularity or continuation-failure claim for the official periodic problem formulated through metric divergence, norm blow-up, or other licensed metric bookkeeping on a realized periodic lineage. Then exactly one of the following holds:*

- (i) $\mathcal{M}$ is only licensed metric skin bookkeeping and carries no standing authority;
- (ii) $\mathcal{M}$ is quotient-visible witness bookkeeping subordinate to the already fixed pre-horizon witness, horizon discriminator, or singular-candidate extractor, and therefore adds no same-scope gate;
- (iii) $\mathcal{M}$ is asserted as an independent same-scope continuation discriminator, in which case it is an illicit metric gate / scope change.

There is no fourth same-scope use of metric divergence.

Proof. The trichotomy is forced by Proposition 13.23A.3 together with the ATS packet. If the metric statement is used without changing any same-scope standing-relevant consequence, then by Theorem 3.39 and Corollary 3.40 it is skin-level bookkeeping and falls under item (i). If it is used on the live route only to package the witness content already fixed by Lemma 19F, Theorem 20.7A, or Definition 21.6, then it is quotient-visible witness bookkeeping by Definition 13.23A.1 and falls under item (ii). If it is asserted to change same-scope continuation status independently of those fixed witnesses, then Theorem 3.35, Theorem 3.37, and Theorem 20B force it into item (iii): an illicit extra gate or scope change. No further same-scope role remains by Theorem 3.39. $\square$

Proposition (13.23A.5 (Boundary licensing compatibility for the periodic closure route)). *On the fixed periodic carrier, boundary licensing of metric descriptors does not generate the missing analytic bridge inequality, a new coercive estimate, or a continuation criterion. Nevertheless, once the official periodic same scope is admitted, same-scope admissibility still governs which purported singularity representations are licit: a metric divergence claim is same-scope load-bearing only insofar as it is quotient-visible witness bookkeeping subordinate to the already fixed horizon witness. Therefore there is no contradiction between saying that licensed metric bookkeeping is non-generative and saying that the same-scope closure route excludes licit singular images.*

Proof. The non-generative part is already the content of Axiom 3.47, the Claim type and method class box on page 1, Remark 17.7, Proposition 13K, and Theorem 20.5B: the paper does not claim that admissibility or boundary licensing produces the missing coercive estimate. The second clause is exactly the content of Propositions 13.23A.3 and 13.23A.4: once the same scope is fixed, a metric divergence claim is licit only as bookkeeping subordinate to the already fixed same-scope witness packet, and if it is promoted beyond that role it becomes an illicit extra gate or scope change. Hence the two statements are compatible and describe different burdens rather than conflicting proof postures. $\square$

Definition (13.23A.M (Three roles of metric quantities)). *On the fixed periodic carrier, a metric quantity may appear in three distinct roles. First, it may be an analytic diagnostic: valid PDE bookkeeping such as an H^m norm, a BKM integral, a critical norm, a profile descriptor, or a vortex-stretching observable. Second, it may be a continuation-obstruction witness: a pre-horizon record of why the standard local continuation radius is not being asserted uniformly. Third, it may be promoted to same-scope endpoint authority: a purported independent rule deciding the official endpoint status. The first two roles are legitimate bookkeeping / witness roles; the third is forbidden unless it factors through the already fixed finite-horizon endpoint condition and horizon discriminator.*

Theorem (13.23A.6 (Metric Diagnostic Non-Authority Theorem for the periodic carrier)). *Let $\mathcal{M}$ be any norm divergence, BKM-type observable, scale-critical threshold, profile concentration descriptor, or Θ_x -derived growth statement formed from lawful periodic Navier–Stokes data on the fixed carrier. Such a quantity remains a valid analytic diagnostic and may serve as a continuation-obstruction witness in the sense of Definition 13.23A.M. What it cannot do, merely because it is metric, large, divergent, or classically familiar, is become an independent same-scope endpoint authority. On the live same-scope route it has exactly one of the following endpoint-status roles:*

- (i) *licensed metric bookkeeping with no endpoint authority;*
- (ii) *quotient-visible witness bookkeeping subordinate to Lemma 19F, Theorem 20.7A, Definition 20C.0, or Definition 21.6; or*
- (iii) *an illicit metric gate / scope change if promoted as an independent continuation discriminator.*

In particular, metric divergence may record the pre-horizon visibility of a purported failure, but it cannot create a same-scope endpoint status.

Proof. Proposition 13.23A.3 proves that lawfully formed metric quantities on the fixed periodic carrier are licensed interior bookkeeping and have no independent standing authority. Proposition 13.23A.4 proves the complete same-scope trichotomy for metric divergence claims: bookkeeping, quotient-visible witness bookkeeping, or illicit extra gate / scope change. Proposition 13.23A.5 records the compatibility point needed here: non-generative metric licensing does not supply a coercive estimate, while same-scope admissibility still governs the licit status of singularity representations. Definition 13.23A.M separates this non-authority claim from any denial of ordinary PDE diagnostic value. The listed observables are all metric descriptors or metric-derived descriptors; therefore they fall under that trichotomy and have no fourth same-scope endpoint-authority role. $\square$

13.2B. Carrier-specific structural exhaustion on the standard periodic carrier

This subsection is now downstream of the finite decision packet 13.5E–13.5L and of the carrier-certified Structural Exhaustion packet of Theorem 13.28 and Corollary 13.29. Its role is not to be the first place where finite same-scope exhaustion appears, and on the narrow horizon-detectability route used for the official periodic theorem it is not a separate live node. Rather, it repackages the earlier typed classification in the continuation-mechanism language used for translation and local bookkeeping in the later branch-forcing and terminality discussion. In that local language, the only surviving same-scope continuation mechanisms are: lawful-skin bookkeeping, horizon factorization through D_T , or bundle deformation forcing one of (S1)–(S5). Thus 13.2B is a carrier-local continuation reformulation of already-certified standing-positive same-scope closure, not as a fresh proof hinge.

Definition (13.23B (Same-scope continuation mechanism on the standard periodic carrier)). *A same-scope continuation mechanism on the standard periodic carrier is any rule, predicate, or family of predicates that purports to distinguish smooth continuation behavior on the fixed torus carrier without changing:*

- (i) *the carrier $\mathbb{T}^3$,*
- (ii) *the smooth divergence-free category,*
- (iii) *the mean-zero pressure normalization,*

- (iv) the lawful skins of the standard carrier, or
- (v) the same-scope continuation notion fixed in Part III.

Definition (13.23C (Three continuation kinds)). *A same-scope continuation mechanism is of:*

- (i) bookkeeping kind if it depends only on lawful presentation / skin choice and factors through the standing quotient;
- (ii) horizon kind if its continuation-relevant content factors through the family of horizon discriminators D_T ;
- (iii) deformation kind if, to change continuation status, it must alter one of the fixed carrier-side ingredients and thereby realize one of the forbidden moves (S1)–(S5).

Proposition (13.23D (Lawful-skin factorization)). *Any same-scope continuation mechanism on the standard periodic carrier that distinguishes only among lawful presentations of the same carrier state is of bookkeeping kind.*

Proof. By Definition 13.2 and Lemma 13.4, the velocity, vorticity, pressure, strain/kernel, and standard frequency presentations are lawful skins of one and the same carrier object. If a same-scope continuation mechanism distinguishes only among those lawful presentations, then it changes neither the underlying carrier state nor its continuation behavior on the fixed torus carrier. By Definition 13.23C, such a mechanism is of bookkeeping kind. $\square$

Proposition (13.23E (Undeformed continuation mechanisms factor through horizons)). *Let $\mathfrak{M}$ be a same-scope continuation mechanism on the standard periodic carrier. If $\mathfrak{M}$ changes continuation status without changing any of the fixed carrier-side bundle ingredients, then $\mathfrak{M}$ factors through the horizon discriminators D_T .*

Proof. By Theorem 19.6, a standing-positive time-zero carrier class determines a unique maximal smooth periodic trajectory on the fixed torus carrier. By Proposition 20.4, smooth continuation status at a fixed horizon depends only on the standing class, and by Proposition 20.7 a first failure of continuation is recorded by the horizon discriminator attached to that class. If $\mathfrak{M}$ changes continuation status while leaving the fixed carrier-side bundle ingredients unchanged, then the only same-scope continuation-relevant datum available to it is whether the trajectory survives a given horizon. Hence $\mathfrak{M}$ factors through the family D_T . $\square$

Proposition (13.23F (Non-factorizing continuation mechanisms force bundle deformation)). *Let $\mathfrak{M}$ be a same-scope continuation mechanism on the standard periodic carrier. If $\mathfrak{M}$ is neither of bookkeeping kind nor of horizon kind, then any continuation-relevant use of $\mathfrak{M}$ must alter one of the fixed carrier-side ingredients and therefore realizes one of the forbidden moves (S1)–(S5).*

Proof. Fix the torus carrier, smooth divergence-free class, mean-zero pressure normalization, lawful skins, and same-scope continuation notion of Definition 13.23B. If $\mathfrak{M}$ is not of bookkeeping kind, it does not merely distinguish lawful presentations of the same carrier state. If it is not of horizon kind, its continuation content does not factor through D_T . Then, to change continuation status on the fixed carrier, $\mathfrak{M}$ must do something more than bookkeeping or horizon discrimination. On this carrier the only remaining same-scope possibilities are exactly the forbidden bundle deformations isolated in the branch-forcing package: weakening a bundle clause, isolating a coupled component, introducing a tolerance or selector, using unlawful redescription, or using composition order to rehabilitate failure. These are precisely the moves (S1)–(S5). Hence any such $\mathfrak{M}$ is of deformation kind. $\square$

Theorem (13.23G (Exhaustive trichotomy of standing-positive same-scope continuation mechanisms)). *Every standing-positive same-scope continuation mechanism on the standard periodic carrier is exactly one of:*

- (i) *bookkeeping kind,*
- (ii) *horizon kind,*
- (iii) *deformation kind.*

No fourth standing-positive same-scope continuation-relevant kind exists.

Proof. The earlier finite standing-positive same-scope decision packet 13.5E–13.5L has already shown that every standing-positive continuation-relevant candidate determination at fixed periodic scope either reduces to existing carrier content, reduces to bookkeeping-only variation, is forbidden as a parameterized gate or competing discriminator, or else changes scope. Propositions 13.23D, 13.23E, and 13.23F now translate those surviving same-scope possibilities into the continuation-mechanism language needed later: existing carrier content / bookkeeping become bookkeeping kind, undeformed continuation content factors through the horizon family D_T , and every non-factorizing same-scope attempt becomes deformation kind. No fourth standing-positive same-scope continuation-relevant kind exists. $\square$

Corollary (13.23H (Architectural continuation-mechanism reformulation of the standing-positive carrier route)). *For the standard periodic Navier–Stokes carrier, the bookkeeping / horizon / deformation trichotomy of Theorem 13.23G is an architectural continuation-mechanism reformulation of the carrier-certified Structural Exhaustion packet already verified in Theorem 13.28 and Corollary 13.29. On the narrow horizon-detectability route used for the official periodic theorem, later load-bearing steps cite those carrier-certified labels together with the finite packet 13.5E–13.5L and Proposition 13K rather than this corollary.*

Proof. Theorem 13.23G reformulates the already-certified standing-positive same-scope closure of the standard periodic carrier in continuation-mechanism language: bookkeeping, horizon factorization through D_T , or deformation forcing one of $(S1)$ – $(S5)$, with no fourth standing-positive same-scope continuation-relevant kind. This is the same carrier-side no-residue content already verified on the live route by Theorem 13.28 and Corollary 13.29; the present corollary records that equivalence for architectural translation only. $\square$

13.2D. Same-scope existence on the certified periodic carrier

The continuation-mechanism reformulation of Section 13.2B, read together with the carrier-certified Structural Exhaustion packet of Theorem 13.28 and Corollary 13.29, has a fixed-scope existence meaning for the standard periodic carrier. The present subsection records that meaning explicitly. It is interpretive rather than load-bearing: it explains what the already-certified standing-positive same-scope closure says about existence on the fixed carrier, without replacing the proof work carried by Part II and Part III.

Definition (13.23I (Standing-positive same-scope existence claim on the standard periodic carrier)). *A standing-positive same-scope existence claim on the standard periodic carrier is an existence claim that preserves simultaneously:*

- (i) *the torus carrier $\mathbb{T}^3$,*

- (ii) the smooth divergence-free class,
- (iii) zero forcing,
- (iv) the mean-zero pressure gauge,
- (v) the lawful skins of the standard carrier, and
- (vi) the same-scope continuation notion fixed in Part III.

Proposition (13.23J (Standing-positive same-scope existence coincides with admissible-interior membership on the certified carrier)). *For the standard periodic carrier, once Corollary 13.23H has been established, every standing-positive same-scope continuation-relevant existence claim is exhausted by bookkeeping/skin factorization, horizon factorization through D_T , or deformation forcing into one of (S1)–(S5). Hence no standing-positive same-scope remainder exists beyond the unique admissible interior realized by the certified carrier.*

Proof. Let a standing-positive same-scope continuation-relevant existence claim on the standard periodic carrier be given in the sense of Definition 13.23I. By Corollary 13.23H, every standing-positive same-scope continuation mechanism on this carrier is exhausted by the three kinds proved in Theorem 13.23G: bookkeeping kind, horizon kind, or deformation kind. Thus the claim can contribute same-scope content only by lawful bookkeeping/skin factorization, by horizon factorization through the carrier family D_T , or by deformation forcing one of the forbidden moves (S1)–(S5). No fourth standing-positive same-scope continuation-relevant kind exists. Accordingly, there is no standing-positive same-scope continuation-relevant remainder beyond the unique admissible interior realized by the certified carrier. $\square$

Corollary (13.23K (Failure of standing is failure of same-scope existence)). *On the certified standard periodic carrier, a purported singularity that is not standing-positive is not merely “outside the current description”; it fails to exist at that fixed scope.*

Proof. By Proposition 13.23J, same-scope continuation-relevant existence claims on the certified standard periodic carrier are exhausted by bookkeeping/skin factorization, horizon factorization through D_T , or deformation forcing into one of the forbidden moves (S1)–(S5). Part II and Part III eliminate the singular branches of that trichotomy on the certified carrier: lawful bookkeeping does not generate a singularity, horizon factorization is handled by the narrow detectability and no-singular-image route, and deformation forcing into one of (S1)–(S5) is excluded by the branch-forcing and branch-elimination package. Thus a purported singularity that is not standing-positive on the certified carrier is not merely absent from a particular description of the carrier; it fails to be an element of the same-scope object class fixed by Definition 13.23I. $\square$

Remark 13.30 (Interpretive status of Section 13.2D). Section 13.2D is consequence-level only. It records what the already-derived carrier-specific structural exhaustion theorem means for same-scope existence on the fixed carrier; it is not additional proof support for the official periodic theorem.

13.2E. Carrier-identification packet (13I–13L): the standard periodic carrier is the unique admissible interior

The fixed-scope necessity theorem of Section 3 now meets the concrete periodic Navier–Stokes carrier. The point of the present subsection is to state, in theorem-bearing form, that this carrier is not one same-scope standing-positive realization among several. At the fixed torus scope of the paper it

realizes the *unique admissible interior*, and therefore no second same-scope continuation regime remains available behind the later singularity discussion. The packet below is explicit: Theorem 13I identifies the standard carrier with the already-forced interior, Corollary 13J excludes any rival same-scope continuation regime, and Proposition 13K states directly that no additional PDE-side continuation criterion remains inside the fixed periodic scope.

Remark 13.31 (Downstream citation discipline after the structural packet). From this point onward, the structural burden is cited only through theorem numbers already proved in this manuscript: Theorem P1, Corollary P2, Corollary P3, and Theorem P4, Theorem 12G, and Corollary 12H. The four structural source papers not function anywhere in the proof spine as load-bearing references.

Theorem (13I (The standard periodic carrier realizes the unique admissible interior)). *Fix the periodic Navier–Stokes scope on $\mathbb{T}^3$ with positive viscosity, incompressibility, zero forcing, smooth category, the mean-zero pressure normalization, the lawful skins of Definition 13.2, and the same-scope continuation notion used throughout the paper; equivalently, fix the comparison class of Definition 3.4A. Then the standard periodic Navier–Stokes carrier is not merely a disciplined formulation: it is exactly the unique admissible interior for this fixed scope.*

Proof. Corollary 12H already fixes the existence-side fact: at the declared periodic scope there is exactly one admissible interior. It therefore remains only to show that the standard periodic carrier is a standing-positive realization of that already-forced interior and that no rival same-scope interior survives outside it.

The standard carrier is admissibility-positive at the fixed scope. Proposition 13.7 shows that the periodic problem itself determines the relevant anchor / tensor / skin split. By Definition 13.2 together with Lemma 13.4, the velocity, vorticity, pressure, strain/kernel, and standard frequency presentations are lawfully completed into one carrier rather than competing same-scope structures. Theorem 13.20 certifies that carrier as zero-parameter and equivalence-complete, while Theorem 13.28 together with Corollary 13.29 show that, on this fixed carrier, there is no second same-scope continuation gate, no parameterized admissibility level, no repair-by-derived-quantity, and no standing-positive same-scope continuation-relevant residue on the live route outside the carrier commitments. Thus the standard carrier satisfies the existence-side certification required of the already-forced admissible interior.

Now let $\mathcal{I}'$ be a purported rival same-scope interior compatible with the same fixed periodic scope. If $\mathcal{I}'$ changes admissibility outcomes while preserving that scope, then it is either a same-scope admissible variation or a second same-scope continuation gate, contrary to Corollary 12H, Theorem 13.10, and Corollary 13.12. If $\mathcal{I}'$ preserves all standing-relevant outcomes, then by Lemma 13.4 it is only lawful bookkeeping / redescription inside the same carrier and is not a rival interior at all. If it alters any fixed anchor commitment listed in Proposition 13.7 — the torus carrier, incompressibility, viscosity sign, forcing choice, smooth category, mean-zero pressure normalization, or lawful presentation class — then it is a scope change in the sense of Definition 3.2, not a same-scope counterexample. Therefore no rival same-scope interior survives, and the standard periodic carrier realizes the *unique admissible interior*. $\square$

Corollary (13J (No rival same-scope continuation regime exists)). *At the fixed periodic scope of Theorem 13I, any purported same-scope continuation regime distinct from the regime of the standard periodic carrier is either continuation-neutral bookkeeping or a scope change. In particular, no rival standing-positive same-scope continuation regime exists.*

Proof. Any standing-positive same-scope continuation regime must live inside the *unique admissible interior* fixed by Corollary 12H and identified concretely with the standard carrier by Theorem 13I.

If a purported rival regime changes standing-relevant continuation outcomes, then it is a second same-scope gate or parameterized family of gates, contrary to Theorem 13.10 and Corollary 13.12. If it changes no standing-relevant continuation outcomes, then it is bookkeeping / lawful redescription only and does not define a rival regime. The only remaining option is a change in the fixed anchor commitments, which is a scope change rather than a same-scope continuation regime. $\square$

Proposition (13K (No additional PDE-side continuation criterion remains inside fixed scope)). *Let $\mathfrak{C}$ be any criterion advertised as a same-scope PDE-side continuation criterion for the fixed periodic Navier–Stokes problem. Then exactly one of the following holds:*

- (i) $\mathfrak{C}$ factors through the already identified carrier-side admissibility structure and adds no proof power; or
- (ii) $\mathfrak{C}$ changes same-scope admissibility outcomes and therefore constitutes a scope change, not an additional criterion inside the fixed periodic scope.

Thus neither a same-scope continuation gate nor a hidden PDE-side discriminator survives once the standard carrier has been identified with the unique admissible interior.

Proof. Lemma 12F is used here only to fix the same-scope typing / boundary-signature-closure status of any purported additional PDE-side criterion at the declared periodic interface. The exclusion consequence used here is not sourced from Lemma 12F in isolation: Theorem 12G applies Theorem P4 (P4) to the completed fixed-periodic bridge packet and therefore excludes additional same-scope admissibility gates once that scope-fixity classification is in place. By Corollary 13J, no rival standing-positive same-scope continuation regime survives once the standard carrier has been identified with the unique admissible interior. Therefore a purported PDE-side continuation criterion $\mathfrak{C}$ has only two possibilities. If it changes same-scope admissibility outcomes, then it is a second same-scope gate and is excluded by the P4 application already instantiated at fixed periodic scope in Theorem 12G. If it does not change those outcomes, then it factors through the already identified carrier-side admissibility structure and contributes no additional same-scope proof power. This is the explicit Part II answer to the analytic-reader expectation that a further Beale–Kato–Majda-style or Serrin-style continuation ingredient must still be hiding somewhere in the proof spine. $\square$

Proposition (13K.1 (Fixed-domain counterexample burden)). *Every proposed finite-horizon singular counterexample to the official periodic theorem has exactly one of the following statuses relative to the fixed-domain comparison class of Definition 3.4A:*

- (i) *it preserves the fixed-domain comparison class, in which case it is evaluated by the same-scope exhaustion, no-new-criterion, realized-lineage status-trichotomy, and branch-elimination chain already fixed in this manuscript; or*
- (ii) *it alters at least one fixed-domain commitment, in which case it is not a counterexample to the fixed periodic theorem stated in Theorem 21.17.*

Thus a skeptical proposal must exhibit a standing-bearing difference while preserving the declared comparison class; merely supplying a richer analytic regime, a different weak formulation, a new profile language, or a different metric descriptor does not by itself engage the theorem.

Proof. Let a proposed finite-horizon singular endpoint condition be given. If it changes the torus, viscosity, forcing status, smooth periodic datum class, pressure gauge, lawful-redescription class, or continuation question, then by Definition 3.4A it has left the comparison class and is not a

same-scope counterexample. Assume instead that all fixed-domain commitments are preserved. Then Definition 3.4B makes the finite-horizon condition a same-scope endpoint condition on the already identified carrier. Proposition 13K rules out additional PDE-side continuation criteria, Theorem 12G supplies the fixed-scope exhaustion consequence, and Part III later transports the same burden to the realized-lineage status trichotomy and branch-elimination packet. Hence every genuine same-scope counterexample is already inside the local exhaustion chain, while every alleged endpoint outside that chain has changed domain. $\square$

Remark 13.32 (Relative uniqueness under locked scope). The words “unique”, “forced”, and “exhausted” in the carrier-identification packet are relative to the fixed-domain comparison class. They do not assert uniqueness across unconstrained mathematical imagination; they assert that, once the official periodic carrier and continuation question are locked, no standing-preserving same-scope alternative remains.

Corollary (13L (Part II closes the same-scope admissibility burden before classical transfer begins)). *By Corollary 12H, Theorem 13I, Corollary 13J, and Proposition 13K, the same-scope admissibility burden later used by the official endpoint is already closed before Part III begins. In particular, Part III is not used to prove the unique admissible interior, the carrier identification, or the exclusion of additional same-scope continuation criteria.*

Proof. Corollary 12H fixes the unique admissible interior at the declared periodic scope. Theorem 13I identifies the standard periodic carrier with that interior, Corollary 13J excludes rival same-scope regimes, and Proposition 13K rules out any additional same-scope PDE-side criterion. The subsequent Part III statements begin only with local realization, transfer faithfulness, detectability, and endpoint transport. None is cited in the proofs of the carrier-identification packet 13I–13K or the earlier Part II certification chain. Hence the same-scope admissibility burden is closed before the classical transfer cascade starts, and no circular handoff occurs on the endpoint chain. $\square$

Remark 13.33 (Consequence of carrier identification). After the carrier-identification packet 13I–13L, the standard carrier is not presented as one disciplined formalization among several. A reader who wants to resist the later singularity exclusion must reject either the fixed-scope primitive necessity chain, the proof that the periodic problem realizes that unique interior, or the explicit Part II closure recorded in Corollary 13L and Proposition 13K that no additional same-scope PDE-side continuation criterion survives once that identification has been made.

14 Uniqueness-first contradiction inside the unique admissible interior

The decisive contradiction now comes before the branch taxonomy. Once the standard periodic carrier has been identified with the *unique admissible interior* by Theorem 13I, a same-scope singularity cannot be imagined as living in some unmentioned admissible regime. It would have to arise either from a second same-scope interior, from an admissible variation or extra continuation discriminator inside the unique interior, or from a scope change misdescribed as same-scope continuation. The later quotient and branch machinery is diagnostic only: it records how such impossible escape attempts would present themselves concretely on the fixed carrier.

Theorem 14.1 (Uniqueness-first contradiction for same-scope singularity). *Fix the periodic Navier–Stokes scope of Theorem 13I. Any same-scope singularity proposal on that fixed scope would require at least one of the following:*

- (i) a second same-scope interior at the same fixed periodic scope;
- (ii) an admissible variation, additional same-scope continuation discriminator, or second continuation regime within the unique admissible interior;
- (iii) a scope change disguised as a same-scope continuation claim.

Each option is impossible.

Proof. Option (i) is excluded directly by Theorem 13I: at the fixed torus scope the *unique admissible interior* has already been fixed.

For option (ii), first note that admissible variation inside a fixed admitted scope is already ruled out by Lemma 3.19 and Theorem 3.20. In the periodic Navier–Stokes specialization, Corollary 13J says that every standing-positive same-scope continuation regime on the fixed carrier coincides with the regime of the standard periodic carrier, while Theorem 13.10 and Corollary 13.12 eliminate any second same-scope gate or parameterized family of such gates. Hence no admissible singular deviation can arise from an interior variation or extra same-scope discriminator within the unique interior.

Option (iii) is not a same-scope singularity at all. If a purported continuation claim changes the torus carrier, viscosity sign, forcing choice, smooth category, pressure normalization, lawful skin class, or other fixed anchor commitments, then by Definition 3.2 it changes scope rather than refining the existing one. Such a move is therefore an illicit scope change, not a same-scope counterexample.

Therefore a same-scope singularity cannot live in a second same-scope interior, cannot be realized as an admissible variation inside the unique admissible interior, and cannot survive as a disguised scope change. The later S1–S5 taxonomy can only record the finite witness classes of such impossible escape attempts. $\square$

The next layer of the endgame uses a quotient-level uniqueness principle internal to the already fixed *unique admissible interior*. Once Theorem 14.1 has removed the possibility of a second interior or an admissible singular variation, every standing-relevant consequence claim must be invariant under lawful redescription and therefore constant on standing classes (Definition 3.30). Otherwise it distinguishes representatives inside one class and functions as a same-scope selector, hence as an illicit same-scope parameter (Definition 3.4). This yields a factorization discipline for admissible consequence functionals.

Definition 14.2 (Admissible consequence functional). Fix an admitted scope with interior state-set P and standing quotient $q : P \rightarrow Q_S$ as in Definition 3.30. An *admissible consequence functional* is any typed map $F : P \rightarrow Y$ used as standing-relevant within the fixed scope interface (for example a legality predicate, enforcement predicate, or compatibility-class output).

Proposition 14.3 (Factorization discipline). *Within a fixed scope, every standing-relevant admissible consequence functional $F : P \rightarrow Y$ (Definition 14.2) is constant on standing classes and therefore factors uniquely through the standing quotient $q : P \rightarrow Q_S$. Any purported same-scope functional that fails to factor is not a refinement of the existing scope but an illicit scope change.*

Proof. This is the periodic specialization of Theorem 3.35. Once the fixed carrier of Definition 13.2 is in force, any same-scope standing-relevant admissible consequence functional is constant on standing classes and therefore factors uniquely through the standing quotient. If it did not, it would distinguish standing-equivalent lawful carrier presentations and would therefore act either as privileged representative skin or as an illicit second same-scope discriminator, contrary to Theorem 3.35 and Corollary 3.40. $\square$

Corollary 14.4 (No second same-scope singularity gate). *Let F_{sing} be any same-scope proposal to separate Navier–Stokes states into “regular” and “singular” classes while preserving the admitted carrier and standing lineage. Then exactly one of the following holds:*

- (i) *F_{sing} factors through the existing quotient and is already part of the standing-relevant invariant content of the scope; or*
- (ii) *F_{sing} is an inadmissible additional discriminator and collapses by same-scope factorization discipline.*

Proof. Apply Proposition 14.3. If F_{sing} fails to factor, Structural Exhaustion and closure logic treat it as a scope change rather than a same-scope addition. If it factors, it contributes no new standing-bearing content beyond what is already present in Q_S . $\square$

Remark 14.5 (Endgame consequence). Corollary 14.4 is the typed replacement for the informal claim that no extra same-scope lemma remains. Any singularity criterion not already quotient-visible is a scope change rather than a same-scope refinement. The endgame therefore reduces to showing that every quotient-visible singularity mark is already standing-negative.

15 Diagnostic escape patterns for singularity inside the unique admissible interior

Because Theorem 14.1 has already ruled out a second same-scope interior and any admissible variation within the unique interior, the remaining task is diagnostic rather than foundational. A same-scope singularity can only appear as an attempted escape from the unique admissible interior already fixed in Part II. The labels (S1)–(S5) below are the finite witness classes of such attempted escapes. They are not the primitive source of impossibility; they are the concrete bookkeeping vocabulary in which an impossible escape attempt would have to present itself once the Part I vortex-stretching wall leaves no licensed same-scope analytic bridge. That wall shows the incompleteness of same-scope PDE analytic closure; it does not reopen primitive admissibility.

Definition 15.1 (Singularity gate, typed). A *singularity gate* is a standing-relevant admissible consequence functional on the fixed Navier–Stokes scope whose output purports to distinguish same-lineage states that remain admissibly continuable from states that terminate as “singular.”

Remark 15.2. Definition 15.1 is not yet a claim that such a gate exists. It only isolates the exact type such a discriminator would have to have in order to bear inferential weight inside the map. By Corollary 14.4, a gate of this kind is admissible only if it is already quotient-visible.

Proposition 15.3 (No generators, no carriers, no repairs). *Within the fixed standing lineage of an admitted Navier–Stokes description, no singular candidate may be made admissible by:*

- (a) *generating a new authorizer or discriminator independent of the existing gate,*
- (b) *carrying authorization across domains, chains, or lineages without fresh standing,*
- (c) *repairing or re-legalizing a failed object post hoc within the same lineage.*

Proof. This is exactly Axiom 3.45 (no generators, no carriers, no repairs) applied to a fixed admitted Navier–Stokes standing lineage. Any attempt to make a singular candidate admissible by (a)–(c) is a post-hoc repair or a standing transfer and is rejected fail-closed. $\square$

Definition 15.4 (Singular candidate, typed). A singular candidate is a purported standing-positive interior Navier–Stokes description whose continued membership in B_{NS} would require at least one of the following:

- (S1) local weakening or exception to a bundle clause;
- (S2) deformation or isolation of one interior constraint from the rest;
- (S3) a tolerance, selector, or staged repair not already licensed;
- (S4) a non-equivalent redescription used to mask incompatibility;
- (S5) downstream composition used to justify an earlier failure.

Definition 15.4 is not an analytic description of blow-up and it is not the primitive source of impossibility. The five labels are the diagnostic witness classes of attempted departure from the unique admissible interior: if the Part I wall is real and no licensed same-scope bridge is available, then any claim of standing-positive singularity would have to present itself by one or more of these escape patterns.

Remark 15.5 (Diagnostic status of (S1)–(S5)). The force of the endgame does not originate in the labels (S1)–(S5). Their role is downstream and explicit: once Theorem 14.1 has ruled out a second interior, admissible variation, and same-scope scope change, the five labels record the finite menu of ways an impossible escape attempt would have to look on the fixed carrier.

Lemma 15.6 (The Part I wall forces a diagnostic escape taxonomy). *Assume the standard carrier commitments of Definition 13.2 are held fixed as the active same-scope commitments, and assume none of the conventional routes in Part I yields a licensed critical bridge. Then any attempt to treat a classical blow-up episode as standing-positive within this fixed scope must instantiate at least one of (S1)–(S5).*

Proof. Part I established that the standard analytic routes do not furnish an unconditional critical bridge from the admitted energy-class bundle to suppression of the vortex-stretching channel. By Theorem 14.1, that analytic failure does *not* reopen primitive admissibility: it does not create a second same-scope interior, an admissible variation inside the unique interior, or a same-scope scope change. This proof therefore does *not* assert that a PDE blow-up, if it exists, “really uses” weakening or repair as a physical mechanism. It asserts only this: once the scope is held fixed, uniqueness has already blocked any second venue for same-scope singularity, and no licensed critical bridge is available, the map admits no way to classify a blow-up proposal as standing-positive except by one of the typed diagnostic alternatives (S1)–(S5).

Now suppose a classical blow-up episode is asserted and is to be *classified as standing-positive* while the commitments of Definition 13.2 remain the active same-scope carrier commitments. Then every standing-positive same-scope continuation-relevant structure admitted on this route is already interior to the standard carrier or a lawful same-scope functional thereof by Theorem 13.24. The finite decision packet 13.5E–13.5L therefore leaves only the typed possibilities already isolated on the fixed carrier: existing carrier content, bookkeeping-only variation, parameterized gate, competing discriminator, or scope change. Existing carrier content and bookkeeping-only variation add no continuation authority by Propositions 13.5G and 13.5H; parameterized and competing same-scope gates are excluded by Propositions 13.5I and 13.5J; and scope changes are not same-scope rivals by Proposition 13.5K. The carrier-certified Structural Exhaustion packet of Theorem 13.28, Corollary 13.29, and Proposition 13K then removes any residual attempt to reinstall a further

same-scope continuation selector or PDE-side criterion on that standing-positive route. Thus, to keep the asserted event inside the same lineage while changing continuation status, one is forced into a deformation of the fixed carrier-side commitments rather than a new licit same-scope criterion. Concretely, one must therefore do at least one of the following: weaken a bundle clause, isolate one interior compatibility from the rest, insert a tolerance or selector, privilege a non-equivalent redescription, or justify an earlier failure only by downstream composition. These are exactly (S1)–(S5), typed via envelope licensing, lawful redescription, and composition discipline (Axioms 3.48, 3.49, 3.50, and 3.45). In this way the forcing step rests on the carrier-certified exhaustion packet rather than on Rule label 3.52 as an external primitive. $\square$

Proposition 15.7 (Quotient-visible vs illicit singularity). *Assume the Navier–Stokes carrier is evaluated inside a fixed admitted scope. Any purported singularity gate falls into exactly one of two classes:*

- (i) **Quotient-visible:** *it factors through the existing standing quotient and therefore marks an incompatibility already present in the unique admissible interior content;*
- (ii) **Illicit:** *it does not factor through the standing quotient and therefore constitutes an inadmissible same-scope discriminator.*

Proof. This is just Corollary 14.4 restated in the specific language of singularity. The point of restating it is to make the final theorem read as an exhaustive dichotomy rather than a single jump. $\square$

Corollary 15.8 (No third branch). *Within a fixed admitted Navier–Stokes scope there is no third same-scope branch for a singularity claim besides the quotient-visible and illicit branches of Proposition 15.7.*

Proof. A third branch would either introduce another same-scope discriminator not already visible in the quotient (excluded by Proposition 14.3 together with Structural Exhaustion), or refine one of the two listed branches by a selector/tolerance (i.e. by a same-scope parameter), which is excluded by zero-parameter certification (Theorem 13.20) and saturation (Theorem 12.1). $\square$

16 Uniqueness-first structural resolution theorem

The proof order in the endgame is explicit:

$$\begin{aligned}
& \text{unique admissible interior} \\
\implies & \text{no second same-scope gate or admissible variation} \\
\implies & \text{no admissible singular deviation} \\
\implies & \text{no singular image.}
\end{aligned}$$

Proposition 16.1 fixes entry into the unique admissible interior realized by the standard carrier, and Theorem 14.1 states the primary contradiction first. Only after that uniqueness-first obstruction is in place do the later diagnostic branches appear: any same-scope singularity proposal must present itself as one of the two residual branches of Table 5. Lemma 16.2 eliminates the illicit branch by exhaustion and redescription legality; Lemma 16.3 eliminates the quotient-visible branch by saturation and fail-closed enforcement. Corollary 15.8 excludes any third same-scope branch. Theorem 16.5 records the resulting structural closure.

Proposition 16.1 (Scope inclusion gate). *For the standard $\mathbb{T}^3$ Navier–Stokes carrier, Theorem 13.20 and Corollary 13.21 imply that Theorem 12.1 applies to B_{NS} and that the quotient/factorization discipline of Proposition 14.3 applies to all same-scope standing-relevant consequence functionals on that carrier.*

Proof. This is the content of Corollary 13.21 in the Navier–Stokes setting: the standard carrier has been certified as (i) zero-parameter and (ii) equivalence-complete at fixed scope. Therefore the saturation theorem applies to B_{NS} and the factorization discipline applies to every standing-relevant consequence functional at that same scope, with no additional analytic hypothesis. $\square$

Lemma 16.2 (Illicit branch elimination). *If a purported singularity gate is illicit in the sense of Proposition 15.7, then it cannot enter the Navier–Stokes interior as a standing-bearing discriminator.*

Proof. An illicit singularity gate is, by definition, a same-scope discriminator that fails factorization through the standing quotient. By Proposition 14.3, such a proposal is not a same-scope refinement but a scope change. Structural Exhaustion forbids the introduction of a new same-scope gate, and redescription neutrality (Axiom 3.49) forbids smuggling that gate in by non-equivalent redescription. Therefore the illicit branch is not merely unproven; it is inadmissible as a same-scope move. $\square$

Lemma 16.3 (Quotient-visible branch elimination). *If a purported singularity gate is quotient-visible in the sense of Proposition 15.7, then any state it marks as singular is standing-negative.*

Proof. Work under Proposition 16.1. Let G be quotient-visible and let x be any state marked “singular” by G . Quotient-visibility means G factors through the standing quotient and therefore contributes no same-scope discriminator; the mark “singular” is a visible incompatibility already present in the unique admissible interior content. By Lemma 15.6, producing such a mark without a licensed critical bridge forces at least one of the forbidden same-scope moves (S1)–(S5). Inside the certified zero-parameter saturated interior, each of (S1)–(S5) is excluded by Theorem 12.1 together with Propositions 12.3 and 15.3. Hence x cannot be standing-positive: it is standing-negative and is rejected fail-closed within its lineage. $\square$

Proposition 16.4 (Exhaustive branch elimination). *Every same-scope singularity proposal is eliminated: the illicit branch by Lemma 16.2, the quotient-visible branch by Lemma 16.3, and no third branch exists by Corollary 15.8.*

Proof. By Lemmas 16.2 and 16.3, the two branches of Proposition 15.7 are both eliminated. Corollary 15.8 excludes any residual third same-scope branch. Hence no same-scope singularity channel remains once the scope inclusion gate is fixed. $\square$

Theorem 16.5 (Structural regularity on the certified periodic carrier). *For the standard $\mathbb{T}^3$ Navier–Stokes carrier, singular candidates in the sense of Definition 15.4 are standing-negative and cannot occur within the unique admissible interior.*

Proof. Within the unique admissible interior of the Navier–Stokes carrier, any candidate singular evolution must correspond to a same-scope parameter capable of selecting a singular branch. Theorem 13.1 classifies any such discriminator into the gate classes, and Theorem 13.19 eliminates those classes as independent admissibility parameters.

Consequently no same-scope parameter remains capable of selecting singular evolution. The unique admissible interior therefore contains only regular evolutions of the Navier–Stokes flow. Hence the flow on $\mathbb{T}^3$ remains globally regular within the unique admissible interior. $\square$

Lemma 16.6 (Determinacy bridge). *Within the certified unique admissible interior of the Navier–Stokes carrier, any admissible consequence functional that factors through the standing quotient determines the admissible evolution class uniquely.*

Proof. By the Structural Regularity Theorem, the unique admissible interior contains no singular branch of evolution. Consequently the admissible state space consists solely of regular evolutions.

Standing-relevant consequence functionals factor through the standing quotient and therefore depend only on the admissible equivalence class of states.

Because every admissible representative corresponds to the same regular evolution class, the value of any such functional is fixed. Hence the admissible evolution class is uniquely determined. $\square$

Corollary 16.7 (Determinacy is forced, not imported). *Under the hypotheses of Theorem 16.5, determinacy of admissible outcomes is not an independent hypothesis but a consequence of (i) consequence uniqueness at fixed scope, (ii) exhaustive branch elimination for singularity proposals, and (iii) constraint saturation of the interior.*

Proof. Work under Proposition 16.1.

(i) **Fixed-scope consequence uniqueness.** Proposition 14.3 fixes consequence uniqueness at the level of standing: every standing-relevant consequence functional factors through the standing quotient, so no non-equivalent same-scope replacement can alter its value on admitted states.

(ii) **Elimination of all same-scope singularity splits.** Theorem 16.5 excludes standing-positive singular candidates. Equivalently, there is no admissible same-scope singularity gate left through which two distinct continuations could be licensed.

(iii) **Rigidity by saturation (no tunable degrees of freedom).** Theorem 12.1 upgrades (i) from quotient-formal uniqueness to rigidity: admissible continuations are rigid, overdetermined, and fail-closed within their lineage.

Combining (i)–(iii), any purported non-determinacy would require a same-scope split gate. But by (ii) no such gate remains, and by (iii) no tuning, parameter transport, or local repair can simulate one inside the unique admissible interior. Determinacy is therefore a forced elimination consequence of the fixed-scope quotient spine together with saturation; it is not imported as an analytic continuation principle. $\square$

Corollary 16.8 (No admissible same-scope singularity gate remains). *Under Proposition 16.1, every same-scope singularity proposal is either excluded at entry or standing-negative.*

Proof. By Proposition 16.4, every same-scope singularity proposal is either illicit or quotient-visible, and Corollary 15.8 excludes any third branch. Lemma 16.2 excludes the illicit branch at entry; Lemma 16.3 makes the quotient-visible branch standing-negative. This is exactly the claimed dichotomy. $\square$

Remark 16.9 (Typed closure formula). The structural closure is not “saturation, therefore smooth.” It is “entry into the unique admissible interior plus consequence uniqueness plus branch elimination leaves no admissible same-scope singularity gate.”

17 Resolution, scope, and terminality

17.1 Internal resolution

Proposition 17.1 (Internal resolution is structural, not analytic). *Within the carrier already identified with the unique admissible interior, Part II does not supply a new estimate on $\mathcal{T}(u)$. It shows instead that, once the standard $\mathbb{T}^3$ carrier is certified by Theorem 13.20, the uniqueness-first contradiction of Theorem 14.1 removes any second same-scope singular venue and the later diagnostic branch package eliminates every residual admissible same-scope singularity gate by Theorem 16.5 and Corollary 16.8.*

Proof. The unresolved analytic bridge isolated in Part I is retyped by Definition 15.4 as one of (S1)–(S5). Theorem 13.20 certifies the standard carrier as zero-parameter and equivalence-complete. Theorem 16.5 excludes standing-positive singular candidates, and Corollary 16.8 records that no admissible same-scope singularity gate remains. The closure therefore acts by elimination of illicit degrees of freedom rather than by a new PDE estimate. $\square$

17.2 Explicit claim type and non-claim

Proposition 17.2 (Explicit claim type and non-claim for Part II). *Part II, by itself, does not claim a PDE-native continuation-estimate proof as its method. In particular, it does not furnish:*

- (a) *a conventional a priori estimate bounding $\mathcal{T}(u)$ by dissipation;*
- (b) *a standard continuation theorem in the PDE literature;*
- (c) *an operator-licence ledger proving interior inclusion directly from ordinary analysis.*

The result claimed here is primitive-side rather than analytic-side. Within the proof architecture of this manuscript, Part II certifies the fixed periodic carrier as the unique admissible interior at fixed same scope and eliminates every same-scope singularity proposal on that interior. The output of Part II is therefore structural first and classical only after Part III’s realization bridge.

Proof. The distinction is enforced by the admissibility rules governing the argument.

Step 1 (Scope discipline). The reasoning proceeds entirely inside the admissibility map of the self-contained rule package of this paper. The certification of the Navier–Stokes carrier as a zero-parameter interior (Theorem 13.20 and Proposition 12.3) establishes a structural setting rather than a new analytic estimate.

Step 2 (Exhaustion constraint). Structural Exhaustion forbids introducing a same-scope operator, bundle, or admissibility level to create an independent analytic route that closes the tensor wall. Consequently Part II does not and cannot present a new PDE mechanism at the same scope; it instead proves that all admissible same-scope continuations are already exhausted.

Step 3 (Redescription neutrality). Any attempt to reinterpret the internal elimination as a new analytic discriminator would require a redescription that changes either invariants or equivalence of the certified carrier. Such a reformulation violates redescription legality and is standing-negative.

Step 4 (No-repair enforcement). Because enforcement is fail-closed, a rejection under the previous steps cannot be repaired within the same lineage by auxiliary corrections, partial reinterpretations, or local repair.

Step 5 (Conclusion). The result is therefore correctly typed as a primitive-side structural resolution inside the admissibility map: Part II carries the carrier-side certification burden and excludes same-scope singular existence there, rather than supplying a conventional PDE-native continuation-estimate proof. $\square$

17.3 Terminality

Lemma 17.3 (Move classification). *Any same-scope continuation of the Navier–Stokes regularity question must fall into exactly one of the following classes:*

- (i) *introduction of new admissibility structure (a new level, question, operator, or atomic bundle);*
- (ii) *redescription of the unique admissible interior (retyping that alters equivalence or invariants);*
- (iii) *repair of a rejected lineage (post-hoc repair within a failed sequence);*
- (iv) *continuation along a singular branch (a standing-positive singular candidate or its gate).*

Proof. At fixed scope, any continuation must either change the admissibility apparatus, reinterpret the existing certified apparatus, attempt to rehabilitate a rejected move, or proceed through a singularity gate. These exhaust the same-scope options by definition of the admissibility map. $\square$

Theorem 17.4 (Terminality). *Within the unique admissible interior determined by Proposition 16.1 and Theorem 16.5, no further same-scope move remains available that could reopen the regularity question.*

Proof. By Lemma 17.3 we perform a complete classification–elimination over the admissible move types.

(i) Exhaustion prohibition. Any attempt to introduce new admissibility structure violates the carrier-certified Structural Exhaustion packet of Theorem 13.28 and Corollary 13.29, together with Proposition 13K, and is rejected at entry by fail-closed enforcement (Axioms 3.44, 3.51).

(ii) Redescription prohibition. Any redescription that changes equivalence or invariants of the unique admissible interior violates redescription neutrality (Axiom 3.49) and collapses standing.

(iii) No-repair prohibition. Because enforcement is fail-closed, any move rejected under (i) or (ii) cannot be repaired within the same lineage: post-hoc rehabilitation is forbidden by Axiom 3.45 and cannot be salvaged by partial composition (Axiom 3.50).

(iv) Branch prohibition. Theorem 16.5 has already eliminated standing-positive singular candidates. By Corollary 15.8 no additional same-scope branch exists.

Therefore no same-scope continuation remains that can reopen the question. The argument terminates structurally. $\square$

Corollary 17.5 (No same-scope analytic escape for the official periodic theorem). *Any future analytic technique that purports to reopen the periodic regularity question while preserving the same torus carrier, the same smooth category, the same continuation notion, and the same lawful skins is already covered by the certified same-scope continuation space of the standard carrier. Consequently, no genuinely same-scope analytic route can overturn the certified periodic result. Any analytic route that changes the outcome must change scope, and such scope changes do not alter the certified official theorem on the original carrier.*

Proof. Under the hypotheses stated, the proposed analytic technique is a same-scope continuation mechanism on the fixed torus carrier in the sense of Definition 13.23B. Because the carrier commitments are held fixed, the finite typed packet 13.5E–13.5L applies to any standing-positive same-scope continuation-relevant candidate determination it introduces. Hence any such technique either reduces to existing carrier content, reduces to bookkeeping-only variation, is forbidden as a parameterized or competing same-scope gate, or changes scope by Theorem 13.5F through Corollary 13.5L. Therefore the technique is not an external enlargement of the continuation picture but a move already covered by the certified same-scope continuation space of the standard carrier.

By Corollary 13.29, Proposition 13K, and Theorem 17.4, no same-scope continuation move remains available that could reopen the regularity question on the carrier already identified with the unique admissible interior. Therefore any purported analytic reopening that still changes the outcome must alter the carrier, the continuation notion, or the lawful skin structure; that is, it is a scope change rather than a same-scope analytic escape. Such scope changes do not alter the certified official theorem on the original carrier. $\square$

17.4 Endgame spine (for audit)

Proposition 17.6 (Endgame spine). *The internal closure in Part II is the following forced chain at fixed scope:*

- (unique admissible interior at fixed periodic scope)*
- $\implies$ *(no second same-scope gate and no admissible variation)*
- $\implies$ *(uniqueness-first contradiction for singularity proposals)*
- $\implies$ *(diagnostic endpoint-status-trichotomy / forced-escape taxonomy)*
- $\implies$ *(exhaustive branch elimination)*
- $\implies$ *(structural regularity & determinacy).*

No link introduces a new admissibility level, a same-scope discriminator, or a repair-by-redescription.

Proof. Theorem 13I gives the unique admissible interior at fixed periodic scope. Corollary 13J, Theorem 13.10, and Corollary 13.12 provide the no-second-gate / no-admissible-variation stage. Theorem 14.1 is then the primary singularity contradiction before any branch labels appear. Proposition 15.7 gives the residual diagnostic dichotomy, and Lemmas 16.2 and 16.3 eliminate the two branches, with Corollary 15.8 excluding any third same-scope branch; this yields Theorem 16.5. Corollary 16.7 is then the fixed-scope consequence of the factorization discipline plus saturation. $\square$

17.5 Closure ledger

The structural work of Part II establishes the following dependency chain:

- (1) the conventional routes all terminate at the same tensor wall;
- (2) each proposed conventional same-scope supplement corresponds to a prohibited move in the map;
- (3) continuation is not the primitive operator route;
- (4) the Yang–Mills tensor-discrimination move does not transfer conventionally;
- (5) the map is exhausted, so no same-scope route may be added;
- (6) the standard periodic carrier is the unique admissible interior at fixed scope;
- (7) therefore no second same-scope singular venue, continuation gate, or admissible variation exists;
- (8) any singularity proposal is therefore only an attempted escape from the unique interior and so must appear either as an illicit move or as an already quotient-visible incompatibility;
- (9) the illicit branch collapses by exhaustion and redescription legality;

- (10) the quotient-visible branch collapses by saturation together with the no-generators, no-carriers, and no-repairs impossibility classes;
- (11) hence no standing-positive singular candidate survives in the admitted interior.

Part III. Faithful Classical Realization

Burden status at entry. By the end of Part II, the fixed periodic problem already has a unique admissible interior, the standard periodic carrier has already been identified with that interior, and the same-scope admissibility burden has already been closed before any classical transfer begins. Part III therefore carries only the realization-and-transport burden: admit the full official datum class, realize the full official local classical class, keep the same same-scope continuation notion fixed earlier rather than silently redefining it, show that no same-scope continuation criterion appears at the classical layer, and then transport the already certified carrier-side contradiction into the official PDE endpoint.

Definition (18.0 (Same-object convention for realized periodic lineages)). *Fix an admitted carrier state $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ and its maximal classical realization*

$$\mathcal{R}_{\text{NS}}(x) = (u_x, p_x) \quad \text{on} \quad [0, T_*(x)).$$

In Part III the realized periodic lineage of x means the single same-scope determination consisting of this maximal realized object together with its fixed torus carrier, viscosity, zero-forcing Navier–Stokes equations, datum anchor, mean-zero pressure gauge, lawful-redescription class, and same-scope continuation question. The time slices $t < T_(x)$ are licensed presentations of this one realized lineage; they are not independently reclassified admissibility states.*

Definition (18.0A (Realized continuation locus)). *For $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$, the realized continuation locus of x is the fixed same-scope object consisting of the admitted carrier seed x , its maximal realized trajectory $\mathcal{R}_{\text{NS}}(x)$, all smooth pre-horizon slice presentations, the restart / overlap identity relation supplied by local uniqueness, and the fixed horizon continuation question. The locus is a carrier-bound object; it is not a sequence of new admissibility decisions indexed by time.*

Definition (18.0B (Boundary trace of a realized lineage)). *A boundary trace of a realized lineage is a lawful record of pre-horizon behavior approaching a proposed horizon: for example a norm profile, BKM profile, stretching profile, or sequence of smooth slice data. A boundary trace is not a new standing-bearing endpoint unless its standing-bearing content factors through the already fixed horizon discriminator.*

Definition (18.0C (Persistence regime)). *A realized lineage is a persistence regime if its standing-bearing commitments persist under lawful redescription, smooth restart, overlap identification, and pre-horizon slice re-entry. Persistence here means fixed-domain continuation of the same standing-bearing object, not the generation of a new analytic continuation estimate.*

Remark 17.7 (Claim-type discipline for the realized-lineage packet). The live burden of Part III is not to produce the missing analytic bridge inequality or a new coercive continuation estimate. Its burden is the one required by the non-PDE-native impossibility proof class: once the official periodic classical scope and the realized lineage of Definition 18.0 are fixed, Part III must show that finite-horizon failure induces a same-scope endpoint condition, that this condition has only the endpoint statuses later named in the realized-lineage trichotomy / collapse, and that no additional classical-layer selector or continuation gate is introduced on that route.

Part III transfer cascade and claim types

Part III carries the official endpoint by one explicit primitive-to-classical transfer cascade:

- (i) **Transfer soundness:** every official periodic datum enters the admitted carrier class in the unique admissible interior.
- (ii) **Transfer completeness:** the realized local class is exactly the full official local classical class.
- (iii) **Transfer faithfulness / no-new-criterion:** classical realization preserves carrier identity on the single realized lineage of Definition 18.0 and adds no new standing-positive same-scope continuation content beyond the certified carrier class on the live route.
- (iv) **Transfer detectability:** any hypothetical finite-time classical breakdown is already carrier-visible on the fixed torus carrier.
- (v) **Transfer impossibility:** all singular images of the unique admissible interior are excluded.
- (vi) **Transfer endpoint:** the official periodic theorem follows first as a short corollary of structural necessity and then as the classical PDE restatement.

From this point onward the proof burden is local and theorem-explicit. Part III does not reopen Part I or Part II; it isolates the transfer chain that carries the already-certified standard periodic carrier into the official PDE formulation. The only analytic machinery used in Part III is now the narrow local realization packet internalized in Appendix B: pressure reconstruction in the periodic mean-zero gauge, the semigroup/Leray contraction window, the classical upgrade for smooth data, and overlap uniqueness for smooth periodic solutions. Lemma 18.14 and Proposition 18B record the sufficiency of this bridge on theorem-local terms, while Remark 17.7 keeps the claim class narrow: the later endpoint theorem is a short same-scope status contradiction, not an imported analytic coercivity result.

Proposition (18A (Part III is realization transport, not a second proof modality)). *Assume the fixed periodic scope has already reached Corollary 12H and Theorem 13I. Then Part III introduces no same-scope admissibility criterion, no independent PDE continuation principle, and no additional carrier-side discriminator. Its sole role is to realize the already-identified unique admissible interior in the official local classical periodic language and to transport the resulting carrier-side exclusions into that classical image, using only the local analytic packet of Appendix B (Proposition B.1 through Corollary B.8).*

Proof. The scope-level exclusion work is already complete before Part III begins: Corollary 12H forces the unique admissible interior, Theorem 13I identifies the standard periodic carrier with that interior, and Proposition 13K states explicitly that no further same-scope PDE-side continuation criterion survives inside fixed scope. Lemma 18.14 then records the exact role of Part III: Carrier Universality, local classical realization, horizon detectability, and faithful import are sufficient to read that prior structural theorem into the official PDE formulation. Every analytic ingredient used below is theorem-local in Appendix B, and specifically in Proposition B.1 through Corollary B.8, so the later sections realize and transport a theorem already forced on the carrier rather than inventing a second proof modality. $\square$

Proposition (18B (No hidden continuation hypothesis enters the transfer cascade)). *On the live Part III dependency chain, the only datum-side assumption is $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$ and the only analytic inputs are Proposition B.1 through Corollary B.8 of Appendix B. No smallness, symmetry, profile, critical-space, Beale–Kato–Majda, Serrin, Prodi–Serrin, Θ_x , or auxiliary continuation hypothesis is assumed in order to pass from the certified carrier to the official local classical class. Any standing-positive same-scope continuation claim at the classical layer must either factor through the carrier-side structure already fixed in Part II or be excluded by Theorem 20.5B and Theorem 20B.*

Proof. Datum-side entry is exactly Corollary 18.12, with Proposition 18.7 showing that no extra datum hypothesis is introduced. The only analytic ingredients used to construct and identify the local classical realization are the Appendix B results Proposition B.1 through Corollary B.8. Theorem 20.5B states that realization adds no new standing-positive same-scope continuation content on the live route, Corollary 20A and Theorem 20.7A make any hypothetical breakdown carrier-visible, and Theorem 20B excludes an independent classical-layer discriminator in that same standing-positive role. Hence no hidden continuation hypothesis enters the transfer cascade. $\square$

Proposition (18C (Part III uses the same same-scope continuation notion fixed in Part II)). *The same-scope continuation notion used throughout Part III is exactly the continuation notion fixed in Lemma 12A and closed before transfer by Corollary 13L. Part III realizes that already-fixed notion in official local classical language; it does not redefine, weaken, or strengthen it.*

Proof. Lemma 12A fixes the same-scope continuation notion as part of the periodic scope itself: smooth periodic Navier–Stokes continuation on the fixed torus carrier, with the fixed viscosity, zero forcing, smooth category, mean-zero pressure gauge, and lawful-redescription class. Corollary 13L then states theorem-locally that this same-scope admissibility burden is already closed before Part III begins. The realization map, horizon discriminator, and detectability statements of Part III merely express that same continuation question on the realized local classical image supplied by Proposition B.1 through Corollary B.8. If Part III altered the continuation notion, it would either change scope or introduce a same-scope admissibility selector, contradicting Proposition 18A and Proposition 18B. Hence Part III uses the same same-scope continuation notion fixed in Part II and only realizes it in classical PDE language. $\square$

18 Transfer Step I (soundness): every official periodic datum enters the unique admissible interior

Burden status. The unique admissible interior and the standard periodic carrier are already fixed before this section begins. What remains here is datum-side soundness only: prove that every official Fefferman (B) datum enters that already-certified carrier class, with no hidden prefilter and no datum-dependent rival regime. The first point at which a specialist PDE referee will press hardest is whether the standard carrier, even after Theorem 13I, silently restricts the official initial-data class or leaves room for a second same-scope admissible continuation regime chosen by the datum. The present section removes both possibilities exactly. It proves that the standard carrier of Definition 13.2 — already identified in Part II with the unique admissible interior at fixed periodic scope — coincides, at time zero, with the ordinary periodic smooth divergence-free data class, so every official datum enters that unique admissible interior. Since no second same-scope interior exists on the fixed torus carrier, no official datum can select, privilege, or force an alternative admissible continuation regime. Thus Carrier Universality now does three jobs at once: it removes the hidden-subclass objection, shows that every official periodic datum lands in

the *unique admissible interior* for the fixed problem, and blocks datum-dependent appeal to any rival same-scope continuation regime. For the reader in conventional PDE notation, *same scope* in Part III means: the fixed torus carrier $\mathbb{T}^3$, the same viscosity $\nu > 0$, the same incompressible Navier–Stokes system, and the fixed mean-zero pressure normalization. *Lawful redescription* means only deterministic bookkeeping changes within that fixed presentation (velocity, vorticity, strain, pressure gauge, and standard frequency decompositions).

Definition 18.1 (Official periodic data class). Fix $\nu > 0$ and the torus carrier $\mathbb{T}^3$. The *official periodic data class* is

$$\mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu) := \{u_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3) : \nabla \cdot u_0 = 0\}.$$

No further smallness, symmetry, decay, spectral, or profile hypothesis is included in $\mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$.

Remark 18.2 (Exact match with Fefferman (B) hypotheses). Definition 18.1 is exactly the initial-data class appearing in the periodic case (B): smooth periodic divergence-free velocity data on $\mathbb{T}^3$ with no smallness, symmetry, decay, profile, or mean-zero velocity restriction. The mean-zero normalization used later fixes only the pressure gauge.

Definition 18.3 (Carrier seed of an official datum). For $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$, define its *carrier seed*

$$\Sigma(u_0) := (u_0, \omega_0, p_0, S_0, \mathcal{F}(u_0)),$$

where

$$\omega_0 := \nabla \times u_0, \quad S_0 := \frac{1}{2}(\nabla u_0 + \nabla u_0^\top),$$

p_0 is the unique mean-zero solution of

$$-\Delta p_0 = \partial_i \partial_j (u_{0,i} u_{0,j}) \quad \text{on } \mathbb{T}^3,$$

and $\mathcal{F}(u_0)$ denotes the standard frequency / Littlewood–Paley skins used in Part I.

Definition 18.4 (Initial-data face of the standard carrier). The *initial-data face* of the standard $\mathbb{T}^3$ Navier–Stokes carrier is the set $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ of time-zero lawful-redescription classes of the object-class described in Definition 13.2. Concretely, an element of $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ is the time-zero carrier object modulo the licensed bookkeeping equivalences already admitted in Lemma 13.4.

Remark 18.5 (Conventional reading of the initial-data face). On the fixed torus carrier, $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ is nothing more than the quotient of the standard time-zero presentations generated from smooth divergence-free periodic velocity data by the deterministic bookkeeping operations already built into the standard carrier. It does not identify distinct official velocity data, and it introduces no additional side condition at time zero.

Lemma 18.6 (Canonical skin completion of official data). *For every $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$, the carrier seed $\Sigma(u_0)$ is well-defined and determines an element of the initial-data face $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$.*

Proof. Because $u_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$ and $\nabla \cdot u_0 = 0$, the derived fields $\omega_0 = \nabla \times u_0$ and $S_0 = \frac{1}{2}(\nabla u_0 + \nabla u_0^\top)$ are smooth and periodic. The right-hand side of the torus Poisson equation

$$-\Delta p_0 = \partial_i \partial_j (u_{0,i} u_{0,j})$$

has mean zero, so Corollary B.2 supplies the unique mean-zero pressure seed p_0 . The standard frequency skins $\mathcal{F}(u_0)$ are ordinary bookkeeping constructions on smooth periodic data. By Definition 13.2, these objects are precisely the licensed time-zero presentations of the same carrier object. Thus $\Sigma(u_0)$ determines an element of $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$. $\square$

Proposition 18.7 (No hidden prefilter on official data). *The assignment*

$$u_0 \longmapsto [\Sigma(u_0)]$$

introduces no hypothesis beyond membership in $\mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$. In particular, it does not impose smallness, symmetry, profile restrictions, spectral localization, critical-space bounds, a mean-zero velocity condition, or any continuation criterion.

Proof. Definition 18.3 uses only four total deterministic operations on $C_\sigma^\infty(\mathbb{T}^3)$: differentiation, curl, the mean-zero periodic Poisson solver for pressure, and the fixed frequency bookkeeping of Part I. The pressure gauge does not impose any extra condition on the velocity datum itself. None imposes a size bound, symmetry assumption, profile condition, or continuation hypothesis. Hence if some datum in $\mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$ were excluded by $u_0 \mapsto [\Sigma(u_0)]$, the exclusion would have to come from an additional side condition not present in Definition 18.3. But Part II already certified the standard carrier as zero-parameter and equivalence-complete (Theorem 13.20), so no such extra criterion exists on the fixed torus carrier. Therefore the construction does not narrow the official class. $\square$

Lemma 18.8 (Velocity reconstruction on the initial-data face). *Every class in the initial-data face $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ contains a unique velocity representative $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$.*

Proof. By Definition 13.2, the standard time-zero presentations all arise from one smooth periodic divergence-free velocity field together with its deterministically derived pressure, vorticity, strain/kernel, and standard frequency skins. Thus every class in $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ contains at least one velocity representative in $\mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$. If a single class contained two different velocity fields $u_0 \neq v_0$, then the quotient defining the initial-data face would identify two distinct official periodic data sets. That is excluded by equivalence completion for the standard carrier and by the zero-parameter certification already proved in Theorem 13.20. Hence the velocity representative is unique. $\square$

Lemma 18.9 (Every admitted time-zero carrier object has an official datum). *Every element of the initial-data face $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ is lawfully equivalent to a unique carrier seed $\Sigma(u_0)$ with $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$.*

Proof. Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$. By Lemma 18.8, the class of x contains a unique velocity representative $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$. Once u_0 is fixed, the remaining pieces of the carrier seed are determined canonically by Definition 18.3: $\omega_0 = \nabla \times u_0$, the strain field S_0 , the unique mean-zero pressure seed p_0 , and the fixed frequency bookkeeping $\mathcal{F}(u_0)$. Therefore the class of x is represented by $\Sigma(u_0)$, and the datum u_0 is unique. $\square$

Theorem 18.10 (Transfer soundness: Carrier Universality for official periodic data). *The map*

$$\mathcal{U}_{\text{car}} : \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu) \rightarrow \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu), \quad u_0 \longmapsto [\Sigma(u_0)]$$

is bijective.

Equivalently, the standard $\mathbb{T}^3$ Navier–Stokes carrier neither narrows nor enlarges the official periodic initial-data class: every smooth divergence-free periodic datum determines exactly one admitted time-zero carrier object up to licensed redescription, every admitted time-zero carrier object arises from exactly one such datum, and therefore no official datum can select an alternative same-scope admissible continuation regime on the fixed torus carrier.

Proof. By Lemma 18.6, every official datum determines an element of the initial-data face. Proposition 18.7 shows that this assignment imposes no extra hypothesis beyond smoothness, periodicity, and divergence-freeness. Lemma 18.9 gives surjectivity and injectivity: every admitted time-zero carrier object is represented by some $\Sigma(u_0)$, and that velocity datum is unique. Therefore the map $\mathcal{U}_{\text{car}}$ is bijective. $\square$

Corollary 18.11 (Transfer soundness: no smaller admissible data subclass). *There is no proper “admitted” subclass of the official periodic data hidden inside the standard $\mathbb{T}^3$ carrier. The unique admissible interior realized by that carrier applies to all $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$.*

Proof. Immediate from Theorem 18.10 and Proposition 18.7. $\square$

Corollary 18.12 (Transfer soundness: every official periodic datum enters the unique admissible interior at time zero). *For every $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$, the unique class $[\Sigma(u_0)]$ lies in the initial-data face $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ and therefore in the unique admissible interior identified in Theorem 13I. In particular, it is standing-positive on the fixed torus carrier.*

Proof. By Theorem 18.10, every official datum determines a unique element of the initial-data face. By Theorem 13I, the standard periodic carrier is the unique admissible interior at the fixed periodic scope. Hence every official datum lands in that unique admissible interior at time zero on the fixed torus carrier. $\square$

Corollary 18.13 (Transfer soundness: no datum-selected rival regime). *For every official periodic datum $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$, the associated carrier seed lies in the unique admissible interior realized by the standard carrier. Consequently, no official datum can select, privilege, or force an alternative same-scope admissible continuation regime on the fixed torus carrier, and the structural regularity theorem (Theorem 16.5) applies to all official periodic data on that carrier.*

Proof. By Theorem 18.10, every official datum determines an admitted time-zero carrier object. By Theorem 13I, that object lies in the unique admissible interior realized by the standard carrier. Corollary 13J then rules out any second standing-positive same-scope continuation regime on the fixed periodic carrier. Therefore the structural regularity package applies to every official datum, not merely to a proper subclass, and no datum can appeal to a rival same-scope regime. $\square$

Lemma 18.14 (Sufficiency of the Part III realization bridge). *Assume the full Part II rule package has been verified for the standard periodic carrier, so that the structural non-singularity theorem on that carrier is an in-paper consequence rather than an external primitive. Then the structural non-singularity theorem implies the official periodic global smoothness theorem provided that the following three bridge ingredients hold on the fixed torus carrier:*

- (a) *every official periodic datum determines a standing-positive time-zero carrier state in the unique admissible interior;*
- (b) *the realization map identifies the certified carrier with the full official local classical class;*
- (c) *finite-time breakdown on the realized trajectory is independently proved to be a standing-relevant carrier-side horizon discriminator on the fixed carrier, and every carrier-side branch elimination imports faithfully to the realized class as exclusion of singular images.*

This bridge is a realization-transport statement, not a second proof modality: it carries a prior primitive theorem into the official PDE formulation using only the local analytic packet proved in Appendix B for realization, overlap uniqueness, and pressure reconstruction. Conditions (a)–(c) are proved in Sections 18–21.

Proof. The in-paper status of the structural non-singularity theorem is supplied by Theorem 13.28, which verifies the full Part II rule package on the standard periodic carrier. Condition (a) is exactly Corollary 18.12. Condition (b) is the combined content of Theorems 19.6 and 19.9. Condition (c)

is the independently proved narrow horizon-detectability route: Theorem 20.7A identifies finite-time breakdown with a standing-relevant carrier-side horizon discriminator on the fixed torus carrier, Theorem 20.11 eliminates any additional same-scope Θ_x -based continuation gate, and the constructive branch-forcing / no-singular-image results proved in Sections 19–21 transport the resulting carrier-side breakdown content into branch exclusion. Once (a)–(c) hold, every official datum enters the unique admissible interior at time zero, every realized local classical trajectory is exactly the official local classical trajectory on the fixed torus carrier, and finite-time breakdown is already a standing-relevant carrier-side event whose branches are excluded by Part II. Therefore the structural non-singularity theorem on the carrier already identified with the unique admissible interior is sufficient for the official periodic global smoothness theorem on a fully self-contained in-paper spine. $\square$

Remark 18.15 (The Part III bridge is transport, not a second proof modality). Lemma 18.14 does not prove the official periodic theorem by a new analytic regularity argument. Its role is only to say that once the unique admissible interior and the carrier-side no-singular-image theorem are already in place, Carrier Universality, local classical realization, horizon detectability, and faithful import read that prior primitive result back into the official PDE presentation. The analytic bridge used there is exactly the self-contained local realization packet recorded later in Proposition H.1.

Lemma 18.16 (Architectural bridge from strong carrier completeness to realized PDE completeness). *Assume the standard carrier is strongly continuation-complete for the official periodic class in the standing-positive carrier-admitted sense of Theorem 13.26. Then on the realized local class no standing-positive continuation-relevant PDE phenomenon remains outside carrier-side representation. Equivalently: once Theorem 19.9 identifies the realized local class with the full official local classical class, every standing-positive same-scope continuation-relevant PDE phenomenon visible in the realized formulation is already represented by a standing-relevant discriminator on the carrier side.*

Proof. Let Ξ be a standing-positive same-scope continuation-relevant PDE phenomenon visible on the realized local class. By Theorem 19.9, the realized local class is exactly the full official local classical class on the fixed torus carrier. Therefore Ξ is a standing-positive same-scope continuation-relevant PDE phenomenon for the official periodic class. By Theorem 13.26, Ξ is represented by a standing-relevant discriminator on the standard carrier. Hence no standing-positive continuation-relevant PDE phenomenon visible in the realized formulation remains outside carrier-side representation. $\square$

Remark 18.17 (Architectural status of the strong-completeness bridge). Lemma 18.16 records how the broader continuation-completeness package would descend to the realized local class if one chooses to read the manuscript through that wider architectural lens. On that same narrow horizon-detectability route, this lemma is explanatory rather than load-bearing for the official periodic theorem: the official periodic theorem uses the horizon-detectability chain through Theorem 20.7A instead of the broader strong-completeness bridge. It is therefore retained explicitly as broader conceptual architecture, not as a necessary ingredient in the official-theorem spine.

19 Transfer Steps II–III (soundness and completeness): local classical realization of the unique admissible interior

Burden status. Datum entry has already been settled. What remains here is realization completeness: construct the local classical image of the admitted carrier class and show that the realized local class is exactly the full official local classical class, without changing same-scope admissibility

content. Part II identified the standard $\mathbb{T}^3$ carrier with the unique admissible interior and certified it internally. The next task is to realize that already-certified interior in the ordinary local classical Navier–Stokes presentation used by the official periodic theorem. Local classical realization supplies the classical image of the prior primitive theorem; it does not generate that theorem and it does not repair the analytic wall by a new estimate. For the official theorem, the crucial detectability step is the narrow horizon route through Theorem 20.7A, not any broader auxiliary continuation package. This section records the constructive realization map and the faithful fixed-carrier local theory needed for that route. No external corpus theorem is added at this stage: the bridge is constructed inside the paper using only the manuscript-local analytic packet proved in Appendix B and recorded later in Proposition H.1.

Definition 19.1 (Classical realization datum). A *classical realization datum* is a carrier seed $\Sigma(u_0)$ arising from some $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$, viewed in the ordinary PDE presentation as the smooth periodic divergence-free initial velocity u_0 together with its canonically induced mean-zero pressure gauge and the licensed skins of Definition 13.2.

Definition 19.2 (Constructive mild realization operator). Fix an integer $m \geq 3$ and $T > 0$. Let

$$H_\sigma^m(\mathbb{T}^3) := \{v \in H^m(\mathbb{T}^3; \mathbb{R}^3) : \nabla \cdot v = 0\},$$

and define

$$X_T^m := C([0, T]; H_\sigma^m(\mathbb{T}^3)) \cap L^2([0, T]; H_\sigma^{m+1}(\mathbb{T}^3))$$

with norm

$$\|v\|_{X_T^m} := \sup_{0 \leq t \leq T} \|v(t)\|_{H^m} + \left(\int_0^T \|v(t)\|_{H^{m+1}}^2 dt \right)^{1/2}.$$

For a classical realization datum $\Sigma(u_0)$, let $\mathbb{P}$ denote the Leray projector on $\mathbb{T}^3$ and define the *constructive mild realization operator*

$$(\Phi_{u_0, T} v)(t) := e^{\nu t \Delta} u_0 - \int_0^t e^{\nu(t-s)\Delta} \mathbb{P} \nabla \cdot (v \otimes v)(s) ds.$$

Definition 19.3 (Pressure reconstruction operator). For $v \in X_T^m$, define $p[v](t)$ to be the unique mean-zero periodic solution of

$$-\Delta p[v](t) = \partial_i \partial_j (v_i v_j)(t) \quad (0 \leq t \leq T).$$

Lemma 19.4 (Constructive local contraction window). *Let $\Sigma(u_0)$ be a classical realization datum and fix an integer $m \geq 3$. Then there exist $T_0 = T_0(\nu, \|u_0\|_{H^m}) > 0$ and a closed ball $B_{T_0}^m \subset X_{T_0}^m$ such that $\Phi_{u_0, T_0} : B_{T_0}^m \rightarrow B_{T_0}^m$ is a contraction. Its unique fixed point $u^{(0)}$ is smooth on $\mathbb{T}^3 \times [0, T_0]$, satisfies $u^{(0)}|_{t=0} = u_0$, and the associated pressure $p[u^{(0)}]$ is smooth with the fixed mean-zero gauge.*

Proof. Lemma B.3 and Proposition B.4 give the exact specialized semigroup / Leray contraction estimate needed for the mild map on X_T^m . Theorem B.5 then provides $T_0 = T_0(\nu, \|u_0\|_{H^m}) > 0$, a closed ball $B_{T_0}^m \subset X_{T_0}^m$, and the unique fixed point $u^{(0)} \in X_{T_0}^m$ of Φ_{u_0, T_0} . Because the datum in the present manuscript is smooth and periodic, Proposition B.6 upgrades that fixed point to a smooth classical solution on $\mathbb{T}^3 \times [0, T_0]$ and gives the uniquely normalized smooth pressure $p[u^{(0)}]$ in the fixed mean-zero gauge. $\square$

Lemma 19.5 (Initial-data extraction from the unique admissible interior). *Every standing-positive time-zero state in the initial-data face $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ determines a unique classical realization datum in the sense of Definition 19.1.*

Proof. By Theorem 18.10, every element of $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ is represented by a unique carrier seed $\Sigma(u_0)$ with $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$. That seed is exactly a classical realization datum by Definition 19.1. Uniqueness is the injective half of Theorem 18.10. $\square$

Theorem 19.6 (Transfer soundness: constructive local classical realization of admitted data). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be a standing-positive time-zero carrier state, and let $\Sigma(u_0)$ be its unique classical realization datum from Lemma 19.5. Then there exists a maximal time $T_*(x) > 0$ and a unique classical solution pair*

$$(u_x, p_x) \in C^\infty(\mathbb{T}^3 \times [0, T_*(x)]; \mathbb{R}^3) \times C^\infty(\mathbb{T}^3 \times [0, T_*(x)]; \mathbb{R})$$

satisfying

$$\partial_t u_x + (u_x \cdot \nabla) u_x + \nabla p_x = \nu \Delta u_x, \quad \nabla \cdot u_x = 0, \quad u_x|_{t=0} = u_0,$$

with periodic boundary conditions and the fixed mean-zero pressure gauge. Moreover, (u_x, p_x) is constructed inside the paper by the mild operator of Definition 19.2: its local segments are unique fixed points of $\Phi_{u_0, T}$ (or of the same operator started from later smooth time slices), and the maximal trajectory is the unique gluing of all overlapping fixed-point segments.

Proof. By Lemma 19.5, x determines a unique smooth periodic divergence-free velocity datum u_0 . Lemma 19.4 gives $T_0 > 0$ and a unique fixed point $u^{(0)}$ of Φ_{u_0, T_0} on $[0, T_0]$; define $p^{(0)} := p[u^{(0)}]$. This is the initial local realization segment.

Let $\mathcal{T}(x)$ be the set of all $T > 0$ for which there exists a smooth classical solution pair on $\mathbb{T}^3 \times [0, T]$ obtained by finitely many overlapping fixed-point constructions of the same form. The set $\mathcal{T}(x)$ is nonempty because $T_0 \in \mathcal{T}(x)$. If $T \in \mathcal{T}(x)$ and $t_1 < T$, then the time slice $u_x(t_1)$ is again a smooth divergence-free periodic datum on the same carrier. Applying Lemma 19.4 to that shifted datum produces a new fixed-point segment on $[t_1, t_1 + \delta]$. The overlap uniqueness theorem of Appendix B (Theorem B.7) implies that any two such segments agree where both are defined, so they glue uniquely.

Set

$$T_*(x) := \sup \mathcal{T}(x).$$

By overlap uniqueness, the union of all fixed-point segments with endpoint below $T_*(x)$ defines a single smooth classical solution pair (u_x, p_x) on $[0, T_*(x))$. If there were a smooth classical continuation beyond $T_*(x)$ on the same torus carrier, its restriction to a short interval beyond some $t_1 < T_*(x)$ would provide an additional fixed-point segment and hence a time larger than $T_*(x)$ in $\mathcal{T}(x)$, contradicting the definition of $T_*(x)$. Therefore (u_x, p_x) is the unique maximal realized trajectory. Since every step uses only the fixed official datum and the fixed carrier geometry, the construction introduces no auxiliary side condition, scale cut-off, or extra continuation criterion. $\square$

Lemma (19A (Local classical realization is faithful)). *For every standing-positive time-zero carrier state $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$, the construction of Theorem 19.6 produces a corresponding local classical periodic realization. If two realization segments issued from the same carrier state overlap in time, then overlap uniqueness identifies them as segments of one and the same realized trajectory. Consequently local classical realization does not branch inside a fixed carrier state and preserves carrier identity across local segments.*

Proof. Existence of a local classical realization is exactly the first conclusion of Theorem 19.6. The same theorem constructs the realized trajectory by gluing fixed-point segments started from later smooth time slices of the already realized solution. Whenever two such segments overlap, Theorem B.7 forces them to agree on the common interval. Hence the local realization attached

to a fixed carrier state is faithful: the construction cannot branch into two different local classical trajectories, and every later local segment is recognized as the same realized lineage rather than a second same-scope evolution. $\square$

Definition 19.7 (Realization map). The *realization map*

$$\mathcal{R}_{\text{NS}} : \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu) \rightarrow \mathcal{C}_{\text{loc}}(\mathbb{T}^3, \nu)$$

sends each standing-positive time-zero carrier state x to the unique maximal classical trajectory $\mathcal{R}_{\text{NS}}(x) := (u_x, p_x)$ constructed in Theorem 19.6, modulo the licensed bookkeeping skins of Definition 13.2. Here $\mathcal{C}_{\text{loc}}(\mathbb{T}^3, \nu)$ denotes the class of maximal local classical periodic Navier–Stokes realizations on the fixed torus carrier.

Definition 19.8 (Horizon continuation discriminator). For $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ and $T \geq 0$, define

$$\mathfrak{D}_T(x) := \begin{cases} 1, & T < T_*(x), \\ 0, & T \geq T_*(x), \end{cases}$$

where $T_*(x)$ is the maximal realization time from Theorem 19.6. Thus $\mathfrak{D}_T(x) = 1$ exactly when the constructed maximal trajectory $\mathcal{R}_{\text{NS}}(x)$ extends smoothly past the horizon T on the fixed torus carrier.

Theorem 19.9 (Transfer completeness: realized local class equals the full official local classical class). *The image of the realization map $\mathcal{R}_{\text{NS}}$ on $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ is exactly the class of all unique maximal smooth local periodic Navier–Stokes solution pairs issued from data in $\mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$. Equivalently, the realization map neither narrows nor enlarges the official local classical problem.*

Proof. Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$. By Theorem 19.6, $\mathcal{R}_{\text{NS}}(x)$ is a unique maximal smooth local periodic Navier–Stokes solution pair issued from the official datum attached to x . Thus the image of $\mathcal{R}_{\text{NS}}$ is contained in the official local classical class.

Conversely, let (u, p) be any smooth periodic Navier–Stokes solution pair on $\mathbb{T}^3 \times [0, T)$ with initial datum $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$. By Theorem 18.10, u_0 determines a unique admitted time-zero carrier class $x = [\Sigma(u_0)]$. Theorem 19.6 then yields the unique maximal local classical realization $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ with the same initial datum u_0 . The overlap uniqueness theorem of Appendix B (Theorem B.7) implies that (u, p) agrees with the restriction of (u_x, p_x) to its interval of definition. Hence every official local classical solution lies in the realized class. Therefore the two classes coincide exactly. $\square$

Theorem (19B) (Transfer completeness restatement: the realized local class equals the full official local classical class). *Part III neither shrinks nor enlarges the official local classical periodic problem. Every standing-positive carrier state realizes as an official maximal local classical periodic trajectory, and every official maximal local classical periodic trajectory is exactly the realized image of a carrier state in the unique admissible interior.*

Proof. The forward direction is the local realization statement of Theorem 19.6, sharpened by Lemma 19A: each admitted carrier state yields one well-defined maximal local classical trajectory. The reverse direction is Theorem 19.9: every official maximal local classical solution arises from the unique admitted carrier state determined by its initial datum. Therefore the realized local class is exactly the full official local classical class, so Part III is faithful transport and not a hidden narrowing or enlargement of the classical PDE problem. $\square$

19.1 PDE-native admissibility packet on the maximal classical interval

Before the standing-conservation packet is allowed to act on realized periodic lineages, the manuscript must record explicitly what is already forced by ordinary local periodic Navier–Stokes theory on the maximal classical interval itself. The next packet is narrow and local-in-time. It does *not* claim global regularity, a new coercive estimate, or the missing bridge inequality. It records only the minimum fixed-reference / same-object / pre-horizon witness content that is already present on $[0, T_*(x))$ once a maximal classical realization exists.

Theorem (19C (PDE-native admissibility on the maximal classical interval)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be standing-positive, and let*

$$\mathcal{R}_{\text{NS}}(x) = (u_x, p_x) \quad \text{on} \quad [0, T_*(x))$$

be the maximal realized lineage from Theorem 19.6. Then, on every compact subinterval $[0, T] \subset [0, T_(x))$, the realized lineage itself — together with any same-scope continuation-relevant determination that merely reads off its licensed carrier presentations on that interval — already satisfies the four standing-positive clauses of Definition ST1:*

- (i) **Fixed reference.** *With the mean-zero pressure gauge fixed, the pair (u_x, p_x) is uniquely determined on $[0, T]$, and hence so are its vorticity, strain, pressure, and standard frequency skins.*
- (ii) **Domain closure.** *Every such determination is read on the fixed torus carrier, with the same viscosity, zero forcing, smooth category, mean-zero pressure gauge, and same-scope continuation notion fixed earlier in Part II and carried into Part III by Definition 18.0 and Proposition 18C.*
- (iii) **Lawful-redescription invariance.** *Passage among the licensed presentations of Definition 13.2 changes only deterministic bookkeeping and does not alter continuation semantics on $[0, T]$.*
- (iv) **Identity stability.** *Smooth restart from any interior time reproduces the same realized lineage on overlap and cannot split it into a second same-scope object.*

In particular, the minimum admissibility / intelligibility content used later on the realized interval is forced directly by standard local periodic theory and the fixed carrier conventions themselves, not by the eventual global endpoint.

Proof. Fixed reference on $[0, T]$ is part of the maximal realization theorem already proved in-body: Theorem 19.6 constructs one unique maximal classical velocity field from the admitted carrier seed, and Proposition B.6 together with Corollary B.2 fixes the corresponding mean-zero pressure uniquely. The derived vorticity, strain/kernel, and standard frequency skins are then deterministic functions of that same classical pair on the fixed torus carrier, so they inherit the same determinate truth conditions.

Domain closure is fixed by the object convention already declared for Part III: Definition 18.0 and Proposition 18C state theorem-locally that the realized lineage is judged on the same torus carrier, viscosity, forcing status, smooth category, mean-zero gauge, and same-scope continuation question that were fixed in Part II.

Lawful-redescription invariance on the realized interval follows because the presentations named in Definition 13.2 are deterministic bookkeeping skins of the same classical velocity evolution:

vorticity and strain are obtained by differentiation, the pressure is reconstructed in the fixed mean-zero gauge, and the standard frequency skins are fixed Fourier bookkeeping on the same torus carrier. None changes the underlying classical continuation question on $[0, T]$.

Finally, identity stability on interior intervals is exactly the content of the restart consistency statement recorded next in Theorem 19D: any smooth restart from time $t_0 < T_*(x)$ reproduces the same lineage on overlap. Thus all four clauses of Definition ST1 are already forced by the classical periodic realization structure on $[0, T_*(x)]$. $\square$

Theorem (19D (Restart consistency and same-object persistence)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be standing-positive and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ be its maximal realized lineage on $[0, T_*(x))$. Fix any $t_0 < T_*(x)$ and regard the smooth slice $u_x(t_0)$ as a new periodic datum on the same torus carrier. Then the local classical realization started from $u_x(t_0)$ is defined on some interval $[t_0, t_0 + \delta)$ and agrees there with the restriction of (u_x, p_x) . Consequently a smooth interior restart does not create a second same-scope object; it persists as the same realized lineage.*

Proof. Because (u_x, p_x) is smooth on $[0, T_*(x))$, the slice $u_x(t_0)$ lies in $C^\infty_\sigma(\mathbb{T}^3)$. Theorem B.5 and Proposition B.6 therefore give a classical realization window of length $\delta > 0$ from this slice on the same torus carrier, together with the uniquely normalized mean-zero pressure reconstructed from that restarted velocity field. On the other hand, the time-shifted restriction of the already existing lineage,

$$(t, x) \mapsto (u_x(t), p_x(t)) \quad (t \in [t_0, \min\{t_0 + \delta, T_*(x)\})),$$

is itself a smooth classical periodic solution with the same datum $u_x(t_0)$ at time t_0 . The overlap-uniqueness theorem of Appendix B (Theorem B.7) therefore forces the restarted realization to coincide with the original lineage on the common interval. Hence interior restart is consistent and preserves the same realized object rather than generating a second same-scope trajectory. $\square$

Corollary (19E (Pre-horizon slice re-entry)). *Let $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$, let $x = [\Sigma(u_0)] \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be its unique admitted carrier seed, and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ be the maximal realized lineage on $[0, T_*(x))$. Then the realized lineage begins from the standing-positive admitted carrier state x in the unique admissible interior, and for every $t_1 < T_*(x)$ the slice $u_x(t_1)$:*

- (i) *belongs to the official periodic data class $\mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$,*
- (ii) *determines a unique carrier seed $\Sigma(u_x(t_1))$ in $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ with the same same-scope continuation semantics, and*
- (iii) *is standing-positive on the fixed torus carrier.*

Thus every smooth pre-horizon slice re-enters the same classical carrier before the horizon, and this re-entry is local in time rather than a disguised appeal to global regularity.

Proof. By Corollary 18.12, the official datum u_0 determines a unique admitted carrier seed $x = [\Sigma(u_0)]$ in $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$, and by Theorem 13I that seed already lies in the unique admissible interior. This gives the initial standing-positive entry claim.

Now fix $t_1 < T_*(x)$. Smoothness of the realized lineage on $[0, T_*(x))$ gives $u_x(t_1) \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$, and incompressibility is preserved by the classical equation, so $u_x(t_1) \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$. Theorem 19D shows that restarting from $u_x(t_1)$ preserves the same realized lineage and the same same-scope continuation question on overlap. Applying Corollary 18.12 to the smooth slice datum $u_x(t_1)$ then yields a unique carrier seed $[\Sigma(u_x(t_1))] \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$, and this seed is standing-positive on the fixed torus carrier. Hence every smooth pre-horizon slice re-enters the same carrier before the horizon. $\square$

Lemma (19E.1 (Continuation-locus fixity)). *For a standing-positive admitted carrier state x , smooth restart and overlap uniqueness preserve the realized continuation locus of Definition 18.0A. In particular, a restart from any smooth time $t_0 < T_*(x)$ is a presentation of the same locus and not a same-scope continuation object.*

Proof. Theorem 19D shows that restarting from a smooth slice produces a local classical realization that agrees with the original trajectory on overlap. Corollary 19E then shows that the slice re-enters the same carrier and remains standing-positive. These two facts preserve every component of Definition 18.0A: the admitted seed, the realized trajectory up to lawful presentation, the restart / overlap identity relation, and the fixed continuation question. Hence the locus is fixed. $\square$

Lemma (19E.2 (Boundary-trace nonpromotion)). *A boundary trace in the sense of Definition 18.0B cannot become an independent same-scope endpoint gate. If it is used in the live route, it is either licensed bookkeeping, quotient-visible witness bookkeeping subordinate to the horizon discriminator, or an illicit metric / representative gate.*

Proof. A boundary trace is built from pre-horizon lawful presentations of one realized lineage. By Corollary 3.40, representative-only data do not carry standing authority. By Theorem 13.23A.6, norm profiles, BKM profiles, stretching profiles, and other metric trends are bookkeeping unless subordinate to an already fixed witness; if promoted independently they are illicit metric gates. Thus a trace has no endpoint authority apart from the fixed horizon discriminator. $\square$

Lemma (19E.3 (Persistence-regime realization)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be standing-positive. Then its maximal realized periodic lineage is a persistence regime in the sense of Definition 18.0C on $[0, T_*(x))$.*

Proof. Lawful redescription persistence follows from Lemma 20.2 and Lemma 20.3. Smooth restart and overlap persistence follow from Theorem 19D. Pre-horizon standing re-entry follows from Corollary 19E. The same continuation question is fixed by Proposition 18C. These are exactly the persistence clauses of Definition 18.0C. $\square$

Lemma (19F (Finite-horizon witness lemma)). *Fix any integer $m \geq 3$. Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be standing-positive and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ be its maximal realized lineage on $[0, T_*(x))$. If $T_*(x) < \infty$, then*

$$\sup_{0 \leq t < T_*(x)} \|u_x(t)\|_{H^m(\mathbb{T}^3)} = \infty.$$

Equivalently, any same-scope finite-horizon continuation failure is already witnessed before the horizon by divergence of an in-scope continuation norm on the smooth realized lineage; no mysterious terminal object at $T_(x)$ is needed to state the failure.*

Proof. Assume for contradiction that $T_*(x) < \infty$ but

$$M := \sup_{0 \leq t < T_*(x)} \|u_x(t)\|_{H^m(\mathbb{T}^3)} < \infty.$$

Choose any $t_0 < T_*(x)$. The slice $u_x(t_0)$ is smooth, divergence-free, and periodic, so Theorem B.5 applies to it. Because the local realization time in Theorem B.5 depends only on ν and the H^m norm of the datum, there exists $\delta = \delta(\nu, M) > 0$ such that the datum $u_x(t_0)$ generates a classical realization on $[t_0, t_0 + \delta)$ whenever $\|u_x(t_0)\|_{H^m} \leq M$. By the bound defining M , this holds for every $t_0 < T_*(x)$.

Now choose $t_0 > T_*(x) - \delta/2$. The restarted local realization from $u_x(t_0)$ therefore exists on an interval extending strictly beyond $T_*(x)$. By Theorem 19D, this restarted realization agrees on overlap with the original lineage and hence extends the same realized object past $T_*(x)$ on the fixed torus carrier. That contradicts the maximality of $T_*(x)$ in Theorem 19.6. Therefore the assumed uniform H^m bound is impossible, so

$$\sup_{0 \leq t < T_*(x)} \|u_x(t)\|_{H^m(\mathbb{T}^3)} = \infty.$$

The final sentence is just the same statement read as a continuation witness built entirely from the smooth pre-horizon lineage. $\square$

20 Transfer Steps IV–V (faithfulness, no-new-criterion, and detectability): realization on the full official local class

Burden status. The official local classical class has already been realized. What remains here is to show that realization is faithful, that no same-scope continuation criterion appears at the classical layer, and that any hypothetical finite-time breakdown is already carrier-visible on the fixed torus carrier. The realization map is useful only if deterministic bookkeeping changes at the carrier level do not create different PDE continuation behavior. The present section records the first faithfulness facts needed for that bridge: lawful redescription commutes with realization, the constructed horizon continuation discriminator $\mathfrak{D}_T$ depends only on the carrier class, and classical local continuation is therefore already visible on the carrier side.

Definition 20.1 (Realized lineage and PDE-equivalent presentation). A *realized lineage* is the image under $\mathcal{R}_{\text{NS}}$ of a standing-positive time-zero carrier state together with its maximal local classical continuation. Two presentations of a realized lineage are *PDE-equivalent* if they differ only by the licensed skins of Definition 13.2 (velocity, vorticity, pressure, strain/kernel, or standard frequency bookkeeping) and therefore represent the same classical velocity evolution on the same maximal interval.

Lemma 20.2 (Lawful redescription commutes with realization). *If $x, y \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ are lawfully equivalent time-zero carrier states, then $\mathcal{R}_{\text{NS}}(x)$ and $\mathcal{R}_{\text{NS}}(y)$ are PDE-equivalent in the sense of Definition 20.1.*

Proof. By Lemma 13.4 and Theorem 18.10, lawfully equivalent time-zero carrier states determine the same official datum u_0 up to the already licensed bookkeeping skins. Theorem 19.6 assigns to that datum a unique maximal local classical solution pair. Hence the realized velocity evolutions coincide, and any difference between the two presentations is confined to the licensed skins of Definition 13.2. This is exactly PDE-equivalence. $\square$

Lemma 20.3 (Continuation status is skin-invariant). *Within the realized class, passage among the licensed skins of Definition 13.2 does not alter maximal local existence interval or continuation status.*

Proof. For a realized lineage (u, p) , the vorticity, strain/kernel, and standard frequency skins are derived from u by differentiation or by fixed linear bookkeeping on the same torus carrier, while the pressure is recovered at each time by the mean-zero reconstruction of Corollary B.2. None of these moves changes the underlying classical velocity evolution. Consequently they cannot create or destroy local classical continuation, nor can they alter the maximal existence interval attached

to that evolution. Therefore continuation status is invariant under passage among the licensed skins. $\square$

Proposition 20.4 (Local continuation cannot split a standing class). *Let $x, y \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be lawfully equivalent time-zero carrier states. Then the realized local solutions $\mathcal{R}_{\text{NS}}(x)$ and $\mathcal{R}_{\text{NS}}(y)$ have the same maximal local existence interval; equivalently, $\mathfrak{D}_T(x) = \mathfrak{D}_T(y)$ for every horizon $T \geq 0$.*

Proof. By Lemma 20.2, the two realized lineages are PDE-equivalent, so they represent the same classical velocity evolution. By Lemma 20.3, licensed skin changes do not alter maximal interval or continuation status. Hence no lawfully equivalent time-zero carrier states can split into distinct local continuation behaviors in the classical realization, and their horizon continuation discriminators agree at every T . $\square$

Theorem 20.5 (Transfer faithfulness: local classical continuation on the realized class). *On the realized class, the horizon continuation discriminator $\mathfrak{D}_T$ is determined by the underlying time-zero carrier class and is invariant under all deterministic bookkeeping skins. Equivalently, the realization map creates no extra local continuation criterion at the PDE level.*

Proof. Theorem 19.6 constructs a unique maximal local classical solution from any standing-positive time-zero carrier state and therefore a concrete horizon continuation discriminator $\mathfrak{D}_T$. Proposition 20.4 shows that lawful equivalence classes are not split by local classical continuation, so $\mathfrak{D}_T$ depends only on the standing class of the time-zero carrier state. Since continuation status is skin-invariant by Lemma 20.3, no additional same-scope continuation discriminator is introduced by the realization map. $\square$

Corollary 20.6 (Transfer no-new-criterion on the full official local class). *On the full official local classical class, continuation and finite-time breakdown are determined solely by the underlying carrier class on the fixed torus carrier. In particular, the passage from carrier data to the PDE formulation introduces no auxiliary side condition, critical threshold, or extra continuation criterion beyond the data already present in $\mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$.*

Proof. Combine Theorem 19.9 with Theorem 20.5. The first identifies the realized local class with the full official local classical class, and the second shows that continuation behavior on that class is carried entirely by the underlying carrier class and is invariant under the licensed bookkeeping skins. $\square$

Theorem (20.5B) (Transfer no-new-criterion: classical realization adds no new standing-positive same-scope continuation content). *On the full official local classical class realized from standing-positive admitted carrier states, the passage from the certified carrier to its realized classical image is standing-positive same-scope continuation-faithful: every continuation-relevant distinction used as a standing-positive classifier in the classical formulation is already determined by the underlying carrier class on the fixed torus carrier. Equivalently, classical realization adds no new standing-positive same-scope continuation content beyond the certified carrier class.*

Proof. Theorem 19B identifies the realized local class with the full official local classical class. Theorem 20.5 and Corollary 20.6 show that continuation status on that class is determined solely by the carrier class and is invariant under the licensed bookkeeping skins. The no-new-criterion statement used here is inherited from the fixed-periodic P4 application: Proposition 13K uses Lemma 12F only for scope-fixity, while the actual exclusion of additional same-scope criteria is sourced from Theorem 12G. Therefore the passage from the certified carrier to its classical realization

introduces no new standing-positive same-scope continuation content; any purported additional classical-layer distinction used as a standing-positive classifier would either be bookkeeping only or would define a forbidden second same-scope discriminator. $\square$

Corollary (20A (Transfer detectability: any finite-time classical breakdown would already be carrier-visible)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ and let its official local classical realization break down at a finite time $T_*(x)$. Then that finite-time breakdown already corresponds to a standing-relevant carrier-side event inside the unique admissible interior identified with the standard periodic carrier. In particular, no standing-positive same-scope classical failure mode can hide outside the carrier image while remaining inside the official periodic problem.*

Proof. By Theorem 19B, the realized local class is exactly the full official local classical class on the fixed torus carrier, so any official finite-time breakdown occurs inside the realized image of a carrier state. By Theorem 13I, that carrier state already lies in the unique admissible interior identified with the standard periodic carrier. Therefore a finite-time classical breakdown cannot remain external to the carrier while still claiming the same fixed periodic scope; if it did, the realized classical class would be split from the identified carrier by an extra same-scope discriminator, contrary to carrier identification and the no-rival corollary. Hence any finite-time classical breakdown is already carrier-visible. $\square$

Proposition 20.7 (Finite-time breakdown is a continuation discriminator on the fixed carrier). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ have maximal interval $[0, T_*(x))$ with $T_*(x) < \infty$. Then the statement*

“the realized trajectory issued from x admits no classical continuation past $T_(x)$ on the fixed torus carrier”*
is invariant under PDE-equivalent presentation and depends only on the underlying carrier class.

Proof. Maximality of $[0, T_*(x))$ is exactly the statement that the constructed horizon discriminator $\mathfrak{D}_T(x)$ drops from 1 to 0 at the first non-extendable horizon. By Lemma 20.3, passage among the licensed skins does not change the underlying velocity evolution and therefore cannot change $\mathfrak{D}_T(x)$. By Proposition 20.4, lawfully equivalent time-zero carrier states have the same maximal local interval and hence the same family of values $\mathfrak{D}_T$. Thus finite-time breakdown, when it occurs, defines a continuation discriminator attached to the carrier class itself, not to a particular bookkeeping presentation. $\square$

Theorem (20.7A (Horizon detectability on the fixed periodic carrier)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ have maximal interval $[0, T_*(x))$ with $T_*(x) < \infty$. Then the horizon discriminator*

$$\mathfrak{D}_{T_*(x)}(x) = 0$$

is a well-typed standing-relevant continuation discriminator on the fixed torus carrier. Equivalently, finite-time breakdown at the first non-extendable horizon is already carrier-side standing-detectable.

Proof. By Theorem 19.6, the realized trajectory $\mathcal{R}_{\text{NS}}(x)$ is constructed as the unique maximal smooth periodic Navier–Stokes solution issued from the admitted carrier class x on the fixed torus carrier. Proposition 20.7 shows that the statement “no classical continuation exists past $T_*(x)$ on the fixed torus carrier” depends only on the underlying carrier class and is invariant under lawful presentation. Thus finite-time breakdown determines the carrier-side horizon discriminator $\mathfrak{D}_{T_*(x)}$.

By Proposition 20.4, horizon continuation status is constant on standing classes. In particular, if y is lawfully equivalent to x , then $\mathfrak{D}_{T_*(x)}(y) = \mathfrak{D}_{T_*(x)}(x) = 0$. Hence the first non-extendable horizon is not merely a classical fact about one presentation of the trajectory; it is a same-scope carrier-side continuation discriminator that is constant on standing classes and therefore standing-relevant on the fixed carrier. $\square$

Lemma 20.8 (Architectural realization completeness for continuation discriminators). *On the full official local classical class, every standing-positive same-scope continuation discriminator of a realized trajectory is already represented on the carrier side. Equivalently, if a continuation or finite-time breakdown statement distinguishes two realized local trajectories on the fixed torus carrier in a standing-positive same-scope way, then the induced distinction is attached to the underlying carrier classes and therefore determines a standing-relevant continuation discriminator in the certified carrier.*

Proof. By Theorem 19.9, every local classical solution in the official periodic class is the realization of some admitted time-zero carrier state. Thus no official local trajectory lies outside the range of $\mathcal{R}_{\text{NS}}$. By Lemma 18.16, every standing-positive same-scope continuation-relevant PDE phenomenon visible on that realized local class is already represented by a standing-relevant discriminator on the carrier side.

By Proposition 20.7, continuation and finite-time breakdown are attached to the carrier class itself and are invariant under PDE-equivalent presentation. By Theorem 20.5 and Corollary 20.6, the realization map introduces no additional same-scope continuation criterion on that full official local class. Therefore any standing-positive same-scope continuation distinction visible in the realized PDE formulation is already represented on the carrier side as a standing-relevant continuation discriminator. $\square$

Logical structure of realization completeness. Proposition 20.7 establishes that any finite-time breakdown statement defines a carrier-side continuation discriminator on the fixed torus carrier. Theorem 13.26 together with Lemma 18.16 records the broader architectural claim that no standing-positive continuation-relevant PDE phenomenon visible on the realized local class lies outside carrier-side representation. Theorem 20.5 establishes that the realization map introduces no additional same-scope continuation criterion. Lemma 20.8 combines these directions at the explanatory level. For the official periodic theorem, however, the load-bearing detectability step is the narrower route through Theorem 20.7A, not this broader completeness package.

Definition 20.9 (Trajectory stretching observable). Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ and let

$$\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$$

be its maximal realized classical trajectory on $[0, T_*(x))$. Define the trajectory stretching observable by

$$\Theta_x(t) := \mathcal{T}(u_x(t)) = \int_{\mathbb{T}^3} \omega_x(\cdot, t) \cdot S(u_x(\cdot, t)) \omega_x(\cdot, t) dx, \quad 0 \leq t < T_*(x).$$

Lemma 20.10 (Skin invariance of the trajectory stretching observable). *For a realized trajectory $\mathcal{R}_{\text{NS}}(x)$, the function $t \mapsto \Theta_x(t)$ is independent of PDE-equivalent presentation. In particular, passage among the licensed velocity / vorticity / pressure / strain / frequency skins does not alter the trajectory stretching observable on its interval of definition.*

Proof. By Theorem 19.6, the maximal realized trajectory $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ is uniquely attached to the carrier class x . By Definition 20.1, PDE-equivalent presentations differ only by the licensed skins of Definition 13.2 and therefore represent the same classical velocity evolution on the same interval. The vorticity $\omega_x = \nabla \times u_x$ and the strain tensor $S(u_x) = \frac{1}{2}(\nabla u_x + \nabla u_x^\top)$ are therefore unchanged as carrier-attached fields under such passage, and so is the scalar functional $\Theta_x(t)$. Hence $t \mapsto \Theta_x(t)$ is independent of PDE-equivalent presentation. $\square$

Definition 20.11A (Θ_x -based continuation gate).

A Θ_x -based continuation gate is a map

$$G_\Theta : \{\text{realized local classical trajectories on } \mathbb{T}^3\} \rightarrow \{0, 1\}$$

such that:

- (i) for each realized trajectory $\mathcal{R}_{\text{NS}}(x)$, the value $G_\Theta(\mathcal{R}_{\text{NS}}(x))$ depends only on the evolution of $\Theta_x(t)$ on its realized interval; and
- (ii) G_Θ is standing-relevant in the sense that it distinguishes continuation behavior on the full official local classical class.

Equivalently, a Θ_x -based continuation gate is a same-scope continuation predicate extracted purely from the trajectory stretching observable.

Theorem 20.11 (No Θ_x -based continuation gate; dynamic compatibility with zero-parameter certification). *Every Θ_x -based continuation gate factors through the carrier-side horizon continuation discriminator and therefore contributes no independent same-scope continuation criterion. In particular, the evolution of Θ_x respects the zero-parameter certification of Theorem 13.20.*

Proof. By Theorem 19.6, every standing-positive admitted time-zero carrier class determines a unique maximal realized trajectory and therefore a well-defined stretching observable $t \mapsto \Theta_x(t)$. By Lemma 20.10, this observable is attached to the realized trajectory itself and not to a chosen PDE skin.

Let G_Θ be a Θ_x -based continuation gate in the sense of Definition 20.11A. By clause (ii), G_Θ is standing-relevant: it distinguishes continuation behavior on the full official local classical class. By Corollary 20.6, the realization map introduces no auxiliary side condition or extra continuation criterion beyond the underlying carrier class on the fixed torus carrier. Thus any same-scope continuation distinction registered by G_Θ must already be attached to carrier classes.

By Proposition 20.4, continuation status is constant on standing classes. By Theorem 20.7A, finite-time breakdown at the first non-extendable horizon is already a standing-relevant carrier-side continuation discriminator. Hence if G_Θ contributed more than the carrier-side horizon continuation content, it would define a second same-scope continuation discriminator on the fixed torus carrier.

But Theorem 13.20 and the exhaustion package of Part II forbid any such second same-scope continuation discriminator on the certified carrier. Therefore every Θ_x -based continuation gate factors through the carrier-side horizon continuation discriminator and contributes no independent same-scope continuation criterion. This is exactly the claim that the evolution of Θ_x respects the zero-parameter certification. $\square$

Theorem (20B (Transfer no-new-criterion: no independent standing-positive same-scope continuation discriminator at the classical layer)). *On the full official periodic local class realized from standing-positive admitted carrier states, no additional PDE-side continuation criterion, horizon parameter, auxiliary regularity gate, or continuation discriminator survives as a standing-positive same-scope classifier once the standard carrier has been identified with the unique admissible interior. Any proposed classical-layer criterion used in that standing-positive role either factors through the carrier-side admissibility structure already identified in Part II or constitutes a scope change.*

Proof. Theorem 20.5B states the general classical-layer closure on the live route: realization adds no new standing-positive same-scope continuation content beyond the certified carrier class. The no-new-criterion statement used here is inherited from the fixed-periodic P4 application already

carried by Theorem 12G: Proposition 13K imports that result into the carrier-identification packet, and Theorem 20.5B then carries the same consequence across the realization map. Hence any purported PDE-side criterion that changes continuation outcomes would define a second same-scope discriminator between official classical trajectories and the already identified carrier-side admissibility structure, contrary to Proposition 13K and Corollary 13J; if it does not change continuation outcomes, then it is bookkeeping only. Theorem 20.11 gives the explicit in-paper Θ_x -based instance of this dichotomy. Therefore no independent standing-positive same-scope classical-layer continuation discriminator survives. $\square$

Remark 20.12 (Architectural status of realization completeness). Lemma 20.8 remains in the manuscript as a broader explanatory statement about continuation content on the realized class. On the narrow horizon-detectability route, the official periodic theorem does not require this broader lemma: for the official periodic theorem it is enough that horizon failure is standing-detectable on the fixed carrier by Theorem 20.7A.

Corollary 20.13 (No stretching-based continuation gate). *Within the full official periodic local class, no threshold, growth criterion, or horizon selector built from the evolution of $\mathcal{T}(u)$ defines an independent same-scope continuation criterion.*

Proof. Immediate from Theorem 20.11, which is now subsumed conceptually by the general classical-layer closure statement of Theorem 20B. $\square$

20.1 Carrier-to-lineage binding, no admissibility dynamics, and representational commitment on realized lineages

The remaining live objection at the realization layer is not whether smooth pre-horizon slices are theorem-locally standing-positive; the PDE-native admissibility packet already settled that. The remaining question is whether carrier identification is merely static while the realized periodic evolution is allowed to acquire a new admissibility status as time advances. The next packet closes that exact route. It proves that once a standing-positive admitted carrier seed is realized, the maximal periodic evolution is one standing-bearing same-scope determination, that admissibility on that determination is global rather than time-indexed, and that the representational commitment fixed at entry is inherited by the full realized lineage.

Theorem (20B.1 (Carrier-to-lineage binding on the fixed periodic carrier)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be standing-positive and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ be its maximal realized lineage on $[0, T_*(x))$. Then this realized periodic evolution is one standing-bearing same-scope determination on the fixed torus carrier. More precisely:*

- (i) *its standing-bearing source is the admitted carrier state x in the unique admissible interior identified with the standard periodic carrier;*
- (ii) *every smooth slice and every smooth restart presentation on $[0, T_*(x))$ is a licensed presentation of that same determination rather than a fresh same-scope admissibility state; and*
- (iii) *the same-scope continuation notion used to judge the realized lineage is exactly the continuation notion fixed earlier in Part II and carried into Part III without redefinition.*

Hence the realized periodic lineage is not a time-indexed family of independently reclassified admissibility states.

Proof. Corollary 18.12 sends every official datum into a unique standing-positive admitted carrier state, and Theorem 13I identifies that state with the unique admissible interior on the fixed torus carrier. Definition 18.0 and Definition 18.0A then fix the realized maximal classical object as one same-scope continuation locus, while Proposition 18C states theorem-locally that the same continuation question used in Part III is exactly the one fixed in Part II.

For smooth interior times, Theorem 19D shows that restarting from $u_x(t_0)$ reproduces the same realized lineage on overlap rather than creating a second same-scope object, Corollary 19E shows that every such smooth slice re-enters the same admitted carrier and is standing-positive there, and Lemma 19E.1 packages these facts as continuation-locus fixity. Proposition 3.41 together with Corollary 3.42 then rules out treating later restart representatives or later lawful presentations as fresh admissibility decisions inside the same fixed scope. Therefore the maximal realized periodic evolution issued from x is one standing-bearing same-scope determination on the fixed carrier, not a stack of independently reclassified admissibility states indexed by time. $\square$

Theorem (20B.2 (No admissibility dynamics on realized periodic lineages)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be standing-positive and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ be its maximal realized lineage on $[0, T_*(x))$. Within the fixed periodic scope, admissibility / standing on this realized lineage is global and quotient-visible; it cannot vary with time, slice number, restart stage, or horizon proximity. In particular, a proposal of the form “the lineage is admissible before $T_*(x)$ but non-admissible at $T_*(x)$ ” is not a neutral same-scope alternative. If read as a same-scope proposal, it is ill-typed: it either makes admissibility depend on history or stage inside one realized lineage, promotes representative-only skin to standing authority, or introduces a same-scope discriminator / gate.*

Proof. By Theorem 20B.1 and Lemma 19E.3, the realized periodic evolution is one standing-bearing same-scope persistence regime rather than a sequence of fresh admissibility states. Proposition 3.41 and Corollary 3.42 then force all same-scope standing-bearing verdicts on that lineage to depend only on the standing class reached, not on which lawful history, restart representative, elapsed stage, or horizon-proximity presentation reached it. Theorem 3.35 says that any same-scope standing-relevant classifier must factor through the fixed quotient, and Corollary 3.40 says that representative-only or metadata-only features cannot become a hidden second gate. Finally, Theorems 20.5B and 20B say theorem-locally that realization introduces no additional same-scope classical-layer discriminator on the official local class.

Therefore a claim that admissibility changes with time on this one realized lineage has only three possible readings. If it depends on slice index, elapsed time, restart stage, horizon proximity, or a boundary trace while the same-scope quotient-visible content is unchanged, then it is a path- or history-dependent same-scope admissibility functional, contrary to Corollary 3.42 and Lemma 19E.2. If it depends only on representative bookkeeping skin, then it is non-authoritative by Corollary 3.40. If it changes standing-bearing outcomes anyway, then it is a same-scope discriminator / gate, contrary to Theorems 3.35 and 20B. Hence admissibility does not evolve along a realized periodic lineage. In particular, the sentence “admissible before $T_*(x)$, non-admissible at $T_*(x)$ ” is not a licit same-scope status claim. $\square$

Definition 20.14 (Standing-bearing representational commitment on a realized lineage). Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ be its realized lineage. The *standing-bearing representational commitment* of this lineage is the fixed package consisting of:

- (i) the torus carrier, viscosity sign, zero-forcing equations, smooth category, and mean-zero pressure gauge fixed by the same periodic scope;
- (ii) the admitted time-zero carrier class x and the official datum it determines; and

- (iii) the lawful-skin equivalence class under which later slices, horizon statements, and continuation claims are read as presentations of that same lineage.

A *global rollback* of this commitment is any purported same-scope admissible move that claims to retract, erase, or neutralize the standing-bearing consequences of this package across the realized lineage rather than merely changing licensed bookkeeping skin.

Proposition (20B.3 (Standing and commitment inheritance on same-scope realized lineages)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be standing-positive and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ be its maximal realized lineage on $[0, T_*(x))$. Then the standing-positive status of the admitted carrier seed and the fixed same-scope commitments of Definition 20.14 are inherited by the whole realized lineage. In particular, every smooth pre-horizon slice is a licensed presentation / re-entry of that same standing-bearing determination, and any purported same-scope later re-evaluation or rollback of those commitments is either bookkeeping only, a scope change, or standing-negative inadmissibility.*

Proof. Corollary 19E says that every smooth slice $u_x(t_1)$ with $t_1 < T_*(x)$ re-enters the same admitted carrier and is standing-positive there. Theorem 20B.1 identifies those slices and their restarts as presentations of one standing-bearing same-scope determination, and Definition 20.14 records exactly which carrier, datum, gauge, lawful-skin, and continuation commitments are fixed on that determination. By Theorem 20B.2, those commitments cannot be re-evaluated later by time index, restart stage, or horizon proximity.

A purported move that changes only lawful presentation is bookkeeping by Lemma 20.2, Lemma 20.3, and Corollary 3.40; a move that alters one of the fixed periodic commitments is a scope change by Proposition 13.7 and Lemma 12A; and a move that claims to change standing-bearing consequences later while preserving the same scope is standing-negative inadmissibility by Theorem ST3 together with Lemma 12E. Hence the standing-positive carrier seed and its fixed representational commitment are inherited by the full realized lineage in the only same-scope way allowed locally in the manuscript. $\square$

Theorem (ST4: Standing conservation / no standing-neutral drift on realized lineages). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ be its maximal realized lineage on $[0, T_*(x))$. Then every smooth pre-horizon slice remains standing-positive on the fixed carrier, and no same-scope realized lineage can drift from standing-positive status to a purportedly neutral non-standing same-scope terminal status. Equivalently, same-scope realized evolution admits no standing-neutral drift.*

Proof. Corollary 19E gives the first claim locally in time. The theorem-local reason this does not become a time-indexed reclassification claim is Theorem 20B.1: the realized maximal trajectory is one standing-bearing same-scope determination. Theorem 20B.2 then rules out any same-scope proposal in which that one determination drifts from standing-positive status to a purportedly neutral non-standing status as time advances. Proposition 20B.3 records the same conclusion as inheritance of the admitted carrier seed and its fixed commitments by the full realized lineage. Finally, Theorem ST3 and Corollary 13.9 say that once standing-positive status is lost no neutral same-scope middle status remains. Therefore the realized lineage cannot leave standing-positive status by drifting into a neutral same-scope endpoint. $\square$

Proposition (ST5: Irreversibility of representational commitment on realized lineages). *Once a realized lineage has been undertaken under the standing-bearing representational commitment of Definition 20.14, there exists no admissible same-scope sequence that globally retracts, erases, or neutralizes that commitment. Any purported same-scope de-commitment is either bookkeeping only, a scope change, or standing-negative and therefore inadmissible.*

Proof. Proposition 20B.3 already fixes the standing-bearing commitments of Definition 20.14 as inherited by the full realized lineage. A purported de-commitment that changes only lawful presentation is bookkeeping only by Lemma 20.2, Lemma 20.3, and Corollary 3.40. A purported de-commitment that alters the torus carrier, viscosity sign, forcing status, smooth category, gauge, datum anchor, or same-scope admissibility interface changes one of the fixed periodic commitments identified in Proposition 13.7 and Lemma 12A; it is therefore a scope change rather than a same-scope rollback.

It remains to consider a purported same-scope rollback that claims to neutralize an earlier standing-bearing commitment while preserving the same carrier and lineage. By Lemma 12E, no failed same-scope admissibility move can be retroactively repaired by later metadata, relabeling, reparameterization, or reinterpretation. By Theorem 20B.2, the lineage does not admit a later time-indexed re-evaluation of its admissibility or commitment status. Therefore any purported rollback that genuinely changes standing-bearing consequences must incur standing loss. By Theorem ST3, that standing loss is inadmissibility. Hence no admissible same-scope global rollback, erasure, or de-commitment exists once realization is underway. $\square$

Corollary 20.15 (No temporary standing-bearing commitment on a realized lineage). *On a realized lineage, the notion of a merely temporary standing-bearing representational commitment is incoherent.*

Proof. A temporary standing-bearing commitment would require a later admissible same-scope de-commitment. Proposition ST5 excludes such a de-commitment. $\square$

20.2 Same-scope endpoint-status trichotomy and collapse on realized periodic lineages

The standing/intelligibility packet ST1–ST5, the PDE-native admissibility packet, the quotient / history / ATS packet, and the carrier-to-lineage / no-dynamics packet now allow the manuscript to say something sharper than the earlier two-branch rhetoric. On a realized periodic lineage there are only three theorem-locally typable ways to keep a finite-horizon endpoint claim alive while still pretending to remain inside the fixed same scope: as a standing-positive in-scope witness, as terminal standing failure, or as an illicit same-scope gate / scope move. The next packet makes that trichotomy explicit and then collapses the non-witness branches before witness exhaustion is invoked in Section 21.

Horizon bridge object and local finite-horizon reductio

The endpoint closeout can now be read locally. A finite-horizon assumption is opened as the exact complement of global smooth continuation for one fixed realized lineage. The bridge object below packages the data forced by that assumption; it does not add a continuation estimate and it does not contain global smoothness by definition.

Definition (20B.4 (Horizon bridge object)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be standing-positive and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ be its realized lineage. For a proposed finite horizon $T = T_*(x) < \infty$, the Navier–Stokes horizon bridge object*

$$B_{\text{hor}}^{\text{NS}}(x, T)$$

records the following same-scope data:

B1. *the official datum $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$ determined by x ;*

- B2.** *the fixed periodic carrier: $\mathbb{T}^3$, $\nu > 0$, zero forcing, smooth divergence-free class, and the fixed mean-zero pressure gauge;*
- B3.** *the same smooth periodic continuation question fixed in Part II and realized in Part III;*
- B4.** *the admitted carrier seed $x = [\Sigma(u_0)]$ as the source of the realized lineage;*
- B5.** *the identity of $\mathcal{R}_{\text{NS}}(x)$ as one same-scope maximal classical lineage, not a sequence of independently reclassified time slices;*
- B6.** *the pre-horizon witness packet: every $t < T$ gives a smooth slice that re-enters the official carrier as a lawful presentation of the same realized lineage;*
- B7.** *the carrier-side horizon discriminator $\mathfrak{D}_T(x) = 0$ supplied by finite maximality at the first non-extendable horizon;*
- B8.** *metric-witness subordination: BKM, critical-norm, vorticity, strain, frequency, and metric-divergence reports attached to the horizon are only quotient-visible witness bookkeeping unless promoted to an illicit gate;*
- B9.** *the branch data $(\beta(x), \mathfrak{B}(x))$, where $\beta(x)$ is the branch tag and $\mathfrak{B}(x) \subseteq \{S1, S2, S3, S4, S5\}$ is the induced branch-forcing set; and*
- B10.** *no independent endpoint authority: the bridge object introduces no new PDE estimate, continuation criterion, hidden selector, or second same-scope endpoint gate.*

Proposition (20B.5 (Horizon bridge object is non-circular)). *The object $B_{\text{hor}}^{\text{NS}}(x, T)$ does not contain the conclusion $T_*(x) = \infty$ by definition. It packages the finite-horizon countercase data forced by the fixed carrier, realized-lineage identity, horizon discriminator, metric-bookkeeping discipline, and branch extraction.*

Proof. Clauses B1–B6 fix the official datum, carrier, continuation question, carrier seed, realized-lineage identity, and pre-horizon witness availability. Clause B7 records the finite-horizon discriminator produced by the local assumption $T_*(x) < \infty$. Clauses B8–B10 assign the role of metric diagnostics and branch data without adding endpoint authority. None of these clauses asserts global continuation. The conclusion $T_*(x) = \infty$ is obtained only after the trichotomy, collapse of the non-witness branches, and elimination of the remaining standing-positive witness branch. $\square$

Theorem (20B.6 (Horizon bridge object induces the endpoint-status trichotomy)). *If $B_{\text{hor}}^{\text{NS}}(x, T_*(x))$ is in force, then the associated finite-horizon endpoint condition has exactly one of the three statuses later stated in Theorem 20C: standing-positive admissible in-scope witness, terminal standing failure, or illicit same-scope gate / scope move.*

Proof. The bridge object fixes the same carrier and realized lineage, records the horizon discriminator, and keeps metric diagnostics subordinate to the fixed witness packet unless they are promoted to an illicit gate. These are precisely the inputs used by Theorem 20C. The object therefore enters that trichotomy without adding another endpoint status. $\square$

Definition (20B.7 (Exact-complement finite-horizon annotation)). *The notation*

$$[T_*(x) < \infty]_i$$

denotes the proof-act annotation generated when the sole local assumption $T_(x) < \infty$ is opened as the exact-complement countercase against global smooth continuation of the fixed realized lineage $\mathcal{R}_{\text{NS}}(x)$. The annotation records proof role only; it is not an additional object-level premise.*

Theorem (20B.8 (Annotation is not a premise)). *The exact-complement annotation does not transform the discharged assumption into*

$$T_*(x) < \infty \wedge \text{ReductioCounterCase}_{NS}(T_*(x) < \infty; T_*(x) = \infty).$$

The only object-level assumption opened and later discharged is $T_(x) < \infty$.*

Proof. The bracket $[\cdot]_i$ marks the assumption's local proof role: it is the branch whose survival blocks the endpoint for the fixed realized lineage. The proof does not add a second mathematical assertion to the assumption. All later bridge, trichotomy, and branch-extraction roles are consequences of using that sole finite-horizon assumption as the local exact complement inside the fixed same-scope endpoint proof. $\square$

Theorem (20B.9 (Local finite-horizon counterexample induces the horizon bridge)). *For a standing-positive admitted carrier state x ,*

$$[T_*(x) < \infty]_i \implies B_{\text{hor}}^{NS}(x, T_*(x)).$$

Proof. The assumption $T_*(x) < \infty$ supplies the proposed finite horizon. The fixed datum and carrier clauses are supplied by Corollary 18.12 and Theorem 13I. Realized-lineage identity and pre-horizon re-entry are supplied by Theorem 20B.1, Theorem 19D, and Corollary 19E. The carrier-side horizon discriminator is supplied by Theorem 20. Metric-witness subordination is supplied by Propositions 13.23A.3 and 13.23A.4. The branch tag and branch-forcing set are supplied by Lemma 21.5 and Definition 21.3. Thus all clauses of Definition 20B.4 are forced by the local finite-horizon counterexample. $\square$

Theorem (20B.10 (Local finite-horizon counterexample contradiction)). *For every standing-positive admitted carrier state x ,*

$$[T_*(x) < \infty]_i \implies \perp.$$

Proof. By Theorem 20B.9, the local finite-horizon counterexample induces $B_{\text{hor}}^{NS}(x, T_*(x))$. By Theorem 20B.6, the associated endpoint condition enters the realized-lineage trichotomy of Theorem 20C. The terminal standing-failure branch is excluded by Proposition 20C.2; the illicit gate / scope-move branch is not a licit same-scope endpoint status by Corollary 20C.3. The remaining standing-positive witness branch is exhausted by Proposition 21.6A and converted into a typed singular candidate by Lemma 21.7. Propositions 21.8 and 21.9, together with Corollary 15.8, eliminate the resulting branches. Hence the local finite-horizon assumption yields contradiction. $\square$

Theorem (20B.11 (Annotation discharge gives global smoothness for the lineage)). *For every standing-positive admitted carrier state x , the local contradiction from $[T_*(x) < \infty]_i$ discharges the original finite-horizon assumption itself. Consequently $T_*(x) = \infty$.*

Proof. By Theorem 20B.8, the only object-level assumption in the local subproof is $T_*(x) < \infty$. Theorem 20B.10 gives a contradiction from that assumption, so reductio yields $\neg(T_*(x) < \infty)$. Since Theorem 19.6 gives a maximal classical interval $[0, T_*(x))$ with $T_*(x) > 0$ and the only alternative to global time is finite maximal time, the result is $T_*(x) = \infty$. $\square$

Definition (20C.0 (Finite-horizon endpoint condition)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ and let $\mathcal{R}_{NS}(x) = (u_x, p_x)$ be its maximal realized lineage on $[0, T_*(x))$. If $T_*(x) < \infty$, the finite-horizon endpoint condition is the same-scope continuation condition determined by failure of smooth extension beyond $T_*(x)$ on the fixed periodic carrier. It is not optional descriptive surplus: it is the theorem-bearing obstruction that a counterexample to Fefferman (B) would have to supply.*

Theorem (20C.0A (Finite-horizon failure induces a same-scope endpoint condition)). *If $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ has $T_*(x) < \infty$, then finite maximality of $\mathcal{R}_{\text{NS}}(x)$ induces the finite-horizon endpoint condition of Definition 20C.0. This condition belongs to the fixed-domain comparison class unless one of the declared periodic commitments is altered, in which case the alleged failure is not a same-scope counterexample to the official periodic theorem.*

Proof. By Theorem 19.6, $T_*(x)$ is the maximal classical realization time on the fixed torus carrier. If $T_*(x) < \infty$, maximality says exactly that no smooth continuation on the same carrier, with the same viscosity, forcing status, pressure gauge, datum anchor, and continuation notion, is available past $T_*(x)$. That is the continuation condition named in Definition 20C.0. If a later description changes any of those commitments, Definition 3.4A classifies it outside the fixed-domain comparison class. Otherwise the condition is same-scope and is governed by the endpoint-status trichotomy below. $\square$

Proposition (20C.1 (Exterior same-scope endpointhood is terminal standing failure)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ and assume $T_*(x) < \infty$. If a purported same-scope terminal endpoint of the realized lineage $\mathcal{R}_{\text{NS}}(x)$ can be formulated only by leaving admissible representability on the fixed torus carrier, then that purported endpoint is standing-negative. Equivalently, an “outside-admissible” same-scope endpoint is terminal standing failure rather than a neutral uncaptured status.*

Proof. Endpoint-status classification here relies on the fixed-scope admissibility structure transported from the 12G packet: 12F classifies scope-fixity at the fixed periodic interface, Theorem 12G then applies P4 to that packet, ST2 and ST3 internalize the resulting fixed-scope standing consequences, and only after that one-directional chain is in place does the present proposition evaluate realized-lineage endpointhood. By Theorem 20B.1, the realized maximal periodic evolution issued from x is one standing-bearing same-scope determination, so any purported terminal endpoint still claiming the same carrier, datum anchor, gauge, lawful skin, and continuation question must be evaluated as a continuation-relevant determination on that fixed object. If such an endpoint can be stated only by leaving admissible representability, then it abandons the fixed reference, domain-of-application closure, lawful-redescription invariance, or identity stability required by Definition ST1. By Theorem ST2, failure of any one of those fixed-scope standing conditions is standing loss in the periodic setting. By Theorem ST3, that standing loss is standing-negative inadmissibility, not a neutral third status. Therefore an “outside-admissible” same-scope endpoint is exactly terminal standing failure. $\square$

Proposition (20C.2 (Terminal standing failure cannot complete a same-scope realized endpoint)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be standing-positive and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ be its maximal realized lineage on $[0, T_*(x))$. Then terminal standing failure cannot complete the same-scope endpoint of this realized lineage. In particular, a standing-positive admitted carrier seed cannot realize its finite-horizon endpoint by collapsing into terminal standing failure while still claiming the same periodic scope.*

Proof. Assume for contradiction that the same realized lineage completed its same-scope terminal endpoint through terminal standing failure. By Proposition 20C.1, this means that the purported endpoint can be stated only by leaving admissible representability while still claiming the same torus carrier and realized lineage.

That completion is impossible for three independent theorem-local reasons. First, Theorems 20B.1 and 20B.2, packaged for the standing/intelligibility route in Theorem ST4, rule out any same-scope reading in which the already-fixed realized lineage is standing-positive before the horizon but acquires a fresh neutral or non-admissible status only at the endpoint. Second, Proposition 20B.3, packaged operationally in Proposition ST5, rules out any admissible same-scope rollback, erasure, or

de-commitment of the standing-bearing representational commitment that anchors the lineage, so the endpoint cannot shed that anchor and still count as the same realized object. Third, Theorem 20B together with Proposition 13.23A.4 rules out any independent classical-layer continuation criterion or metric-only gate that could repackage a standing-negative endpoint as a licit same-scope completion. Therefore terminal standing failure cannot complete the same-scope endpoint of a standing-positive realized lineage. $\square$

Theorem (20C (Same-scope realized-lineage status trichotomy and collapse)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ and let $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ have maximal interval $[0, T_*(x))$ with $T_*(x) < \infty$. Then the finite-horizon endpoint condition induced by Definition 20C.0 has exactly one of the following theorem-local statuses (equivalently, any attempted same-scope description of that condition must fall into one of these statuses):*

- (i) *it is represented on the fixed torus carrier by a standing-positive admissible in-scope witness, namely the carrier-side horizon discriminator supplied by Theorem 20.7A together with the equivalent carrier-side witness bookkeeping built from it;*
- (ii) *it is terminal standing failure; or*
- (iii) *it is an illicit same-scope gate / scope move, meaning that it keeps the endpoint alive only by path-dependent reclassification, representative-only or metric-only gate authority, a new selector / tolerance / discriminator, or an undeclared scope change.*

Moreover, clause (ii) cannot complete the same-scope endpoint by Proposition 20C.2, and clause (iii) is not a licit same-scope endpoint status at all. Consequently, among licit same-scope endpoint statuses only the standing-positive witness branch of clause (i) remains.

Proof. If $T_*(x) < \infty$, Lemma 19F already gives an in-scope pre-horizon continuation witness on the smooth realized lineage itself. By Corollary 20A, any such finite-time breakdown is carrier-visible on the fixed torus carrier. By Theorem 20.7A, the first non-extendable horizon determines the carrier-side discriminator $\mathfrak{D}_{T_*(x)}(x) = 0$, and this discriminator is standing-relevant on the fixed torus carrier. If the continuation failure is presented through that carrier-side standing-relevant determination, then it lies in the standing-positive witness branch of clause (i).

Assume instead that the finite-horizon endpoint condition is not presented through a standing-positive admissible in-scope witness. Because Theorem 20B.1 fixes the realized evolution as one standing-bearing same-scope determination and Theorem 20B.2 rules out time-indexed admissibility change on that determination, the endpoint condition cannot survive as a fresh stage-indexed status. There are then only two remaining ways to keep the endpoint claim alive while still pretending to stay in the same scope.

First, the condition may be described as completing the same realized lineage only by leaving admissible representability while preserving the same torus carrier and lineage identity. Proposition 20C.1 identifies that case exactly with terminal standing failure, giving clause (ii).

Second, the condition may be kept alive only by smuggling in a same-scope discriminator. That is exactly what the factorization, history, role, metric, and classical-layer packets now forbid theorem-locally: Theorem 3.35 rules out standing-relevant functionals that fail to factor through the fixed quotient, Theorem 3.37 together with Corollary 3.42 rules out path-, order-, or history-representative-dependent admissibility, Corollary 3.40 rules out representative-only skin as standing authority, Proposition 13.23A.3 rules out raw metric divergence as an independent same-scope gate, Proposition 13.23A.4 rules out a third metric-gate status, and Theorem 20B rules out any surviving

classical-layer continuation discriminator. Any such use is therefore clause (iii): an illicit same-scope gate / scope move rather than a licit same-scope endpoint status.

The three clauses are exhaustive for finite-horizon endpoint conditions because carrier-to-lineage binding and no-admissibility-dynamics remove time-indexed reclassification, Theorem ST3 together with Corollary 13.9 remove any neutral same-scope middle status, and the factorization / role / metric packet removes any fourth same-scope standing-relevant proposal form. Finally, Proposition 20C.2 excludes clause (ii) as a licit same-scope completion, while clause (iii) is already non-licit by the just-cited factorization, role, metric, and classical-layer theorems. Hence only clause (i) remains as a licit same-scope endpoint status. $\square$

Corollary (20C.3 (No third same-scope endpoint status; collapse of the “outside admissible representability” objection)). *For a standing-positive admitted carrier state, no third licit same-scope endpoint status exists beyond the standing-positive admissible witness branch and the already-excluded terminal standing-failure branch. Any purported same-scope endpoint condition not in those two branches is an illicit same-scope gate / scope move rather than a same-scope endpoint status. In particular, the objection that finite-time breakdown might exist only “outside admissible representability” collapses into the already excluded non-witness branches and does not remain as a neutral remainder.*

Proof. Immediate from Theorem 20C together with Proposition 20C.2. $\square$

Theorem (20C.4 (Counterexample soundness for fixed same-scope endpoints)). *In a non-degenerate same-scope theorem-bearing proof/refutation regime, a counterexample to a universal continuation theorem must supply a licit same-scope endpoint condition contradicting the asserted continuation. Applied to Fefferman’s official periodic Navier–Stokes statement (B), this requirement is not an extra AASC filter added to the official problem and does not redefine Fefferman counterexamplehood: it states the role conditions already required for an ambient mathematical object to function as a counterexample to Fefferman (B). Theorem 3.K3 and Theorem 3.K5 make that role-equivalence explicit. Therefore, if every finite-horizon endpoint condition is classified as terminal standing failure, an illicit same-scope gate / scope move, or a standing-positive witness branch eliminated by Section 21, then no same-scope counterexample to Fefferman (B) exists.*

Proof. We separate ordinary universal-theorem logic, ambient proof-system standing, and fixed-scope endpoint status.

1. Fefferman (B), in the periodic form restated in Theorem 21.17, is universal over the fixed official datum class of Definition 18.1. A counterexample must therefore start from an official datum and preserve the same equations, carrier, viscosity, forcing status, gauge, and smooth periodic continuation question.
2. A counterexample in ordinary mathematics is not merely the object-level phrase $T_* < \infty$. It is an ambient proof/refutation construction showing that an object satisfying the official hypotheses also satisfies the negation of the asserted conclusion. By Lemma 3.K4, such counterexamplehood requires admissible object formation, proof-standing, fixed reference, same-target fidelity, and final theorem consequence.
3. The appeal to ZFC, classical analysis, or any other ambient proof system does not bypass those roles. By Theorem 3.K3 and Corollary 3.K6, any ambient proof system capable of establishing a theorem-bearing counterexample instantiates the fixed-scope kernel roles at the level of proof/refutation use.

4. If a purported counterexample changes any official commitment, it is outside the fixed-domain comparison class of Definition 3.4A; it may describe a different problem, but it is not a counterexample to the theorem stated here.
5. If a purported counterexample preserves those commitments, then the only way it can contradict the universal global-continuation conclusion is by finite maximality on the same realized lineage: $T_*(x) < \infty$ for the carrier state associated to the datum. By Theorem 20C.0A, that finite maximality induces a finite-horizon endpoint condition for the same carrier, same lineage, and same continuation question.
6. Because the official periodic proof/refutation regime is non-degenerate by Theorem 12A.0, because ambient Fefferman counterexamplehood is subsumed by the kernel by Theorem 3.K5, and because the fixed-scope kernel applies by Corollary 12A.0.1, a theorem-bearing same-scope counterexample must complete through a licit same-scope endpoint status. A standing-negative object, terminal standing failure, hidden selector, metric-only gate, or undeclared scope move cannot refute the universal theorem while also preserving the declared comparison class.
7. Theorem 20C exhausts the endpoint-status options for that endpoint condition. Proposition 20C.2 and Corollary 20C.3 collapse the non-witness branches, and Section 21 eliminates the remaining standing-positive witness branch.

Thus a same-scope counterexample would have to supply a licit same-scope endpoint condition, but no such condition survives. Therefore no same-scope counterexample to Fefferman (B) exists. $\square$

21 Transfer Steps VI–VII (impossibility and endpoint): no singular image and the official periodic theorem

Burden status. Every earlier burden has already been narrowed to one final local task. By entry to this section, official data admission, realized-class completeness, no-new-criterion, horizon detectability, and the realized-lineage status trichotomy/collapse are all settled theorem-locally. What remains is to act on the exhaustive standing-positive same-scope witness class, eliminate every standing-positive singular image branch, and compress that contradiction into the official periodic theorem. The non-witness proposals have already been collapsed: by Theorem 20C, any purported same-scope finite-horizon endpoint is either a standing-positive in-scope witness, terminal standing failure, or an illicit same-scope gate / scope move; Proposition 20C.2 excludes terminal standing failure as a licit same-scope completion, and Corollary 20C.3 records that no third licit same-scope endpoint status remains. Part II eliminated every standing-positive same-scope singular candidate on the carrier already identified with the unique admissible interior, and §20 showed that the realization map introduces no new standing-positive same-scope continuation criterion on the fixed torus carrier. The metric/tensor/witness localization packet makes the role split theorem-local for the actual Navier–Stokes observables used here: Proposition 13.23A.2 localizes the anchor / tensor / skin roles, Proposition 13.23A.3 states that raw metric divergence has no same-scope standing authority by itself, and Proposition 13.23A.4 says that any standing-bearing metric use must already factor through the fixed horizon witness or else becomes an illicit extra gate. In particular, Theorem 20B rules out any surviving standing-positive same-scope classical-layer continuation discriminator, and Theorem 20.11 records the explicit Θ_x -based instance built from the evolution of the vortex-stretching observable. Here the vortex-stretching wall does only one thing: it marks the incompleteness of same-scope PDE analytic closure. It does not reopen primitive admissibility, it does not create a second same-scope interior, and it does not supply a same-scope continuation

gate. By Lemma 19E, every smooth interior time slice of a realized trajectory re-enters that unique admissible interior and remains standing-positive; thus the breakdown witness below is built entirely from carrier states already inside the unique admissible interior up to the first non-extendable horizon. Part III now upgrades that implication from a factorization statement to an explicit construction on the standing-positive branch: finite-time breakdown yields a concrete breakdown witness, a horizon continuation discriminator that is already standing-detectable on the fixed carrier, and a singular-candidate extractor whose output lands in the typed singular-candidate class already eliminated in Part II. Proposition 21.6A below makes the local completeness burden explicit for the standing-positive branch actually used later: no standing-positive same-scope continuation-relevant witness remains outside the extracted carrier-side horizon discriminator and its induced branch data. The extractor and the branch-forcing theorem below are therefore bookkeeping devices for how a classical breakdown would have to try to leave the unique admissible interior; they are not the primitive source of uniqueness itself. This section carries out that construction and derives the official periodic theorem. The branch-forcing theorem below is organized in three explicit stages: first the uniqueness-first contradiction is recalled, then the quotient-visible branch is ruled out for standing-positive breakdown representations, and finally the fixed-scope branch-forcing lemma from Section 15 extracts the nonempty set of forbidden same-scope witness moves that any such representation would have to instantiate.

Definition 21.1 (Classical singular image). A *classical singular image* is a realized lineage $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ with maximal local existence interval $[0, T_*(x))$ such that $T_*(x) < \infty$ and the realized trajectory admits no classical continuation on the same torus carrier beyond $T_*(x)$.

Definition 21.2 (Breakdown witness). Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ with $T_*(x) < \infty$. The *breakdown witness* of x is

$$\mathfrak{W}(x) := (x, T_*(x), \mathfrak{D}_{T_*(x)}, \mathcal{R}_{\text{NS}}(x)|_{[0, T_*(x))}).$$

It records the carrier class, the first non-extendable horizon, the associated continuation discriminator, and the realized trajectory up to that horizon.

Definition 21.3 (Branch-forcing set of a breakdown witness). Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ with $T_*(x) < \infty$ and breakdown witness $\mathfrak{W}(x)$. Define the *branch-forcing set*

$$\mathfrak{B}(x) \subseteq \{S1, S2, S3, S4, S5\}$$

by the following membership conditions:

- (S1) $S1 \in \mathfrak{B}(x)$ iff any same-scope standing-positive representation of $\mathfrak{W}(x)$ requires a local weakening, exception, or relaxation of at least one clause of the bundle B_{NS} at or before the horizon $T_*(x)$;
- (S2) $S2 \in \mathfrak{B}(x)$ iff any such representation requires isolating one interior compatibility, diagnostic, or subbundle from the coupled bundle B_{NS} and letting it govern continuation independently of the rest;
- (S3) $S3 \in \mathfrak{B}(x)$ iff any such representation requires a threshold, tolerance, selector, staged repair, or horizon rule not already licensed, including any continuation gate built from the evolution of the trajectory stretching observable Θ_x ;
- (S4) $S4 \in \mathfrak{B}(x)$ iff any such representation requires a non-equivalent redescription, presentation-dependent masking, or skin-dependent reformulation of the realized trajectory in order to classify breakdown as standing-positive;

(S5) $S5 \in \mathfrak{B}(x)$ iff any such representation requires downstream continuation, restart data, or later compositional information to legalize an earlier failure at the horizon $T_*(x)$.

Theorem 21.4 (Constructive diagnostic branch forcing from breakdown witnesses). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ with $T_*(x) < \infty$, and let $\mathfrak{W}(x)$ be its breakdown witness. If $\mathfrak{W}(x)$ is to be represented as standing-positive on the fixed torus carrier, then:*

- (i) *its branch tag $\beta(x)$ is illicit rather than quotient-visible;*
- (ii) *its branch-forcing set $\mathfrak{B}(x)$ is nonempty;*
- (iii) *every same-scope standing-positive representation of $\mathfrak{W}(x)$ instantiates at least one branch in $\mathfrak{B}(x)$.*

Equivalently: once uniqueness has ruled out a second interior and any admissible singular variation, a classical finite-time breakdown can only appear as a same-scope attempted escape that forces at least one specific diagnostic witness move among (S1)–(S5) on the carrier.

Proof. Stage 0: uniqueness-first obstruction. Because x is already a standing-positive state of the unique admissible interior, Theorem 14.1 has already fixed the deepest contradiction: a same-scope singularity cannot live in a second same-scope interior, cannot survive as an admissible variation inside the unique interior, and cannot be rescued by a disguised scope change. The only remaining question is therefore diagnostic: how would an impossible same-scope escape attempt have to present itself on the fixed carrier?

Stage 1: the quotient-visible branch collapses immediately. By Lemma 21.5, the carrier-side continuation discriminator attached to $\mathfrak{W}(x)$ has a unique branch tag

$$\beta(x) \in \{\text{illicit}, \text{quotient-visible}\}.$$

If $\beta(x) = \text{quotient-visible}$, then the corresponding singular candidate is standing-negative by Lemma 16.3. Hence a standing-positive same-scope representation of $\mathfrak{W}(x)$ is impossible in that branch. Therefore any standing-positive representation of the breakdown witness is forced into the illicit branch:

$$\beta(x) = \text{illicit}.$$

This is already subordinate to the uniqueness-first obstruction: the quotient-visible branch cannot host a standing-positive representation of finite-time breakdown on the fixed carrier.

Stage 2: constructive extraction of the diagnostic witness moves. By Lemma 19E, every smooth slice of the realized trajectory prior to $T_*(x)$ determines a standing-positive carrier seed in the unique admissible interior. By Theorem 20.7A, finite-time failure at the first non-extendable horizon already yields a well-typed standing-relevant continuation discriminator on the fixed torus carrier. By Theorem 20.11, no additional continuation content can be created by the evolution of Θ_x . Thus the breakdown witness is standing-positive on the fixed carrier precisely because horizon failure is already standing-detectable there, with no licensed critical bridge added beyond the commitments already frozen in Part II.

The abstract branch-forcing result now applies directly on the live carrier-certified route: by Lemma 15.6, together with the carrier-certified Structural Exhaustion packet of Theorem 13.28 and Corollary 13.29, any attempt to treat a classical blow-up episode as standing-positive under the fixed carrier commitments and in the absence of a licensed critical bridge must instantiate at least one of (S1)–(S5). Applied to $\mathfrak{W}(x)$, this yields only diagnostic content: the breakdown witness

forces a nonempty branch set of forbidden same-scope moves that witness how the impossible escape attempt would have to appear.

Therefore $\mathfrak{B}(x) \neq \emptyset$, and every standing-positive representation of $\mathfrak{W}(x)$ instantiates at least one branch in $\mathfrak{B}(x)$. $\square$

Lemma 21.5 (Branch assignment for breakdown witnesses). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ with $T_*(x) < \infty$. Then the continuation discriminator $\mathfrak{D}_{T_*(x)}$ determined by the breakdown witness $\mathfrak{W}(x)$ has a unique branch tag*

$$\beta(x) \in \{\text{illicit}, \text{quotient-visible}\}$$

in the dichotomy of Proposition 15.7.

Proof. By Theorem 20.7A, the failure of continuation at $T_*(x)$ determines the horizon discriminator $\mathfrak{D}_{T_*(x)}$ as a well-typed standing-relevant continuation discriminator on the fixed carrier. Proposition 13.23A.2 and Proposition 13.23A.3 make explicit that no raw metric mark can compete with that role: any standing-bearing use of such a mark must already factor through the fixed horizon witness, while Proposition 13.23A.4 rules out a third same-scope metric-gate status. Theorem 20.5 then shows that realization introduces no additional continuation criterion, while Theorem 20.11 rules out any additional continuation content built from Θ_x . Proposition 15.7 and Corollary 15.8 then force a unique branch tag: illicit or quotient-visible, with no third possibility. $\square$

Definition 21.6 (Singular-candidate extractor). For $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ with $T_*(x) < \infty$, define the *singular-candidate extractor*

$$\mathfrak{X}(x) := (x, T_*(x), \mathfrak{D}_{T_*(x)}, \beta(x), \mathfrak{B}(x)),$$

where $\beta(x)$ is the branch tag from Lemma 21.5 and $\mathfrak{B}(x)$ is the branch-forcing set of Definition 21.3.

Proposition (21.6A (Standing-positive same-scope witness exhaustion for finite-time breakdown)). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ with $T_*(x) < \infty$. On the fixed torus carrier, every standing-positive same-scope continuation-relevant witness of finite-time breakdown is exhausted by the carrier-side horizon discriminator $\mathfrak{D}_{T_*(x)}$ together with the induced branch data $(\beta(x), \mathfrak{B}(x))$. More precisely, once Table 3, Corollary 13.23A, Theorem 13.28, Corollary 13.29, Proposition 13K, Corollary 20A, Theorem 20.7A, Proposition 13.23A.2, Proposition 13.23A.3, Proposition 13.23A.4, Theorem 20B, and Theorem 20.11 are in force, no additional standing-positive same-scope continuation witness remains outside this extracted carrier-side representation. Combined with Theorem 20C, this leaves only one other same-scope branch — terminal standing failure — and that branch is already excluded by Proposition 20C.2.*

Proof. Theorem 20C shows that any same-scope finite-horizon continuation failure has only two statuses: a standing-positive carrier witness or terminal standing failure. Proposition 20C.2 removes the terminal standing-failure branch for realized lineages issued from standing-positive admitted carrier states. It therefore suffices here to exhaust the standing-positive witness branch.

Table 3 and Corollary 13.23A record the carrier-side translation fact used on the live route: every continuation-relevant classical Navier–Stokes observable appearing there as standing-positive same-scope witness content already has a carrier-side representative. The carrier-certified Structural Exhaustion packet of Theorem 13.28 and Corollary 13.29, together with Proposition 13K, then remove same-scope residue and any additional PDE-side continuation criterion beyond the certified carrier structure. At the classical layer, Corollary 20A and Theorem 20.7A say that a finite-time breakdown is already the standing-relevant carrier-side horizon discriminator on the fixed

torus carrier, while Theorem 20B and Theorem 20.11 eliminate any additional classical-layer or trajectory-level continuation gate.

It therefore remains only to record the diagnostic witness data forced by a standing-positive representation of that already-extracted horizon event. Lemma 21.5 assigns the unique branch tag $\beta(x)$, and Theorem 21.4 forces the nonempty branch set $\mathfrak{B}(x)$ for every same-scope standing-positive representation of the breakdown witness. Hence every standing-positive same-scope continuation-relevant witness of finite-time breakdown is exhausted by $\mathfrak{D}_{T_*(x)}$ together with the induced branch data $(\beta(x), \mathfrak{B}(x))$, and no standing-positive same-scope continuation-relevant residue exists outside this representation. Together with the already-excluded terminal standing-failure branch, this closes the same-scope endpoint burden locally. $\square$

Lemma 21.7 (Constructive singular-candidate extraction). *If a realized lineage admits a classical singular image in the sense of Definition 21.1, then its singular-candidate extractor $\mathfrak{X}(x)$ is a typed singular candidate in the sense of Definition 15.4. More precisely, the extracted singularity gate is tagged by $\beta(x)$ as either illicit or quotient-visible in the sense of Proposition 15.7, and whenever the breakdown is represented as standing-positive on the fixed carrier the extractor carries a nonempty forced branch set $\mathfrak{B}(x) \subseteq \{S1, S2, S3, S4, S5\}$ in the sense of Theorem 21.4. In the realized-lineage endpoint-status trichotomy / collapse of Theorem 20C, this extractor therefore captures the entire standing-positive continuation content of the breakdown; the complementary proposal branches are terminal standing failure and illicit same-scope gate / scope moves, already collapsed by Proposition 20C.2 and Corollary 20C.3.*

Proof. Let x be the carrier class underlying the classical singular image. By Definition 21.2, finite-time breakdown produces the concrete witness $\mathfrak{W}(x)$ carrying the first non-extendable horizon and the carrier-side continuation discriminator $\mathfrak{D}_{T_*(x)}$. Proposition 21.6A, using Table 3, Corollary 13.23A, Theorem 13.28, Corollary 13.29, Proposition 13K, Corollary 20A, Theorem 20.7A, Theorem 20B, and Theorem 20.11, already makes the completeness burden local: every standing-positive same-scope continuation-relevant witness of this breakdown is exhausted by the carrier-side horizon discriminator together with its induced branch data. Thus the extractor records the entire standing-positive continuation content of the breakdown witness on the fixed carrier. By Theorem 20C and Proposition 20C.2, there is no third same-scope remainder besides this extracted standing-positive content and the already-excluded terminal standing-failure branch.

Lemma 21.5 assigns to that discriminator the unique branch tag $\beta(x)$. Because a classical singular image is the same-scope breakdown of a realized lineage issued from a standing-positive admitted carrier state, and because Lemma 19E shows that every smooth prehorizon slice of that lineage re-enters the unique admissible interior and remains standing-positive, the breakdown witness is a standing-positive representation on the fixed carrier throughout the realized interval. By Theorem 20.7A, this standing-positivity is triggered specifically by horizon failure itself: the horizon discriminator $\mathfrak{D}_{T_*(x)}$ is already a well-typed standing-relevant continuation discriminator on the fixed torus carrier. Therefore Theorem 21.4 applies in its overt two-stage form: first the quotient-visible branch collapses, forcing $\beta(x) = \text{illicit}$, and then the abstract Part I branch-forcing result yields a nonempty forced branch set $\mathfrak{B}(x)$, with every standing-positive representation of $\mathfrak{W}(x)$ instantiating at least one branch in $\mathfrak{B}(x) \subseteq \{S1, S2, S3, S4, S5\}$.

Definition 21.6 packages these data into $\mathfrak{X}(x)$. This extraction is a standing-positive admissibility act on the fixed carrier, not a merely representational re-description: prehorizon standing-positivity is preserved along the realized lineage, horizon failure itself is carrier-visible, and no additional same-scope continuation gate survives beyond the extracted discriminator. Since $\mathfrak{D}_{T_*(x)}$ is a same-scope continuation discriminator on the fixed carrier, $\beta(x)$ records its illicit/quotient-visible status, and

$\mathfrak{B}(x)$ records the explicit forced (S1)–(S5) content required for standing-positive representation, the extracted tuple has exactly the form of a typed singular candidate from Definition 15.4. Hence $\mathfrak{X}(x)$ is a typed singular candidate. $\square$

Proposition 21.8 (No illicit singular image). *No realized lineage admits a classical singular image whose induced singularity gate is illicit.*

Proof. By Lemma 21.7, an illicit singular image would produce a singular-candidate extractor $\mathfrak{X}(x)$ tagged as illicit. Because the breakdown is represented as standing-positive on the fixed carrier, Theorem 21.4 yields a nonempty forced branch set $\mathfrak{B}(x) \subseteq \{S1, S2, S3, S4, S5\}$; every standing-positive representation of the breakdown witness must instantiate at least one branch in $\mathfrak{B}(x)$. But Lemma 16.2 eliminates precisely those same-scope branches on the certified carrier. Hence no illicit singular image exists. $\square$

Proposition 21.9 (No quotient-visible singular image). *No realized lineage admits a classical singular image whose induced singularity gate is quotient-visible.*

Proof. By Lemma 21.7, a quotient-visible singular image would produce a singular-candidate extractor $\mathfrak{X}(x)$ tagged as quotient-visible. But because a classical singular image is the same-scope breakdown of a realized lineage issued from a standing-positive admitted carrier state, Theorem 21.4 forces the standing-positive branch tag to be illicit rather than quotient-visible. This already contradicts the assumed quotient-visible tag. Equivalently, Lemma 16.3 shows that every quotient-visible candidate is standing-negative, so the quotient-visible branch cannot occur as a classical singular image. $\square$

At this point the branch-elimination machinery acts on the exhaustive standing-positive same-scope witness class addressed locally in Proposition 21.6A: by that proposition every standing-positive same-scope continuation-relevant breakdown witness has already been reduced to the carrier-side data $(\mathfrak{D}_{T_*(x)}, \beta(x), \mathfrak{B}(x))$, and by Lemma 21.7 that data is precisely a typed singular candidate on the fixed carrier. The impossibility theorem below therefore eliminates the entire standing-positive same-scope witness class actually used on the live endgame, not a chosen encoding of it.

Theorem 21.10 (Transfer impossibility: no singular image of the unique admissible interior). *No standing-positive admitted time-zero carrier state in $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ has a classical singular image under the realization map $\mathcal{R}_{\text{NS}}$.*

Proof. Suppose $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ had a classical singular image. By Theorem 20C.0A, this finite maximality induces a same-scope finite-horizon endpoint condition on the fixed periodic carrier. Theorem 20C then forces that endpoint condition into the realized-lineage status trichotomy of clauses (i)–(iii).

Terminal standing-failure branch. If the alleged endpoint survives only as terminal standing failure, Proposition 20C.2 excludes it immediately as a licit same-scope completion of the realized lineage.

Illicit scope / gate branch. If the alleged endpoint survives only by path-dependent reclassification, representative-only or metric-only gate authority, an extra selector / tolerance, or an undeclared scope move, then Theorem 20C together with Corollary 20C.3 already says that this is not a licit same-scope endpoint status at all. So the singular image cannot survive on that branch either.

Standing-positive witness branch. The only licit same-scope endpoint status left is the standing-positive witness branch of clause (i). Proposition 21.6A makes that branch theorem-local: the entire standing-positive same-scope continuation content of the breakdown is exhausted by the carrier-side horizon discriminator together with the induced branch data $(\beta(x), \mathfrak{B}(x))$. Lemma 21.7

then converts exactly that exhausted standing-positive content into a typed singular candidate extractor $\mathfrak{X}(x)$ whose singularity gate is either illicit or quotient-visible and whose forced branch set records the explicit witness moves required for standing-positive representation.

At that point the fixed-carrier contradictions are all local. Theorem 14.1 blocks any attempt to reinterpret the candidate as a second same-scope interior, an admissible variation inside the unique interior, or a disguised scope change. Theorem 20.7A grounds the standing-positive witness narrowly by the horizon discriminator itself, and Theorem 20.11 eliminates any additional Θ_x -based continuation gate, so no broader continuation-completeness claim is needed. Proposition 21.8 eliminates the illicit singularity gate and Proposition 21.9 eliminates the quotient-visible singularity gate. By Corollary 15.8, no third standing-positive same-scope singularity branch exists on the fixed carrier.

All theorem-local proposal branches are therefore closed before the endpoint is compressed: terminal standing failure is excluded by Proposition 20C.2, illicit scope / gate completion is excluded by Theorem 20C and Corollary 20C.3, and the remaining standing-positive witness branch is eliminated by exhaustion, extraction, and branch contradiction. Hence no classical singular image can arise from a standing-positive admitted state of the unique admissible interior. $\square$

Corollary 21.11 (Transfer endpoint on the realized class: global smoothness). *Let $x \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ be a standing-positive admitted time-zero carrier state. Then its realized classical Navier–Stokes trajectory $\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$ extends smoothly for all time; equivalently, $T_*(x) = \infty$ in Theorem 19.6.*

Proof. By Theorem 19.6, x admits a unique maximal local classical realization on $[0, T_*(x))$ with $T_*(x) > 0$. If $T_*(x) < \infty$, then this realized trajectory would be a classical singular image in the sense of Definition 21.1, contradicting Theorem 21.10. Hence $T_*(x) = \infty$. $\square$

Proposition 21.12 (Full-class reduction of the official periodic theorem). *Assume that no standing-positive admitted time-zero carrier state in $\mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$ has a classical singular image. Then the official periodic global smoothness theorem holds for the full Fefferman (B) data class $\mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$.*

Proof. Let $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$. By Corollary 18.12, u_0 determines a standing-positive admitted time-zero carrier state $x = [\Sigma(u_0)] \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$. By Theorem 19.9, the official local classical solution issued from u_0 is exactly the realized trajectory $\mathcal{R}_{\text{NS}}(x)$. If that realized trajectory had finite maximal time, then Theorem 20C would place the alleged endpoint into the realized-lineage status trichotomy. The terminal standing-failure branch is excluded by Proposition 20C.2, and any illicit scope / gate proposal is already non-licit by Corollary 20C.3. The only licit same-scope endpoint status left would therefore be the standing-positive witness branch, which would make the realized trajectory a classical singular image of the admitted carrier state x , contradicting the hypothesis of the proposition. Hence the realized trajectory continues for all time. Since u_0 was arbitrary, the official periodic theorem follows for the full Fefferman (B) class with no hidden narrowing. $\square$

Proposition 21.13 (Compressed dependency chain for the official periodic theorem). *For each datum $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$, the proof of Theorem 21.17 consists of exactly the following steps:*

- (i) *Theorem 13I, Corollary 13J, and Proposition 13K identify the standard carrier with the unique admissible interior and state explicitly that no rival standing-positive same-scope continuation regime or extra PDE-side continuation criterion survives inside fixed scope.*
- (ii) *Lemma 18.14 records that the structural non-singularity theorem is sufficient for the official periodic theorem once Carrier Universality, realization/full-class identification, and faithful import of branch elimination are supplied.*

- (iii) Corollary 18.12 identifies u_0 with a unique standing-positive admitted time-zero carrier class $x = [\Sigma(u_0)] \in \mathcal{I}_{\text{car}}(\mathbb{T}^3, \nu)$.
- (iv) Theorem 19.6 constructs the unique maximal local classical trajectory $\mathcal{R}_{\text{NS}}(x)$, and Theorem 19.9 identifies that realized trajectory with the full official local classical solution issued from u_0 .
- (v) Theorem 20.7A shows that finite-time breakdown for that realized trajectory already appears as the standing-relevant carrier-side horizon discriminator, and Theorem 20.11 shows that no formal Θ_x -based continuation gate creates an additional continuation criterion.
- (vi) Theorem 20C, together with Propositions 20C.1 and 20C.2 and Corollary 20C.3, turns any alleged same-scope finite-horizon endpoint into the realized-lineage status trichotomy and collapses the non-witness branches locally.
- (vii) Proposition 21.6A and Lemma 21.7, with Theorem 21.4 grounded in the carrier-certified Structural Exhaustion packet of Theorem 13.28 and Corollary 13.29, convert the entire standing-positive continuation content of the remaining licit branch into an explicit typed singular candidate extractor $\mathfrak{X}(x)$ carrying a nonempty forced branch set $\mathfrak{B}(x) \subseteq \{S1, S2, S3, S4, S5\}$.
- (viii) Propositions 21.8 and 21.9, together with Corollary 15.8 and the uniqueness-first obstruction of Theorem 14.1, eliminate all standing-positive same-scope branches of such a candidate.
- (ix) Therefore Theorem 21.10 excludes finite-time singularity for the realized trajectory, and Proposition 21.12 upgrades that exclusion to the full Fefferman (B) data class.

No additional theorem enters the proof of the official periodic statement.

Proof. Item (i) is the strengthened carrier-identification packet 13I–13K: before the bridge begins, the manuscript states explicitly that the standard carrier is the unique admissible interior and that no rival standing-positive same-scope regime or hidden PDE-side continuation criterion survives inside fixed scope. Item (ii) is the theorem-facing transport claim of Proposition 18A together with the sufficiency statement of Lemma 18.14. Items (iii)–(iv) are the carrier-universality and faithful realization steps of Part III: local classical realization exists, is overlap-faithful, and its realized local class is exactly the full official local classical class. Item (v) is the narrowed classical-layer closure hinge: Corollary 20A and Theorem 20.7A show that finite-time breakdown is already the standing-relevant carrier-side horizon discriminator on the fixed torus carrier, while Theorem 20B and Theorem 20.11 show that no formally proposed PDE-side gate survives as an additional same-scope continuation criterion. Item (vi) is the theorem-local status trichotomy/collapse required by the endpoint-status audit: an alleged endpoint is either a standing-positive carrier witness, terminal standing failure, or an illicit same-scope gate / scope move, and only the standing-positive witness branch remains as a licit same-scope endpoint status after Theorem 20C. Item (vii) is the constructive diagnostic branch-forcing step together with the singular-candidate extractor: because the non-witness branches have already been collapsed, the remaining standing-positive breakdown witness is converted into an explicit typed carrier object carrying a nonempty forced branch set of forbidden same-scope witness moves. Item (viii) is the branch-elimination package already established on the carrier identified with the unique admissible interior in Part II, with Theorem 14.1 blocking any attempt to reinterpret the extracted candidate as a second same-scope interior, an admissible singular variation, or a disguised scope change. Item (ix) is the passage from no singular image to the explicit transport corollary and then to the full official periodic theorem. No diffuse cross-section load remains once these nine steps are recorded explicitly. $\square$

Remark 21.14 (Detailed audit versus short visible proof spine). Proposition 21.13 is the detailed theorem-facing audit for Part III. Theorem 21.15 below is the short theorem-local transfer cascade from which the endpoint is taken as a local corollary.

Theorem 21.15 (Primitive-to-classical transfer cascade for the official periodic theorem). *For the fixed torus carrier $\mathbb{T}^3$ and viscosity $\nu > 0$, the official periodic theorem is a local consequence of the following theorem-local chain:*

- (i) **Transfer soundness:** Corollary 18.12.
- (ii) **Transfer completeness:** Theorem 19B.
- (iii) **Transfer no-new-criterion:** Theorem 20.5B together with Theorem 20B.
- (iv) **Transfer detectability:** Corollary 20A and Theorem 20.7A.
- (v) **Transfer endpoint-status trichotomy/collapse:** Propositions 20C.1 and 20C.2, Theorem 20C, and Corollary 20C.3.
- (vi) **Transfer impossibility on the remaining licit branch:** Theorem 21.10.

Once these six steps are in the theorem-local chain, Proposition 21.12 yields the official periodic theorem for the full Fefferman (B) class.

Proof. Let $u_0 \in \mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$ and let $x = [\Sigma(u_0)]$ be the admitted carrier class from Corollary 18.12. By Theorem 19B, the unique official local classical trajectory issued from u_0 is exactly the realized trajectory $\mathcal{R}_{\text{NS}}(x)$. Theorem 20.5B and Theorem 20B state that realization introduces no new standing-positive same-scope continuation content beyond the certified carrier class on the live route. Therefore if that official local classical trajectory had finite maximal time, Theorem 20C.0A, Corollary 20A, and Theorem 20.7A would already turn the resulting finite maximality into a standing-relevant carrier-side endpoint condition / horizon event. Theorem 20C then places that alleged endpoint into the realized-lineage status trichotomy; Proposition 20C.2 excludes the terminal standing-failure branch, and Corollary 20C.3 excludes any illicit scope / gate proposal as a licit same-scope status. Theorem 21.10 excludes the remaining standing-positive witness branch. Proposition 21.12 then upgrades this theorem-local collapse to the full Fefferman (B) class. Hence the official periodic theorem follows for arbitrary u_0 . $\square$

Corollary (21A (Transfer endpoint corollary: official periodic global smoothness as a corollary of structural necessity)). *For the full official datum class $\mathcal{D}_{\text{off}}(\mathbb{T}^3, \nu)$, the periodic case of the incompressible Navier–Stokes problem on $\mathbb{T}^3$ holds globally and smoothly. Equivalently, no official periodic datum generates a finite-time singular evolution at the declared fixed scope.*

Proof. Immediate from Theorem 21.15: the transfer soundness, completeness, no-new-criterion, detectability, realized-lineage status trichotomy/collapse, and standing-positive impossibility steps have all been proved locally in this manuscript, so the official periodic conclusion follows as a corollary of structural necessity. $\square$

Remark 21.16 (Endpoint numbering convention). Corollary 21A is the claim-type endpoint of the manuscript: at that point the official periodic conclusion has already been obtained as a corollary of structural necessity. Theorem 21.17 is retained immediately afterward only to restate the same conclusion in the official classical PDE wording of Fefferman (B).

Theorem 21.17 (Official periodic theorem on $\mathbb{T}^3$ (Fefferman (B); official classical restatement of Corollary 21A)). *Let $\nu > 0$ and let $u_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$ satisfy $\nabla \cdot u_0 = 0$. Then there exists a unique smooth periodic Navier–Stokes solution pair (u, p) on $\mathbb{T}^3 \times [0, \infty)$ satisfying*

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad u|_{t=0} = u_0,$$

with the fixed mean-zero pressure gauge.

Remark 21.18 (Classical statement, structural claim type). The theorem above is theorem-numbered because Fefferman’s official periodic endpoint is theorem-shaped and classical in statement. Its claim type is already fixed by Corollary 21A: unique admissible interior first, carrier identification second, local classical realization third, carrier-visible breakdown fourth, same-scope endpoint-status trichotomy and collapse fifth, and the remaining witness-branch contradiction sixth.

Five-link proof spine for the official periodic theorem

- (1) **Unique interior.** The fixed periodic scope admits the *unique admissible interior* by Corollary 12H.
- (2) **Carrier identification.** Theorem 13I identifies the standard periodic Navier–Stokes carrier with that unique admissible interior, Corollary 13J rules out any rival same-scope continuation regime, Proposition 13K states explicitly that no extra PDE-side continuation criterion survives inside fixed scope, and Corollary 18.12 says that every official datum enters the identified carrier.
- (3) **Local classical realization / transport only.** Proposition 18A, Theorems 19.6 and 19.9, Lemma 19A, and Theorem 19B realize that admitted carrier state as exactly the official local classical trajectory and make explicit that Part III contributes only realization transport.
- (4) **Carrier-visible breakdown plus endpoint-status trichotomy/collapse.** Theorem 20.5B, Corollary 20A, Theorem 20.7A, Theorem 20B, Theorem 20C, Proposition 20C.2, and Corollary 20C.3 show that any hypothetical finite-time breakdown of that realized trajectory is already carrier-visible and then placed into the realized-lineage status trichotomy, with the non-witness branches collapsed locally.
- (5) **Remaining witness-branch contradiction and endpoint compression.** Proposition 21.6A, Lemma 21.7, Theorem 21.10, Theorem 21.15, Proposition 21.12, Corollary 21A, and Theorem 21.17 eliminate the remaining licit witness branch and compress the resulting contradiction into the official Fefferman (B) statement.

Proof. This is the official classical PDE restatement of Corollary 21A. The corollary is already supplied by the isolated transfer cascade of Theorem 21.15, so no further proof burden remains here beyond restating the same conclusion in Fefferman’s classical PDE language. $\square$

Corollary 21.19 (No finite-time blow-up on the official periodic carrier). *On the official periodic carrier $\mathbb{T}^3$, smooth divergence-free initial data do not produce finite-time singularity in the classical incompressible Navier–Stokes realization.*

Proof. This is an immediate restatement of Theorem 21.17. $\square$

Classical NS observable	Carrier-side object	Standing role	Primary governing result
velocity u	primary lawful carrier skin / state representative	lawful presentation, no second gate	Definition 13.2
vorticity $\omega = \nabla \times u$	vorticity field as lawful skin	standing-bearing only through licensed compatibility laws, not as a second gate	lawful-redescription / skin-invariance package
strain $S(u)$	strain / kernel field as lawful skin	standing-bearing only through licensed compatibility laws, not as a second gate	lawful-redescription / skin-invariance package
mean-zero pressure p	pressure field as lawful skin	lawful reconstructed presentation in the fixed gauge, no second gate	pressure reconstruction in Definition 18.3
energy / dissipation identity	tensor-level bundle compatibility	supports binary continuation status	Proposition 13.5B
pointwise $\mathcal{T}(u)$	pointwise tensor observable	quotient-visible bookkeeping, no independent pointwise gate	Lemma 13.14
trajectory $t \mapsto \mathcal{T}(u(t))$	Θ_x -observable on realized trajectory	no independent continuation gate	Theorem 20.11
continuation past T	horizon discriminator D_T	bivalent gate	Section 20
BKM-type quantity / critical norm bookkeeping	licensed metric skin / witness bookkeeping	no independent gate; if promoted to one, forces forbidden move	Appendix C; Propositions 13.23A.3 and 13.23A.4
concentration / profile selector	candidate continuation selector	forbidden parameterized gate / branch move	branch-forcing package
finite-time breakdown	breakdown witness / extractor $\mathfrak{W}(x), \mathfrak{X}(x)$	typed carrier-side witness packaging	Section 21

Table 3: Table 13.A — Translation dictionary for the standard periodic carrier. Every listed continuation-relevant classical observable used on the standing-positive live route is already represented on the carrier side.

Actual periodic NS object	ATS role on the live route	Standing authority / limit	Primary local label
fixed torus, $\nu > 0$, zero forcing, smooth divergence-free class, mean-zero pressure gauge	anchor	changing any item changes scope or definedness; no same-scope failure may be repaired by moving one of these commitments	Proposition 13.7; Lemma 13.3
realized velocity-pressure pair (u, p) in the fixed gauge	lawful skin presentation	presents one realized lineage without creating a second gate or a continuation criterion	Definition 13.2; Definition 18.0
vorticity ω and strain $S(u)$ as displayed fields	lawful skins with tensor-relevant reconstructions	as displayed fields they are presentations; any standing-bearing force comes only through the licensed curl / strain compatibilities already fixed in the carrier	Proposition 13.5; Proposition 13.23A.2
Biot–Savart / curl / Poisson reconstruction and velocity-vorticity-pressure compatibility	tensor	quotient-visible compatibility content tying the lawful skins to one same-scope carrier state; not a second gate	Proposition 13.6; Proposition 13.23A.2
energy / dissipation ledger	tensor bookkeeping	records continuation-relevant bundle compatibility but does not itself introduce a selector beyond the fixed gate	Proposition 13.5B; Proposition 13.23A.2
critical continuation norms, BKM-type growth quantities, and similar metric divergences	licensed metric skin / quotient-visible witness bookkeeping only	no independent standing authority; on the live route they may only package the pre-horizon witness or the horizon discriminator already fixed elsewhere	Proposition 13.23A.3; Proposition 13.23A.4
horizon discriminator $\mathfrak{D}_T$	tensor-typed quotient-visible witness gate	the actual same-scope bivalent continuation discriminator on the fixed carrier	Theorem 20.7A
singular-candidate extractor $\mathfrak{X}(x)$ and branch data $(\beta(x), \mathfrak{B}(x))$	tensor-typed witness packaging	no independent continuation authority beyond the extracted standing-positive witness content	Definition 21.6; Proposition 21.6A

Table 4: Table 13.B — ATS role localization for the actual periodic Navier–Stokes observables used on the live same-scope route.

Singularity branch	Structural reading	Outcome
Illicit branch	Same-scope discriminator that does not factor through the standing quotient; requires an extra gate, selector, or non-equivalent retyping.	Collapses by exhaustion and redescription legality.
Quotient-visible branch	No gate is introduced; the singularity mark is already a visible incompatibility of the admitted interior.	Collapses by saturation, fail-closed enforcement, and the impossibility of repair in-lineage.

Table 5: The final endpoint-status collapse. No third same-scope singularity gate is admissible.

A Minimal self-contained fixed-scope exhaustion core

This appendix supplies the full proof core for P4. It ports only the minimal fixed-scope exhaustion core required by the Navier–Stokes manuscript. No quotient universal-property material, role-trichotomy machinery, or ATS-style objection map is used here. Fix a fixed-scope reasoning regime in the sense of Definition 3.3. Let $\mathcal{E}$ denote its evaluated fragment and let $L : \mathcal{E} \rightarrow \{0, 1\}$ denote the standing-bearing admissibility judgement already in force at that scope. Throughout this appendix assume the four governance hypotheses named in Theorem P4: bivalent fail-closed admissibility, lawful redescription neutrality, irreversibility / no standing-preserving repair, and boundary signature-closure, meaning that no additional boundary-level discriminator, selector, comparison rule, or admissibility authority may be introduced while still claiming the same scope.

Definition (D.1 (Evaluated fragment / fixed typed domain)). *The evaluated fragment $\mathcal{E}$ of a fixed-scope reasoning regime is the class of same-scope candidates for which the admissibility judgement is well-formed. The pair $(\mathcal{E}, L)$, where $L : \mathcal{E} \rightarrow \{0, 1\}$ is the existing standing-bearing admissibility judgement, is the fixed typed domain. Fail-closed means that any candidate outside $\mathcal{E}$ is rejected as unlicensed rather than carried as a third standing-preserving value.*

Definition (D.2 (Gate on the evaluated fragment)). *A gate on the fixed typed domain $(\mathcal{E}, L)$ is any map $G : \mathcal{E} \rightarrow \{0, 1\}$. The existing admissibility classification is the distinguished gate L itself.*

Definition (D.3 (Standing-relevant discriminator)). *A same-scope discriminator is any extra datum, clause, or rule F presented as refining admissibility while still claiming the same fixed scope. It is standing-relevant if it comes with a candidate revised admissibility classification $L^{(F)} : \mathcal{E} \rightarrow \{0, 1\}$ and $L^{(F)} \neq L$ on at least one element of $\mathcal{E}$.*

Definition (D.4 (Admissibility-relevant parameterization)). *An admissibility-relevant parameterization is a family of gates $\{G_i\}_{i \in I}$ on the same evaluated fragment $\mathcal{E}$ together with a selector rule $\sigma : \mathcal{E} \rightarrow I$ used to choose which gate governs each evaluated candidate. It is admissibility-relevant if there exist $a \in \mathcal{E}$ and $i \neq j$ with $G_i(a) \neq G_j(a)$ so that the selector changes admissibility outcomes while claiming to preserve scope.*

Lemma (D.5 (Gate induction)). *Any standing-relevant same-scope discriminator induces a competing gate on the evaluated fragment. If a purported discriminator claims to change admissibility outcomes but does not determine a bivalent gate on $\mathcal{E}$, then it is ill-typed and terminates fail-closed.*

Proof. By Definition D.3, a standing-relevant discriminator F is presented together with a candidate revised admissibility classification $L^{(F)} : \mathcal{E} \rightarrow \{0, 1\}$. Setting $G^{(F)} := L^{(F)}$ produces a gate in the sense of Definition D.2, and standing-relevance means exactly that $G^{(F)}$ disagrees with the existing gate L somewhere on $\mathcal{E}$. If, instead, a proposal says that admissibility outcomes change but leaves some evaluated candidate without a determinate bivalent output, then it has not defined a gate on the fixed typed domain at all. Under fail-closed admissibility such an under-specified proposal is not a third standing value; it is simply not licensed for downstream use. $\square$

Lemma (D.6 (No admissibility-relevant selector)). *Within a fixed scope, no admissibility-relevant selector is well-typed. Any rule that chooses between two disagreeing gates or admissibility profiles introduces an additional boundary-level selector and therefore fails the boundary signature-closure hypothesis.*

Proof. A selector that changes which gate governs an evaluated candidate is itself acting as admissibility authority. But boundary signature-closure is exactly the prohibition on inserting any

additional boundary-level discriminator, selector, comparison rule, or admissibility authority while still claiming the same scope. Therefore an admissibility-relevant selector is not a same-scope refinement. It is either a scope change or an ill-typed proposal, and in either case it cannot survive inside the fixed typed domain. $\square$

Lemma (D.7 (Evaluated-fragment fixity)). *A same-scope refinement cannot change the evaluated fragment. If a purported admissibility refinement changes which candidates count as evaluated, then it changes the fixed typed domain and is not a same-scope move.*

Proof. Changing the evaluated fragment means that some candidate is now declared evaluable or non-evaluable by a rule not already contained in the fixed typed domain of Definition D.1. That extra rule acts at the same boundary level as the admissibility judgement itself: it decides whether a candidate even enters the standing-bearing fragment. By boundary signature-closure, no such extra boundary-level authority may be inserted while still claiming the same scope. Hence same-scope admissibility refinements must keep the evaluated fragment fixed. $\square$

Lemma (D.8 (No competing gates)). *Let $G, H : \mathcal{E} \rightarrow \{0, 1\}$ be same-scope gates on the fixed typed domain. Then $G = H$ extensionally on $\mathcal{E}$.*

Proof. Assume, for contradiction, that there exists $a \in \mathcal{E}$ with $G(a) \neq H(a)$. Since admissibility on the evaluated fragment is bivalent, the candidate a would then receive two incompatible same-scope classifications. To preserve a determinate standing judgement, one would have to specify which gate governs a . That specification is precisely an admissibility-relevant selector, excluded by Lemma D.6. Nor can the discrepancy be treated as something repaired later, because irreversibility / no standing-preserving repair forbids retroactive rewriting of an already standing-bearing same-scope commitment. Hence no such a exists, and the two gates agree everywhere on $\mathcal{E}$. $\square$

Theorem (D.9 (Structural Exhaustion core)). *Assume bivalent fail-closed admissibility, lawful redescription neutrality, irreversibility, and boundary signature-closure at a fixed scope. Then admissibility classification on the fixed typed domain is unique. Consequently, no second same-scope discriminator, gate, selector, or admissibility-relevant parameterization survives inside that scope.*

Proof. Let F be any purported same-scope discriminator. If F is standing-irrelevant, then it contributes no new admissibility authority and is bookkeeping only. Suppose instead that F is standing-relevant. By Lemma D.5, it induces a competing gate $G^{(F)}$ on the evaluated fragment. By Lemma D.7, a same-scope refinement cannot evade the issue by changing which candidates belong to the evaluated fragment. By Lemma D.8, the induced gate cannot disagree with the existing gate L anywhere on $\mathcal{E}$. Therefore no standing-relevant same-scope discriminator survives.

The same conclusion rules out admissibility-relevant parameterizations: if a family of gates $\{G_i\}_{i \in I}$ genuinely disagreed on some evaluated candidate, then any rule selecting which member governs would be an admissibility-relevant selector, forbidden by Lemma D.6; and if no disagreement occurs, then the family is not admissibility-relevant in the first place. Finally, lawful redescription neutrality prevents presentation-only rephrasings from masquerading as new admissibility authority. Therefore admissibility classification is unique at fixed scope, and no second same-scope discriminator, gate, selector, or admissibility-relevant parameterization survives. $\square$

Corollary (D.10 (Closure logic / singleton meta-necessity)). *Any purported second same-scope admissibility mechanism either agrees extensionally with the existing gate and is bookkeeping only, or else it is not a same-scope refinement.*

Proof. If the mechanism changes admissibility outcomes, then by Theorem D.9 it would be a second same-scope discriminator or gate and is therefore impossible. If it does not change outcomes, then it contributes no new standing-bearing authority and is bookkeeping only. This is exactly the singleton meta-necessity / closure-logic form of P4 used by the main text. $\square$

Remark A.1 (Appendix A in the proof spine). Appendix A is the first appendix-level burden carried beyond the main line of the paper. Its role is narrow: it internalizes the minimal generic gate/exhaustion theorem needed by the main text. The later carrier-specific exhaustion theorem and corollary remain valuable because they show that the standard periodic carrier itself realizes this generic fixed-scope prohibition and therefore closes the official theorem on a fully periodic carrier-side scope.

B Self-contained local analytic realization packet

This appendix internalizes the exact local analytic subroutines used in Part III and nothing more. Its role is narrow and theorem-local: it proves the periodic mean-zero Poisson solver used for pressure normalization, the torus heat-semigroup / Leray-projector bounds used in the mild realization window, the corresponding contraction estimate on X_T^m , the classical upgrade for smooth data, and overlap uniqueness for smooth periodic solutions. Classical references such as Leray, Hopf, and later local well-posedness papers remain historical provenance only; the results actually used by the proof spine are recorded below in their exact specialized form.

Proposition (B.1 (Mean-zero torus Poisson solver)). *Let $s \geq 2$, and let $f \in H^{s-2}(\mathbb{T}^3)$ satisfy*

$$\int_{\mathbb{T}^3} f(x) \, dx = 0.$$

Then there exists a unique mean-zero solution $p \in H^s(\mathbb{T}^3)$ of

$$-\Delta p = f \quad \text{on } \mathbb{T}^3, \quad \int_{\mathbb{T}^3} p(x) \, dx = 0.$$

Moreover,

$$\|p\|_{H^s(\mathbb{T}^3)} \leq C_s \|f\|_{H^{s-2}(\mathbb{T}^3)}.$$

If f is smooth, then p is smooth.

Proof. Write

$$f(x) = \sum_{k \in \mathbb{Z}^3} \widehat{f}(k) e^{ik \cdot x}.$$

The mean-zero hypothesis is exactly $\widehat{f}(0) = 0$. Define

$$\widehat{p}(0) := 0, \quad \widehat{p}(k) := |k|^{-2} \widehat{f}(k) \quad (k \neq 0).$$

Then $-\Delta p = f$ and p has zero mean by construction. Uniqueness in the mean-zero gauge is immediate: if $-\Delta p = 0$ and $\int_{\mathbb{T}^3} p = 0$, then every Fourier mode vanishes. The estimate follows from

$$(1 + |k|^2)^s |\widehat{p}(k)|^2 \leq (1 + |k|^2)^{s-2} |\widehat{f}(k)|^2 \quad (k \neq 0),$$

after summation over $k \in \mathbb{Z}^3$. If f is smooth, the Fourier coefficients decay faster than every power, and the same formula shows the same for $\widehat{p}$. $\square$

Corollary (B.2 (Pressure reconstruction in the periodic mean-zero gauge)). *Let $m \geq 2$ and let $u \in H_\sigma^m(\mathbb{T}^3)$. Then*

$$f_u := \partial_i \partial_j (u_i u_j)$$

has mean zero, and there exists a unique mean-zero pressure $p[u] \in H^m(\mathbb{T}^3)$ satisfying

$$-\Delta p[u] = \partial_i \partial_j (u_i u_j).$$

Moreover,

$$\|p[u]\|_{H^m} \leq C_m \|u\|_{H^m}^2.$$

If u is smooth, then $p[u]$ is smooth.

Proof. Because $u_i u_j$ is periodic, integration by parts on $\mathbb{T}^3$ gives

$$\int_{\mathbb{T}^3} \partial_i \partial_j (u_i u_j) \, dx = 0.$$

Since $m > 3/2$ on $\mathbb{T}^3$, $H^m(\mathbb{T}^3)$ is an algebra, so

$$\|u_i u_j\|_{H^m} \leq C_m \|u\|_{H^m}^2.$$

Hence $f_u \in H^{m-2}(\mathbb{T}^3)$ with

$$\|f_u\|_{H^{m-2}} \leq C_m \|u\|_{H^m}^2.$$

Apply Proposition B.1 with $s = m$. □

Lemma (B.3 (Torus heat semigroup and Leray-projector bounds)). *Let $s \geq 0$ and $\nu > 0$. Then:*

(i) *the Leray projector $\mathbb{P}$ is bounded on $H^s(\mathbb{T}^3; \mathbb{R}^3)$;*

(ii) *the heat semigroup satisfies*

$$\|e^{\nu t \Delta} f\|_{H^s} \leq \|f\|_{H^s} \quad (t \geq 0);$$

(iii) *there exists $C_s > 0$ such that for every $t > 0$,*

$$\|e^{\nu t \Delta} f\|_{H^{s+1}} \leq C_s (\nu t)^{-1/2} \|f\|_{H^s}, \quad \|\nabla e^{\nu t \Delta} \mathbb{P} g\|_{H^s} \leq C_s (\nu t)^{-1/2} \|g\|_{H^s}.$$

Proof. On nonzero Fourier modes,

$$\widehat{\mathbb{P} f}(k) = \left(I - \frac{k \otimes k}{|k|^2} \right) \widehat{f}(k),$$

and the multiplier matrix has operator norm at most 1 uniformly in $k \neq 0$. This gives (i). For the heat semigroup,

$$\widehat{e^{\nu t \Delta} f}(k) = e^{-\nu t |k|^2} \widehat{f}(k),$$

so $|e^{-\nu t |k|^2}| \leq 1$ and (ii) follows immediately. For (iii), use

$$(1 + |k|^2)^{1/2} e^{-\nu t |k|^2} \leq C (\nu t)^{-1/2}$$

uniformly in $k \in \mathbb{Z}^3$, together with the boundedness of the Leray symbol. □

Proposition (B.4 (Mild bilinear estimate on X_T^m)). *Let $m \geq 3$, let $T > 0$, and let*

$$\mathcal{B}(v, w)(t) := \int_0^t e^{\nu(t-s)\Delta} \mathbb{P} \nabla \cdot (v \otimes w)(s) \, ds.$$

Then for all $v, w \in X_T^m$,

$$\|\mathcal{B}(v, w)\|_{X_T^m} \leq C_{m,\nu} T^{1/2} \|v\|_{X_T^m} \|w\|_{X_T^m}.$$

Consequently,

$$\|\Phi_{u_0, T}(v) - \Phi_{u_0, T}(w)\|_{X_T^m} \leq C_{m,\nu} T^{1/2} (\|v\|_{X_T^m} + \|w\|_{X_T^m}) \|v - w\|_{X_T^m}.$$

Proof. Because $m > 3/2$, $H^m(\mathbb{T}^3)$ is an algebra. Hence

$$\|v \otimes w\|_{H^m} \leq C_m \|v\|_{H^m} \|w\|_{H^m}.$$

Applying Lemma B.3 to $g = v \otimes w$ gives

$$\|\mathcal{B}(v, w)(t)\|_{H^m} \leq C_{m,\nu} \int_0^t (t-s)^{-1/2} \|v(s)\|_{H^m} \|w(s)\|_{H^m} \, ds.$$

Taking the time supremum and using $\int_0^t (t-s)^{-1/2} \, ds \leq 2T^{1/2}$ yields

$$\sup_{0 \leq t \leq T} \|\mathcal{B}(v, w)(t)\|_{H^m} \leq C_{m,\nu} T^{1/2} \|v\|_{X_T^m} \|w\|_{X_T^m}.$$

For the $L_t^2 H_x^{m+1}$ part, Lemma B.3 gives

$$\|\mathcal{B}(v, w)(t)\|_{H^{m+1}} \leq C_{m,\nu} \int_0^t (t-s)^{-1/2} \|v(s)\|_{H^m} \|w(s)\|_{H^m} \, ds.$$

Convolution on $[0, T]$ with the kernel $r^{-1/2}$ maps L^∞ to L^2 with norm $O(T^{1/2})$, so the same bound holds for the $L^2([0, T]; H^{m+1})$ term. The displayed Lipschitz estimate follows by writing

$$v \otimes v - w \otimes w = v \otimes (v - w) + (v - w) \otimes w$$

and applying the same estimate. $\square$

Theorem (B.5 (Local mild realization window)). *Let $m \geq 3$ and let $u_0 \in H_\sigma^m(\mathbb{T}^3)$. Then there exist*

$$T_0 = T_0(\nu, \|u_0\|_{H^m}) > 0$$

and a closed ball $B_{T_0}^m \subset X_{T_0}^m$ such that the operator Φ_{u_0, T_0} of Definition 19.2 maps $B_{T_0}^m$ to itself and is a contraction there. Consequently it has a unique fixed point

$$u^{(0)} \in X_{T_0}^m.$$

Proof. Set $M := \|u_0\|_{H^m}$. By Lemma B.3,

$$\sup_{0 \leq t \leq T} \|e^{\nu t \Delta} u_0\|_{H^m} \leq M.$$

Also

$$\int_0^T \|e^{\nu t \Delta} u_0\|_{H^{m+1}}^2 \, dt \leq C_{m,\nu} M^2,$$

because on each Fourier mode

$$\int_0^T (1 + |k|^2)^{m+1} e^{-2\nu t |k|^2} \, dt \leq C_\nu (1 + |k|^2)^m.$$

Hence

$$\|e^{\nu t \Delta} u_0\|_{X_T^m} \leq C_{m,\nu} M.$$

Choose

$$R := 2C_{m,\nu} M$$

and let $B_T^m := \{v \in X_T^m : \|v\|_{X_T^m} \leq R\}$. For $v \in B_T^m$,

$$\|\Phi_{u_0, T}(v)\|_{X_T^m} \leq C_{m, \nu} M + C_{m, \nu} T^{1/2} R^2$$

by Proposition B.4. Choose $T_0 > 0$ so that

$$C_{m, \nu} T_0^{1/2} R \leq \frac{1}{4}, \quad C_{m, \nu} T_0^{1/2} R^2 \leq \frac{1}{2} R.$$

Then Φ_{u_0, T_0} maps $B_{T_0}^m$ into itself and, again by Proposition B.4,

$$\|\Phi_{u_0, T_0}(v) - \Phi_{u_0, T_0}(w)\|_{X_{T_0}^m} \leq \frac{1}{2} \|v - w\|_{X_{T_0}^m}$$

for $v, w \in B_{T_0}^m$. Banach's fixed-point theorem gives the unique fixed point. $\square$

Proposition (B.6 (Classical upgrade and smooth pressure on the local window)). *Assume the hypotheses of Theorem B.5, and suppose in addition that $u_0 \in C^\infty(\mathbb{T}^3; \mathbb{R}^3)$ and $\nabla \cdot u_0 = 0$. Then the unique fixed point $u^{(0)} \in X_{T_0}^m$ from Theorem B.5 is a smooth classical solution on $\mathbb{T}^3 \times [0, T_0]$, and the pressure reconstructed by Definition 19.3 is the unique smooth mean-zero pressure associated to $u^{(0)}$.*

Proof. Because $u^{(0)}$ is a mild fixed point, it satisfies

$$u^{(0)}(t) = e^{\nu t \Delta} u_0 - \int_0^t e^{\nu(t-s)\Delta} \mathbb{P} \nabla \cdot (u^{(0)} \otimes u^{(0)})(s) \, ds.$$

Since $m \geq 3$, the quadratic term belongs to $C([0, T_0]; H^{m-1})$, and differentiation of the mild formula shows that

$$\partial_t u^{(0)} = \nu \Delta u^{(0)} - \mathbb{P} \nabla \cdot (u^{(0)} \otimes u^{(0)})$$

in $C([0, T_0]; H^{m-2})$. Thus $u^{(0)}$ is a classical solution in the usual periodic sense.

For every $\tau > 0$, repeated use of the smoothing estimate in Lemma B.3 on the Duhamel formula upgrades $u^{(0)}$ to $C([\tau, T_0]; H^{m+r})$ for every integer $r \geq 0$. Hence $u^{(0)}$ is smooth on $\mathbb{T}^3 \times (0, T_0]$. At $t = 0$, the datum u_0 is smooth, and Corollary B.2 gives a smooth mean-zero pressure seed. Repeated time differentiation of the equation therefore determines all mixed derivatives at $t = 0$ recursively from smooth spatial data. Thus the smoothness extends to the closed strip $\mathbb{T}^3 \times [0, T_0]$.

Finally, for each $t \in [0, T_0]$, Corollary B.2 applied to the smooth slice $u^{(0)}(t)$ gives the unique mean-zero pressure $p[u^{(0)}](t)$, and the same bootstrap implies that $p[u^{(0)}]$ is smooth on $\mathbb{T}^3 \times [0, T_0]$. $\square$

Theorem (B.7 (Overlap uniqueness for smooth periodic realizations)). *Let (u, p) and (v, q) be smooth periodic Navier–Stokes solution pairs on $\mathbb{T}^3 \times [t_0, t_1]$ with the same viscosity, the same torus carrier, and the same mean-zero pressure gauge. If*

$$u(\cdot, t_0) = v(\cdot, t_0),$$

then

$$u(\cdot, t) = v(\cdot, t) \quad \text{and} \quad p(\cdot, t) = q(\cdot, t) \quad \text{for all } t \in [t_0, t_1].$$

Proof. Set

$$w := u - v, \quad r := p - q.$$

Then w is divergence-free and satisfies

$$\partial_t w + (u \cdot \nabla) w + (w \cdot \nabla) v + \nabla r = \nu \Delta w, \quad w(\cdot, t_0) = 0.$$

Taking the $L^2(\mathbb{T}^3)$ inner product with w and integrating by parts gives

$$\frac{1}{2} \frac{d}{dt} \|w(t)\|_{L^2}^2 + \nu \|\nabla w(t)\|_{L^2}^2 = - \int_{\mathbb{T}^3} (w \cdot \nabla) v \cdot w \, dx,$$

because the transport term with u vanishes by $\nabla \cdot u = 0$ and the pressure term vanishes by $\nabla \cdot w = 0$. Since v is smooth,

$$\left| \int_{\mathbb{T}^3} (w \cdot \nabla) v \cdot w \, dx \right| \leq \|\nabla v\|_{L^\infty} \|w\|_{L^2}^2.$$

Gronwall's inequality and $w(\cdot, t_0) = 0$ imply $w \equiv 0$ on $[t_0, t_1]$. Then

$$-\Delta r = \partial_i \partial_j (u_i u_j - v_i v_j) = 0,$$

and the mean-zero pressure gauge forces $r \equiv 0$. □

Corollary (B.8 (The local analytic packet used in Part III)). *For every smooth divergence-free periodic datum u_0 on $\mathbb{T}^3$ and every fixed viscosity $\nu > 0$, the local realization bridge used in Part III is now fully internalized in the manuscript:*

- (i) *the mean-zero pressure seed and all later pressure slices are reconstructed uniquely by Proposition B.1 and Corollary B.2;*
- (ii) *the mild realization operator admits a unique local fixed point by Theorem B.5;*
- (iii) *for smooth data, that fixed point is a smooth classical periodic solution with smooth mean-zero pressure by Proposition B.6;*
- (iv) *any two smooth local realization segments with the same initial slice agree on overlap by Theorem B.7.*

Accordingly, no statement in Part III depends on unstated local periodic PDE folklore.

Proof. Items (i)–(iv) are exactly the statements proved above. □

Remark B.1 (Appendix B in the proof spine). Appendix B is the only analytic appendix carrying proof load for the endpoint theorem. Its role is local and specialized: it does not contribute any global continuation criterion, blow-up exclusion theorem, or analytic replacement for Part I. It only internalizes the realization machinery already isolated in Part III.

C Worked outsider-facing translation of a BKM-type blow-up heuristic

This appendix is optional and non-load-bearing. It is included only to make visible, in ordinary PDE notation, one explicit instance of the mechanism used abstractly in the main proof: a familiar blow-up heuristic is translated into a forbidden same-scope carrier move. Nothing in this appendix is needed for the proof of Theorem 21.17; its role is explanatory and audit-facing. Appendix D records the broader obstruction ledger; the present appendix is one concrete worked row of that ledger.

C.1 A BKM-type blow-up observable

Let $x \in I_{\text{car}}(\mathbb{T}^3, \nu)$ and let

$$\mathcal{R}_{\text{NS}}(x) = (u_x, p_x)$$

be its maximal realized trajectory on $[0, T_*(x))$. Define the BKM-type vorticity-growth observable

$$\mathcal{I}_x(t) := \int_0^t \|\omega_x(s)\|_{L^\infty(\mathbb{T}^3)} \, ds, \quad 0 \leq t < T_*(x).$$

Remark C.1 (Why this is the right test case). The Beale–Kato–Majda criterion is a standard continuation criterion for smooth solutions: if $\int_0^T \|\omega(s)\|_{L^\infty} \, ds < \infty$, then the solution continues past T . Consequently, any finite-time breakdown hypothesis for a smooth trajectory forces the growth of the observable $\mathcal{I}_x(t)$ as $t \uparrow T_*(x)$. This makes $\mathcal{I}_x$ the cleanest outsider-facing example of a PDE-side quantity that appears to detect breakdown.

Lemma C.2 (BKM growth at a finite breakdown horizon). *Let $(u_x, p_x) = \mathcal{R}_{\text{NS}}(x)$ be a smooth realized trajectory on $[0, T_*(x))$ with $T_*(x) < \infty$. Then*

$$\lim_{t \uparrow T_*(x)} \mathcal{I}_x(t) = \infty.$$

Proof. If $\sup_{0 \leq t < T_*(x)} \mathcal{I}_x(t) < \infty$, then the Beale–Kato–Majda criterion applies on $[0, T_*(x))$ and yields smooth continuation past $T_*(x)$, contradicting maximality of the realized trajectory. Therefore $\mathcal{I}_x(t)$ cannot remain bounded as $t \uparrow T_*(x)$. $\square$

C.2 Threshold selectors and forbidden same-scope gates

The structural point is not that $\mathcal{I}_x$ fails to exist, nor that the BKM criterion is invalid as PDE analysis. The point is role discipline: using $\mathcal{I}_x$ as an independent same-scope endpoint authority requires an extra threshold, tolerance, or horizon selector not contained in the fixed carrier.

Definition C.3 (Finite-threshold BKM gate). Fix $M > 0$ and $T > 0$. The associated finite-threshold BKM gate on realized trajectories is

$$G_{M,T}^{\text{BKM}}(x) := \mathbf{1} \left[\sup_{0 \leq t < T} \mathcal{I}_x(t) \leq M \right].$$

More generally, a BKM-profile gate is any same-scope rule asserting continuation from a comparison of $\mathcal{I}_x(t)$ with a prescribed bound, threshold, tolerance, or growth profile as $t \uparrow T$.

Proposition (C.3A (BKM observable as metric witness bookkeeping, not endpoint authority)). *The BKM-type observable $\mathcal{I}_x$ has two different roles that must not be conflated. As a classical conditional continuation observable it can record a standard pre-horizon blow-up implication and may witness obstruction to a local continuation estimate. As a same-scope endpoint authority, however, it has no independent force: on the fixed periodic carrier it is either licensed metric bookkeeping, quotient-visible witness bookkeeping subordinate to the horizon discriminator, or an illicit metric gate if promoted beyond that role.*

Proof. The observable $\mathcal{I}_x$ is computed from the realized vorticity field and therefore is a lawful metric descriptor on the realized lineage. Lemma C.2 records the standard conditional continuation consequence. But Theorem 13.23A.6 applies to BKM-type observables specifically: metric divergence does not create a same-scope endpoint status. Thus $\mathcal{I}_x$ may serve as witness bookkeeping for an already fixed finite-horizon failure, but any attempt to use it as an independent same-scope gate is illicit. $\square$

Proposition C.4 (Worked translation: BKM-type continuation tests force $S3$). *Let $x \in I_{\text{car}}(\mathbb{T}^3, \nu)$ and suppose a same-scope continuation statement for the realized trajectory $\mathcal{R}_{\text{NS}}(x)$ is certified by a finite-threshold BKM gate or, more generally, by a BKM-profile gate in the sense of Definition C.3. Then the certification instantiates the forbidden move $S3$: it introduces a threshold, tolerance, selector, or horizon rule not already licensed by the fixed carrier.*

Proof. The observable $\mathcal{I}_x(t)$ is built from the realized vorticity field and is therefore a lawful PDE-side metric quantity. Proposition C.3A fixes its licit status: it may be bookkeeping or witness bookkeeping, but not independent endpoint authority. The extra structure enters when one attempts to turn $\mathcal{I}_x$ into a same-scope continuation criterion.

For the finite-threshold gate $G_{M,T}^{\text{BKM}}$, the rule depends on the choice of the external parameter M and on the horizon parameter T . Neither parameter is determined by the carrier seed, by the realization map, or by the horizon continuation discriminator already present in Part III. The same is true for any profile gate depending on a prescribed growth profile $\Phi(T-t)$, tolerance ε , or selector determining what growth rate counts as admissible. Such data are not forced by lawful redescription of the realized trajectory; they are additional same-scope choices.

Therefore the gate does not merely read continuation content already present on the carrier. It adds a threshold / tolerance / selector rule for deciding continuation from the BKM observable. That is exactly the structural role labeled $S3$ in the main text. $\square$

Corollary C.5 (Concrete PDE-to-carrier translation of one blow-up heuristic). *A BKM-style blow-up heuristic does not produce an admissible independent continuation mechanism on the fixed torus carrier. Interpreted as a same-scope continuation gate, it forces the forbidden branch $S3$. If one weakens the gate by allowing local exceptions or selective tolerance to large-vorticity episodes, then one additionally enters the bundle-exemption branches $S1$ or $S2$.*

Proof. The first statement is Proposition C.4. If the gate is weakened by saying, for example, that only selected portions of the trajectory, selected regions of the flow, or selected instances of large vorticity matter, then the rule not treats the bundle clauses uniformly on the fixed carrier. That is precisely the move-class captured by $S1$ or $S2$ in the main theorem package. $\square$

Remark C.6 (What this appendix shows). The main proof does not require any special analytic calculation attached to the BKM observable. The role of this appendix is different: it shows, in one concrete and familiar PDE example, how a standard blow-up heuristic is translated by hand into the carrier language. The contradiction in the main line is therefore not a bare reclassification; it is

the same mechanism exhibited abstractly in the proof spine, made explicit here for one canonical continuation test.

D How the structural route bypasses the standard analytic walls

This appendix is expository and non-load-bearing. Its purpose is to give a Clay-style referee a single ledger showing, row by row, how the structural route replaces the familiar analytic continuation walls. Nothing here adds a new step to the proof of Theorem 21.17; it records in one place how the carrier-side structures already established in the main text absorb or render unnecessary the continuation functions controlled by the standard analytic criteria.

Proposition (C.1 (Structural replacement of the standard analytic walls)). *For each standard analytic continuation wall listed in Table C.1, the continuation function it seeks to control is already represented or rendered unnecessary by the carrier-side structures indicated in the table. Consequently, no row of Table C.1 supplies an independent same-scope escape from the official periodic theorem.*

Proof. Case-by-case by Table C.1. In each row, the continuation-relevant content is either already carried by the horizon discriminator and the strong continuation-completeness package, or else the proposed analytic criterion would amount to an additional same-scope selector, parameterized gate, or branch move already excluded by the zero-parameter certification, branch-forcing package, and terminality results cited in the table. $\square$

Classical analytic wall	What it tries to control	Structural replacement	Primary citations
Serrin–Prodi	Mixed-norm continuation criterion	No same-scope continuation gate beyond the carrier-side horizon discriminator; any additional continuation threshold is absorbed or ruled out by strong continuation-completeness and terminality.	Theorem 13.26; Lemma 18.16; Corollary 17.5.
Beale–Kato–Majda	Vorticity-growth continuation gate	Formal Θ_x -gate collapse in the main line; any BKM-style gate used as an independent same-scope continuation criterion forces a forbidden move.	Theorem 20.11; Proposition C.4; Corollary C.5.
Koch–Tataru / scale-critical smallness	Scale-based continuation selector	The zero-parameter carrier blocks any same-scope scale gate; smallness criteria are large-data external selectors rather than new carrier-side continuation data.	Theorem 13.20; Propositions 13.5A–13.5C; Corollary 13.5D.
Geometric depletion	Special geometric regularization selector	Geometric information is already interior skin/tensor data represented in the carrier dictionary, not a second same-scope gate.	Table 13.A; Theorem 13.24; Theorem 13.26.
Ancient-solution / profile extraction	Asymptotic profile selector	Profile-based continuation requires a forbidden branch move or parameterized gate; no same-scope profile selector reopens the certified carrier.	Theorem 21.4; Lemma 13.18; Corollary 17.5.

Table 6: Table C.1 — Bypassing the classical obstructions. Each standard analytic wall seeks to control a continuation function that is already represented, absorbed, or rendered unnecessary by the carrier-side structures established in the main text.

Remark D.1 (How to read Table C.1). The table is not a proof layer. It is a one-page summary of how the main structural package replaces the continuation work usually assigned to the classical analytic criteria. Appendix C is the worked BKM row in full detail; the present appendix records the broader ledger in compressed form.

E Detailed analytic atlas of standard routes

The sections collected in this appendix are the route-by-route standard analytic atlas compressed out of the main line of the paper. They justify the common bottleneck statement and the atlas-closure theorem used in Part I and Part II, but they are not needed for the first-pass reading of the proof spine.

E.1 The Navier–Stokes system, its bundle, and its scale

We consider the incompressible Navier–Stokes equations on $\mathbb{R}^3$ or $\mathbb{T}^3$:

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad (1)$$

with viscosity $\nu > 0$ and divergence-free initial data $u(0) = u_0$.

E.1.1 A “constraint bundle” viewpoint (standard content, nonstandard name)

In the Yang–Mills manuscript the central objects are *constraint bundles* and the compatibility classes they induce. We introduce the analogous standard content for Navier–Stokes.

Definition E.1 (Navier–Stokes constraint bundle). Fix a domain $M \in \{\mathbb{R}^3, \mathbb{T}^3\}$. The Navier–Stokes constraint bundle $B_{\text{NS}}(M, \nu)$ consists of:

- (i) incompressibility ($\nabla \cdot u = 0$);
- (ii) momentum compatibility (1) in the appropriate sense (distributional for weak solutions);
- (iii) pressure compatibility (e.g. p determined up to constants by $-\Delta p = \partial_i \partial_j (u_i u_j)$ on $\mathbb{T}^3$);
- (iv) the energy inequality (global or local, depending on solution class).

Nothing in Definition E.1 is new; the term “bundle” is simply a mnemonic for the fact that global regularity would follow if one could show that compatibility with B_{NS} forces membership in the smooth class.

E.1.2 Scaling and criticality

Equation (1) is invariant under the scaling

$$u(x, t) \mapsto u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t), \quad p(x, t) \mapsto p_\lambda(x, t) = \lambda^2 p(\lambda x, \lambda^2 t). \quad (2)$$

A norm $\|\cdot\|_X$ is *critical* if $\|u_\lambda\|_X = \|u\|_X$.

Remark E.2 (Where the Clay difficulty lives). The energy class $L_t^\infty L_x^2 \cap L_t^2 \dot{H}_x^1$ is subcritical under (2). The Clay difficulty can be read as the absence of an a priori mechanism that upgrades subcritical control (energy) to critical control (any of the known criteria).

E.2 Energy method, Leray–Hopf theory, and the vortex-stretching tensor

E.2.1 Energy inequality

Formally multiplying (1) by u and integrating yields

$$\frac{1}{2} \frac{d}{dt} \|u(t)\|_{L^2(M)}^2 + \nu \|\nabla u(t)\|_{L^2(M)}^2 = 0. \quad (3)$$

Definition E.3 (Leray–Hopf weak solution). A vector field u is a Leray–Hopf solution on $[0, \infty)$ if $u \in L^\infty([0, \infty); L^2) \cap L^2([0, \infty); \dot{H}^1)$, $\nabla \cdot u = 0$ in distributions, (1) holds in distributions for some pressure p , and the energy inequality corresponding to (3) holds for almost every t .

Theorem E.4 (Leray–Hopf existence [Ler34; Hop51]). *For divergence-free $u_0 \in L^2(M)$, there exists a global Leray–Hopf weak solution.*

E.2.2 Enstrophy and the nonlocal production channel

Let $\omega = \nabla \times u$ be the vorticity and $S(u) = \frac{1}{2}(\nabla u + \nabla u^\top)$ the strain tensor. Formally ω satisfies

$$\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega. \quad (4)$$

Taking L^2 inner product with ω yields the enstrophy balance

$$\frac{1}{2} \frac{d}{dt} \|\omega\|_{L^2}^2 + \nu \|\nabla \omega\|_{L^2}^2 = \int_M \omega \cdot S(u) \omega \, dx. \quad (5)$$

Definition E.5 (Vortex-stretching tensor load). Define the vortex-stretching production functional

$$\mathcal{T}(u) := \int_M \omega \cdot S(u) \omega.$$

Proposition E.6 (The first wall). *Energy control (3) does not, by itself, control the sign or size of $\mathcal{T}(u)$. Any global regularity argument must either:*

- (a) bound $\mathcal{T}(u)$ by dissipation at a scale-critical level, or
- (b) replace (5) by a mechanism that bypasses $\mathcal{T}(u)$ entirely.

E.2.3 Kernel representation: why geometry alone does not close

In $\mathbb{R}^3$ (or on $\mathbb{T}^3$ with periodic kernels), the strain may be written as a Calderón–Zygmund transform of vorticity:

$$S(u) = \mathcal{R}_s(\omega),$$

where $\mathcal{R}_s$ is a matrix of singular integral operators. Symbolically,

$$S(x) = \text{p.v.} \int K(x-y) \omega(y) \, dy, \quad \int_{|z|=1} K(z) \, d\sigma(z) = 0.$$

Kernel cancellation permits the rewriting

$$S(x) = \text{p.v.} \int K(x-y) (\omega(y) - \omega(x)) \, dy.$$

This decomposition is the root of many “geometric depletion” arguments: one splits $\omega(y) - \omega(x)$ into direction-difference and magnitude-difference pieces. The latter is the persistent escape channel (see Section E.9).

E.3 Mild formulation, Fujita–Kato, and the critical fixed-point barrier

Another standard route writes (1) in mild form:

$$u(t) = e^{\nu t \Delta} u_0 - \int_0^t e^{\nu(t-s)\Delta} \mathbb{P} \nabla \cdot (u \otimes u)(s) \, ds. \quad (6)$$

Fixed-point arguments in critical spaces yield global solutions for small data.

Theorem E.7 (Prototype small-data global theory (Fujita–Kato paradigm)). *If the initial data is sufficiently small in a critical norm (e.g. $\dot{H}^{1/2}$ or BMO^{-1}), then the mild solution exists globally and is smooth.*

Proposition E.8 (The fixed-point wall). *For large data, (6) offers no coercive a priori smallness. Any large-data global result must supply a new mechanism that prevents the bilinear term in (6) from building critical norm. This restates the vortex-stretching wall in semigroup language.*

E.4 Scale-critical regularity criteria (Serrin–Prodi, endpoints)

Theorem E.9 (Serrin–Prodi regularity [Pro59; Ser62]). *Let u be a Leray–Hopf solution on $(0, T)$. If $u \in L^p(0, T; L^q(\mathbb{R}^3))$ with $\frac{2}{p} + \frac{3}{q} \leq 1$ and $q > 3$, then u is smooth on $(0, T]$.*

Theorem E.10 (Endpoint $L^\infty L^3$ criterion [ESS03]). *If a suitable weak solution satisfies $u \in L^\infty(0, T; L^3(\mathbb{R}^3))$, then it is smooth on $(0, T]$.*

Proposition E.11 (The Serrin wall). *Theorems E.9–E.10 reduce regularity to showing that every energy-class solution satisfies at least one critical integrability condition. No known a priori mechanism upgrades (3) to any of these critical bounds without extra hypotheses.*

E.5 Weak–strong uniqueness and the uniqueness barrier

Theorem E.12 (Weak–strong uniqueness (standard)). *If u is a Leray–Hopf solution and v is a strong solution with the same initial data on $(0, T)$, then $u = v$ on $(0, T)$.*

Proposition E.13 (The uniqueness wall). *Weak–strong uniqueness shifts the problem: if one could produce any strong solution globally, it would coincide with the weak solution. The remaining obstruction is exactly the construction of a global strong solution for large data, which again reduces to a critical a priori bound on the nonlinear term.*

E.6 Beale–Kato–Majda and vorticity control

Theorem E.14 (Beale–Kato–Majda [BKM84]). *Let u be a smooth solution on $[0, T)$. If*

$$\int_0^T \|\omega(t)\|_{L^\infty} \, dt < \infty,$$

then u extends smoothly past T .

Proposition E.15 (The BKM wall). *Theorem E.14 reduces blow-up prevention to controlling the critical quantity $\int_0^T \|\omega\|_{L^\infty} \, dt$, but the energy inequality does not provide such control. The missing step is a scale-critical upgrade from $L_t^\infty L_x^2 \cap L_t^2 \dot{H}_x^1$ to $L_t^1 L_x^\infty$.*

E.7 Local energy inequality and partial regularity

Definition E.16 (Suitable weak solution). A weak solution is *suitable* if it satisfies the local energy inequality in the sense of Caffarelli–Kohn–Nirenberg.

Theorem E.17 (Partial regularity (CKN) [CKN82]). *The one-dimensional parabolic Hausdorff measure of the singular set of a suitable weak solution is zero.*

Proposition E.18 (The CKN wall). *Theorem E.17 provides epsilon-regularity and a strong geometric measure restriction on possible singularities, but does not rule out their existence. The missing step is to show that every point satisfies the requisite smallness condition; doing so would again yield a critical bound.*

E.8 Blow-up profiles, Type I/II, and ancient solutions

E.8.1 Rescaling and profiles

Assume a smooth solution fails to extend past $T < \infty$. Consider the rescaling around a point (x_0, T) :

$$u(x, t) = \lambda(t) U(\lambda(t)(x - x_0), s(t)), \quad \frac{ds}{dt} = \lambda(t)^2.$$

Type I blow-up corresponds to $\lambda(t) \sim (T - t)^{-1/2}$; Type II corresponds to faster growth.

Theorem E.19 (Nonexistence of nontrivial Leray self-similar profiles [NRS96; Tsa98]). *There are no nontrivial backward self-similar finite-energy solutions corresponding to Leray’s ansatz.*

Proposition E.20 (The profile wall). *Excluding explicit self-similar blow-up leaves the Type II (ancient-solution) regime. To close regularity via profiles one must prove a rigidity theorem for all relevant ancient solutions or extract a contradiction from a scale-critical normalization. This is equivalent to establishing a critical a priori bound (often $L_t^\infty L_x^3$ or equivalent Morrey/Besov control).*

E.9 Geometric depletion of nonlinearity

One line of attack exploits cancellations in $\mathcal{T}(u)$ by imposing geometric structure on vorticity.

Theorem E.21 (Vorticity-direction coherence criterion (Constantin–Fefferman) [CF93]). *If the vorticity direction field $\xi = \omega/|\omega|$ is sufficiently coherent (e.g. Hölder continuous in regions of large vorticity), then the solution remains regular.*

Proposition E.22 (The geometry-to-kernel wall). *Geometric criteria control the direction-difference contribution in kernel representations of $S(u)$, but do not control the magnitude-difference channel without an additional coupling mechanism. Thus geometric depletion is conditional: the missing step is to prove the required coherence (or an equivalent cancellation) from the energy-class invariants.*

E.10 Frequency splitting and Littlewood–Paley bottlenecks

Let $(\Delta_j)_{j \in \mathbb{Z}}$ be a Littlewood–Paley decomposition. Many arguments split

$$u = u_{\leq N} + u_{> N},$$

attempting to show dissipation dominates at high frequencies and nonlinearity is controlled at low frequencies. Formally one seeks differential inequalities of the form

$$\frac{d}{dt} \|u_{> N}\|_{L^2}^2 + \nu \|\nabla u_{> N}\|_{L^2}^2 \leq (\text{controlled}).$$

Proposition E.23 (The frequency wall). *Frequency splitting reorganizes the nonlinear term but does not remove the need for a critical control: one must prevent transfer of energy/enstrophy into arbitrarily high frequencies at the scaling rate. Any successful frequency argument ultimately supplies a critical bound equivalent to Proposition 1.1 below.*

E.11 Critical harmonic analysis (Koch–Tataru and beyond)

Theorem E.24 (Small data global well-posedness in BMO^{-1} [KT01]). *There exists $\delta > 0$ such that if $\|u_0\|_{BMO^{-1}} < \delta$, then the corresponding solution is global and smooth.*

Proposition E.25 (The large-data wall). *Critical-space well-posedness provides global regularity for small initial data in critical norms. The Clay problem concerns large data, where no a priori mechanism yields critical smallness or a coercive monotonicity.*

F Route-locus audit for analytic and endpoint alternatives

This appendix gives a compact route-locus form of the analytic and endpoint alternatives already classified in the main text. It adds no new proof premise. Its purpose is to make the exhaustion surface inspectable in the same format used by the hardened endpoint papers: every same-scope finite-horizon proposal either enters one of the listed loci or does no endpoint work.

Definition F.1 (Navier–Stokes route locus). A Navier–Stokes route locus is a standard analytic, representative, metric, formulation, or endpoint-status route by which a proposed finite-horizon failure attempts to bear consequence for the official periodic theorem while preserving the fixed-domain comparison class of Definition 3.4A.

Locus	Route	Disposition
L1	Energy/enstrophy coercive bridge	Valid analytic support, but as endpoint authority it is either an already-carrier-visible consequence or an unlicensed bridge to vortex-stretching control.
L2	BKM / vorticity-growth route	Diagnostic and witness bookkeeping; if used independently as endpoint authority it becomes an illicit metric gate.
L3	Serrin / critical-norm route	Valid conditional criterion. As unconditional endpoint authority it requires the missing bridge or an unlicensed threshold selector.
L4	Critical smallness route	Lawful small-data theory; not a full-data same-scope endpoint unless bridged to the official datum class.
L5	Geometric-depletion route	Support or bridge material; without invariant bridge to the fixed endpoint it is redescription or selector collapse.
L6	Frequency-cascade route	Lawful skin/bookkeeping unless quotient-visible; otherwise a scale selector or extra continuation gate.
L7	Profile / ancient-solution route	Lawful compactness/profiling support; as endpoint authority it is a profile selector or branch-forcing move unless a same-scope proof is supplied.
L8	Weak or alternative formulation	Carrier/scope shift unless bridged back to the official smooth periodic comparison class.
L9	Local-energy / partial-regularity route	Support only unless upgraded by a same-scope bridge to global smooth continuation.
L10	Time-slice reclassification	Blocked by realized-lineage identity and no admissibility dynamics.
L11	Metric divergence route	Licensed bookkeeping or quotient-visible witness bookkeeping; if promoted independently, illicit metric gate.
L12	Restart / repair route	Blocked by overlap uniqueness, continuation-locus fixity, irreversibility, and no post-hoc repair.
L13	Hidden endpoint classifier	Forbidden second same-scope continuation gate or endpoint-status discriminator.
L14	Scope-change route	Different-domain proposal: not a counterexample to Fefferman (B) on the fixed periodic carrier.

Theorem F.2 (Route-locus exhaustion for finite-horizon proposals). *Every same-scope endpoint-defeating finite-horizon proposal for the official periodic Navier–Stokes theorem enters at least one of L1–L14. If it enters none, it has no continuation-status effect and cannot refute the official endpoint.*

Proof. A finite-horizon proposal that does endpoint work must either proceed through a classical analytic continuation route, a metric diagnostic, a representative or formulation change, a restart/repair pattern, or an endpoint-status classifier. The analytic routes are L1–L9 and are exactly the families isolated by the analytic wall and atlas. Representative, time-slice, metric, and repair routes are L10–L12. Endpoint classifier and scope-change routes are L13–L14. If none of these occurs, the proposal neither changes the analytic support, representative, metric witness, continuation rule, endpoint classifier, nor fixed comparison class; it therefore does no endpoint work. $\square$

Theorem F.3 (Route-locus collapse). *Every route locus L1–L14 collapses into one of the following statuses:*

support/bookkeeping $\vee$ standing-positive horizon witness $\vee$ terminal standing failure
 $\vee$ illicit same-scope gate $\vee$ scope change.

Only the standing-positive horizon-witness status is a licit same-scope finite-horizon endpoint status, and that status is eliminated by Theorem 21.10.

Proof. L1–L9 are legitimate analytic support routes unless promoted to endpoint authority. If they remain support, they are support/bookkeeping; if they supply a finite-horizon witness, they factor through the horizon discriminator; if they require a threshold, profile selector, or hidden gate, they are illicit same-scope gates; if they alter formulation or claim class, they are scope changes. L10 collapses by Theorems 20B.1 and 20B.2. L11 collapses by Propositions 13.23A.3 and 13.23A.4. L12 collapses by Theorem 19D and Proposition ST5. L13 collapses by Theorem 20B. L14 is scope change by Definition 3.4A. The only licit same-scope endpoint branch that can remain is the standing-positive horizon-witness branch, which is exhausted by Proposition 21.6A and eliminated by Theorem 21.10. $\square$

Corollary F.4 (No surviving route-locus counterexample). *No same-scope finite-horizon counterexample to the official periodic theorem survives the route-locus audit.*

Proof. By Theorem F.2, every endpoint-relevant finite-horizon proposal enters a route locus. By Theorem F.3, each locus reduces to a status already classified by the realized-lineage trichotomy or to scope change. The non-witness branches collapse by Proposition 20C.2 and Corollary 20C.3; the witness branch is eliminated by Theorem 21.10. Hence no same-scope finite-horizon counterexample survives. $\square$

G Theorem ladder for the modernized proof spine

This appendix records the proof ladder in compact form. It is an audit guide, not an additional proof route. Each node points back to theorem-local material in the body.

Node	Theorem-local content	Function in the endpoint proof
0	Proof-class / claim-class / source-use locks	Fixes the method as non-PDE-native finite-horizon endpoint impossibility and prevents source-role drift.
1	Ambient counterexamplehood subsumption	Theorems 3.K3–3.K5 show ordinary counterexamplehood performs the kernel roles.
2	Kernel necessity	Theorem 3.K places non-degenerate theorem-bearing regimes under fixed-scope kernel governance.
3	Official periodic regime non-degenerate	Theorem 12A.0 verifies the official periodic proof/refutation regime.
4	Fixed-domain comparison class	Definition 3.4A locks torus, viscosity, forcing, datum class, pressure gauge, lawful skins, and continuation question.
5	Structural packet P1–P4	Theorem P1, Corollary P2, Corollary P3, and Theorem P4.
6	Periodic bridge 12A–12H	Theorem 12G and Corollary 12H.
7	Standard carrier identification	Theorem 13I; Corollary 13J; Proposition 13K.
8	Finite candidate classification	Theorem 13.5F; Corollary 13.5L.
9	PDE local realization	Theorem 19.6; Theorem 19B.
10	Realized-lineage identity	Definition 18.0; Theorem 20B.1.
11	No time-indexed admissibility	Theorem 20B.2; Theorem ST4.
12	Metric diagnostic non-authority	Proposition 13.23A.3; Theorem 13.23A.6.
13	Horizon bridge object	Definition 20B.4; Proposition 20B.5.
14	Local finite-horizon reductio annotation	Definition 20B.7; Theorem 20B.8.
15	Finite-horizon endpoint condition	Theorem 20C.0A; Theorem 20B.9.
16	Endpoint-status trichotomy	Theorem 20C; Theorem 20B.6.
17	Non-witness branch collapse	Proposition 20C.2; Corollary 20C.3.
18	Standing-positive witness exhaustion	Proposition 21.6A; Lemma 21.7.
19	Branch forcing and branch elimination	Theorem 21.4; Propositions 21.8–21.9.
20	Local contradiction and discharge	Theorem 20B.10; Theorem 20B.11.
21	No singular image	Theorem 21.10.
22	Official endpoint	Theorem 21.15; Corollary 21A; Theorem 21.17.
23	Route-locus audit	Appendix F; Theorem F.2; Corollary F.4.
24	Lean / formal audit boundary	Appendix H records the audit boundary and does not replace the manuscript proof.

H Source, hostile-referee, and formal-audit appendix

This appendix is secondary audit material. It adds no theorem used by the official periodic endpoint. The proof is the theorem chain in the main text; the tables below record source governance, bridge-seam dependencies, anti-circularity checks, referee denial burdens, and the Lean support boundary.

H.1 Source governance and live-dependency freeze

The live endpoint route is frozen to the main body, Appendix A, Appendix B, Appendix E for Part I's analytic atlas, the official problem-alignment sources, and the named Lean audit source only as support boundary. Private notes, slide decks, project notes, validation reports, and companion-paper slogans are not proof premises.

Material	Status	Function
Main body theorem chain	Live proof material	Carries every theorem used by Theorem 21.17.
Appendix A	Live proof material	Supplies the fixed-scope gate/exhaustion core used by P4.
Appendix B	Live proof material	Internalizes the local periodic Poisson, mild-realization, classical-upgrade, and overlap-uniqueness packet used by Part III.
Appendix E	Live support for Part I	Records the detailed analytic atlas behind the common-wall and atlas-closure statements.
Official Clay / Fefferman sources	Alignment only	Fix the target statement and method-neutral review frame; they do not carry proof burden.
Appendices C, D, F, G, H	Audit / exposition	Translate, summarize, and audit the proof route; they do not add an alternate endpoint proof.
Companion titles, private notes, validation material	Provenance only / excluded	May orient terminology but cannot substitute for theorem-local proof material.

Proposition H.1 (Audit-only support ledger for the official periodic theorem). *For the official endpoint, the support material separates into three levels. The main text, Appendix A, Appendix B, and Appendix E carry the live theorem-local proof burden; Appendices C, D, F, G, and H supply translation, route-locus, hostile-referee, and theorem-ladder audit; external corpus or project materials remain provenance only unless a theorem is internalized and cited locally.*

Proof. This is the source-governance table above restated as a proposition for cross-reference. It adds no proof premise to Theorem 21.17. $\square$

H.2 Bridge-seam audit

The bridge seam is one-directional. Lemma 12F classifies fixed-periodic scope-fixity; Theorem 12G applies P4 to the completed packet; Proposition 13K carries the no-new-criterion consequence into the carrier packet; Theorems 20.5B and 20B transport that consequence to the realized classical layer; Theorem 20C consumes it in endpoint-status form.

Node	Role	Closure
12F	Fixed-periodic boundary-signature closure / scope typing	Classifies additional criteria as gate proposals or bookkeeping; does not by itself exclude all gates.
12G / P4	First no-second-gate exclusion node	Applies structural exhaustion to the completed periodic bridge packet.
13K	Carrier-level no-additional-criterion node	Imports the 12G/P4 consequence into the standard periodic carrier.
20.5B	Transfer no-new-content node	Shows classical realization adds no new standing-positive same-scope continuation content.
20B	Classical-layer no-independent-discriminator node	Blocks PDE-side continuation gates after realization.
ST2–ST3	Standing/intelligibility transport	Converts fixed-scope failure of reference, domain closure, redescription, or identity into standing loss.
20C	Endpoint-status trichotomy	Exhausts finite-horizon endpoint condition statuses on realized lineages.
21.x	Witness exhaustion and branch elimination	Eliminates the only remaining licit same-scope witness branch.

H.3 Anti-circularity audit

Potential circle or overreach	Audit response
The proof assumes global smoothness through admissibility.	No. The endpoint subproof opens $[T_*(x) < \infty]_i$ and discharges that finite-horizon assumption after trichotomy, witness exhaustion, and branch elimination.
The horizon bridge object contains $T_*(x) = \infty$.	No. B_{hor}^{NS} packages finite-horizon counterexample data only; non-circularity is Proposition 20B.5.
Metric divergence is declared illicit by fiat.	No. It remains valid diagnostic or witness bookkeeping. Only independent endpoint authority is excluded.
Finite T_* is automatically retyped as counterexample failure.	No. It must enter theorem-bearing counterexamplehood and the finite-horizon endpoint condition; see Theorem 20C.4.
Part III reopens admissibility.	No. Corollary 13L closes same-scope admissibility before transfer; Part III is realization transport.
Local PDE theory is imported silently.	No. Appendix B internalizes the local realization, pressure reconstruction, classical upgrade, and overlap uniqueness used by Part III.
Vortex stretching is ignored.	No. It is isolated as the analytic wall and localized as diagnostic/witness material unless promoted to an endpoint gate.
The trichotomy redefines Fefferman (B).	No. It classifies the endpoint status of a proposed counterexample to the fixed official periodic statement.

H.4 Referee burden worksheet

A same-mode objection must break a named theorem link or produce a same-domain object that survives the declared classification.

Target	Successful objection must produce
Proof-class lock	A failed theorem link in the non-PDE-native fixed-scope impossibility route, not merely absence of a continuation estimate.
Ambient counterexamplehood	A theorem-bearing counterexample to Fefferman (B) that does not require admissible object formation, standing, reference, same-target fidelity, or final consequence.
Fixed-domain comparison class	A finite-horizon proposal that changes no official carrier commitment yet remains outside the comparison class.
Unique admissible interior	A same-scope second interior or gate compatible with Corollary 12H and Theorem 13L.
Part III transfer	A hidden datum restriction, extra continuation criterion, or changed continuation notion in the realization map.
Realized-lineage identity	A smooth restart or pre-horizon slice that is not a presentation of the same realized lineage.
No time-indexed admissibility	A same-scope lineage whose admissibility varies by time, slice, restart stage, or horizon proximity without introducing a second gate.
Metric diagnostic non-authority	A BKM, critical norm, profile, or metric-divergence report that becomes endpoint authority without factoring through the horizon witness and without becoming an illicit gate.
Horizon bridge object	A clause of B_{hor}^{NS} that smuggles global smoothness or changes the official finite-horizon counterexample interface.
Local reductio annotation	A proof that $[T_*(x) < \infty]_i$ adds an object-level premise beyond $T_*(x) < \infty$.
Endpoint-status trichotomy	A fourth licit same-scope finite-horizon endpoint status outside standing-positive witness, terminal standing failure, and illicit gate/scope move.
Witness exhaustion	A standing-positive finite-horizon witness not represented by the carrier-side horizon discriminator and branch data.
Branch elimination	A surviving branch among $S1$ – $S5$ compatible with the certified carrier and same-scope closure.
Route-locus audit	A same-scope endpoint-defeating route not entering Appendix F.
Official endpoint correspondence	A reason Corollary 21A and Theorem 21.17 fail to state Fefferman (B).

H.5 Lean / A+ formal audit boundary

The manuscript-facing structural surface is supported by the dedicated Lean4 A+ audit archive *AASC-NavierStokes-Lean-Audit*, Version 1.0.2, archived at DOI 10.5281/zenodo.20524517 [Mal26]. The audit layer packages theorem routing, primitive route, continuation and obstruction layers, obligation ledgers, endpoint audit, and manuscript correspondence for the proof spine.

The recorded checks include `lake build`, the clean axiom check for the paper-surface summary, and the A+ audit checkpoint script. The release records Lean `leanprover/lean4:v4.28.0`; the audited surface reports no live project-level `axiom`, `sorry`, `admit`, or `unsafe` declarations in the audited `MaLeYLean / Checks` files. It also records the compact route as 36/36 load-bearing obligations routed or classified as imported analytic boundary, with the expanded manuscript inventory classifying 492 theorem / definition / label rows.

This formal layer is support, not a substitute for the manuscript proof. It does not claim to reprove every classical PDE theorem inside Lean. The local periodic analytic inputs used in the endpoint proof are either internalized in Appendix B or kept inside the explicitly named analytic-background boundary.

H.6 Modernization patch ledger

This edition modernizes the defense format without changing the proof strategy. The front matter is compressed to a proof-class lock, counterexamplehood lock, horizon proof spine, vortex-stretching seam lock, and fast denial table. The local endpoint closeout now explicitly uses the horizon bridge object $B_{\text{hor}}^{NS}(x, T)$ and the exact-complement annotation $[T_*(x) < \infty]_i$. The analytic and endpoint alternatives are summarized in Appendix F, and the theorem ladder is summarized in Appendix G.

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Data Availability: Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

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